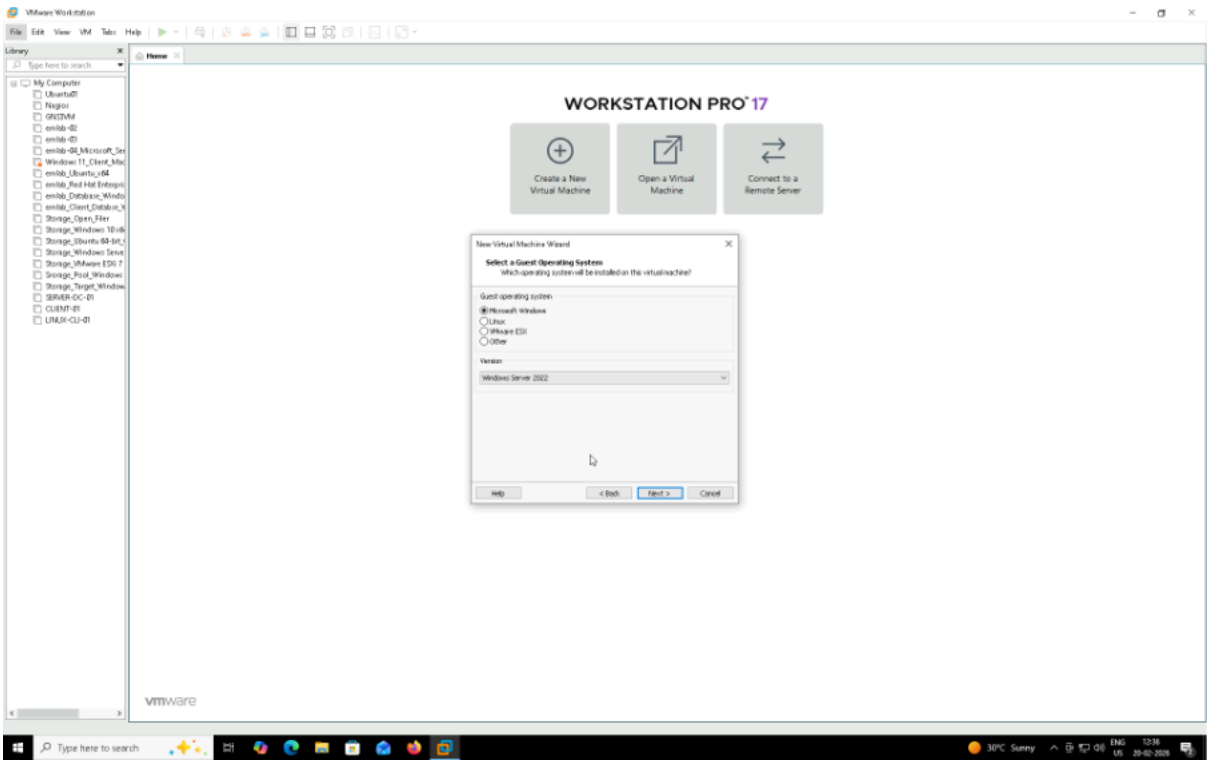
TOPIC – MONITORING

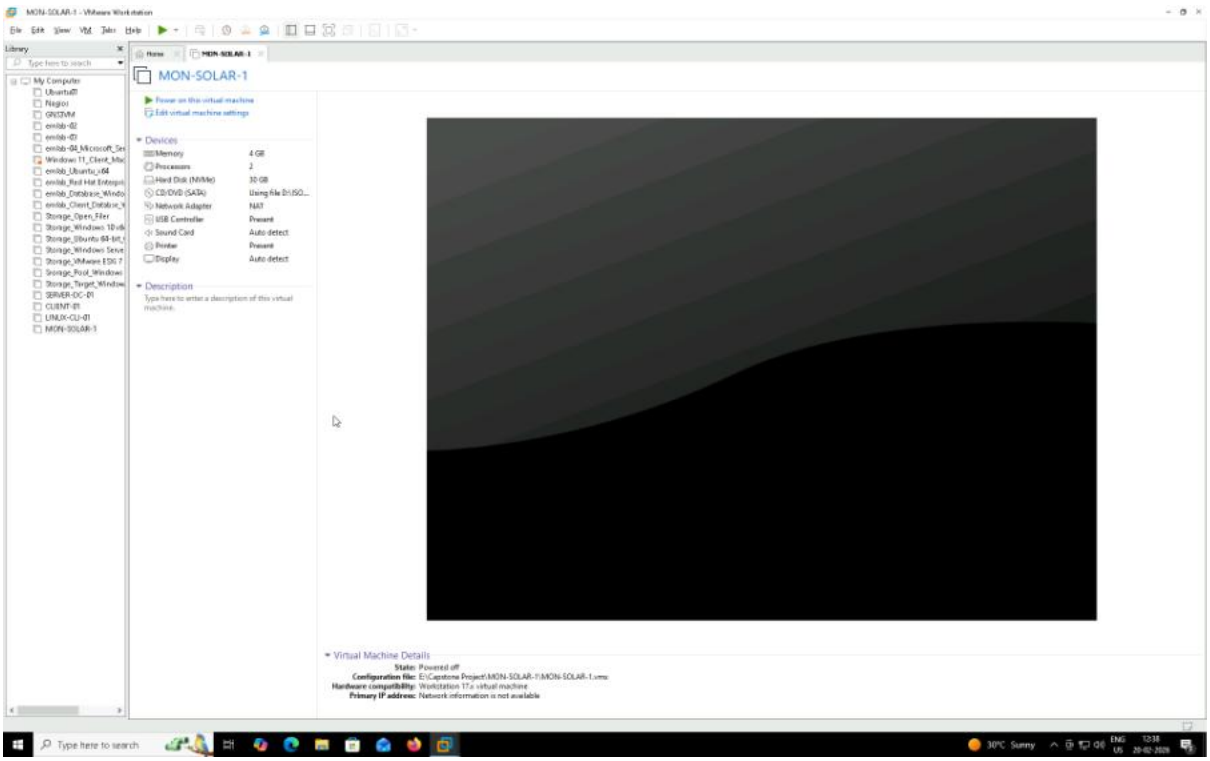
SolarWinds

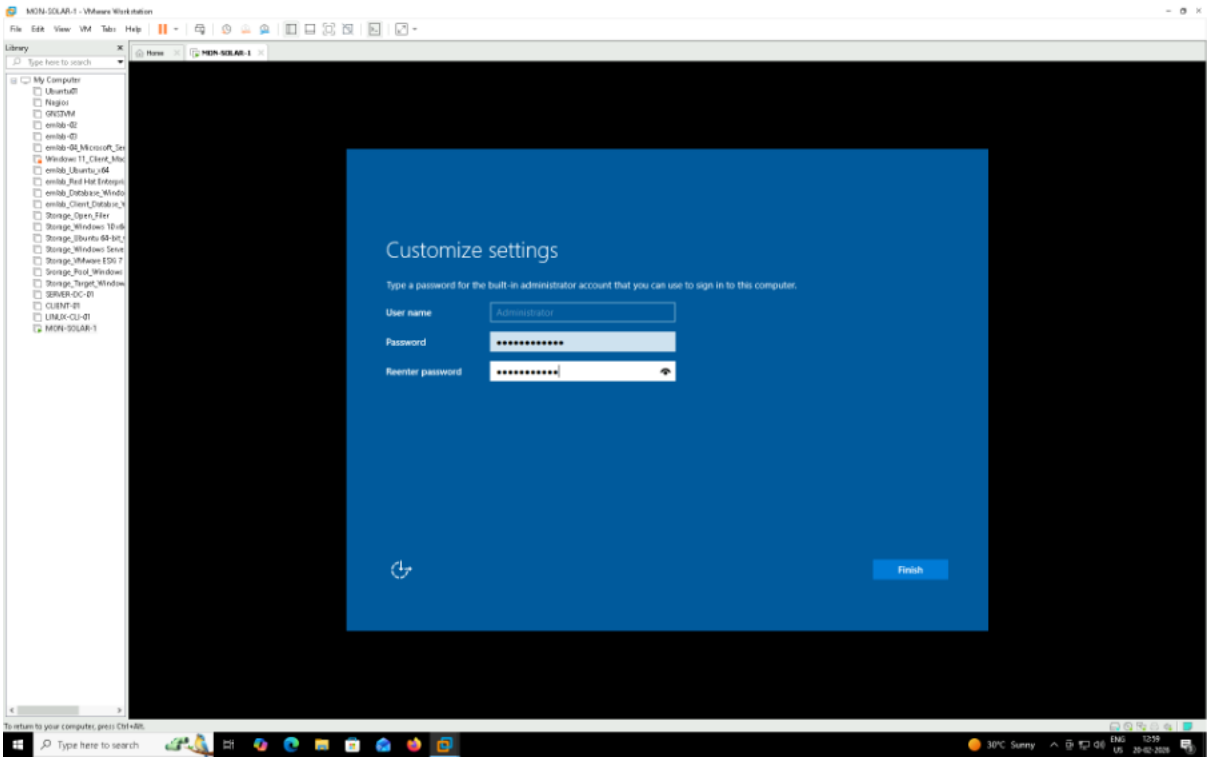
1. Installation and Configuration:
   - Install SolarWinds Orion on a virtual machine or server.
   - Configure the initial setup and connect it to a sample network.

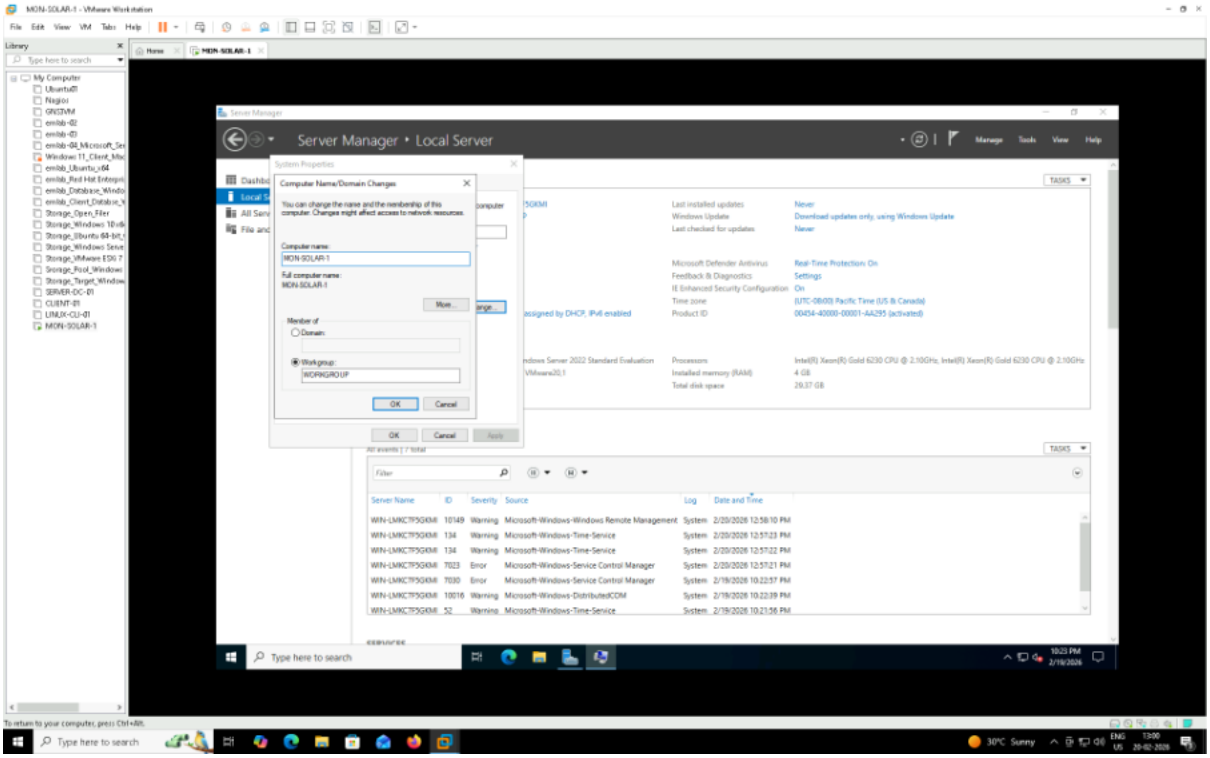
2. Virtual Machine Monitoring:
   - Connect a VM to the monitoring network.
   - Configure alerts and notifications for network events.

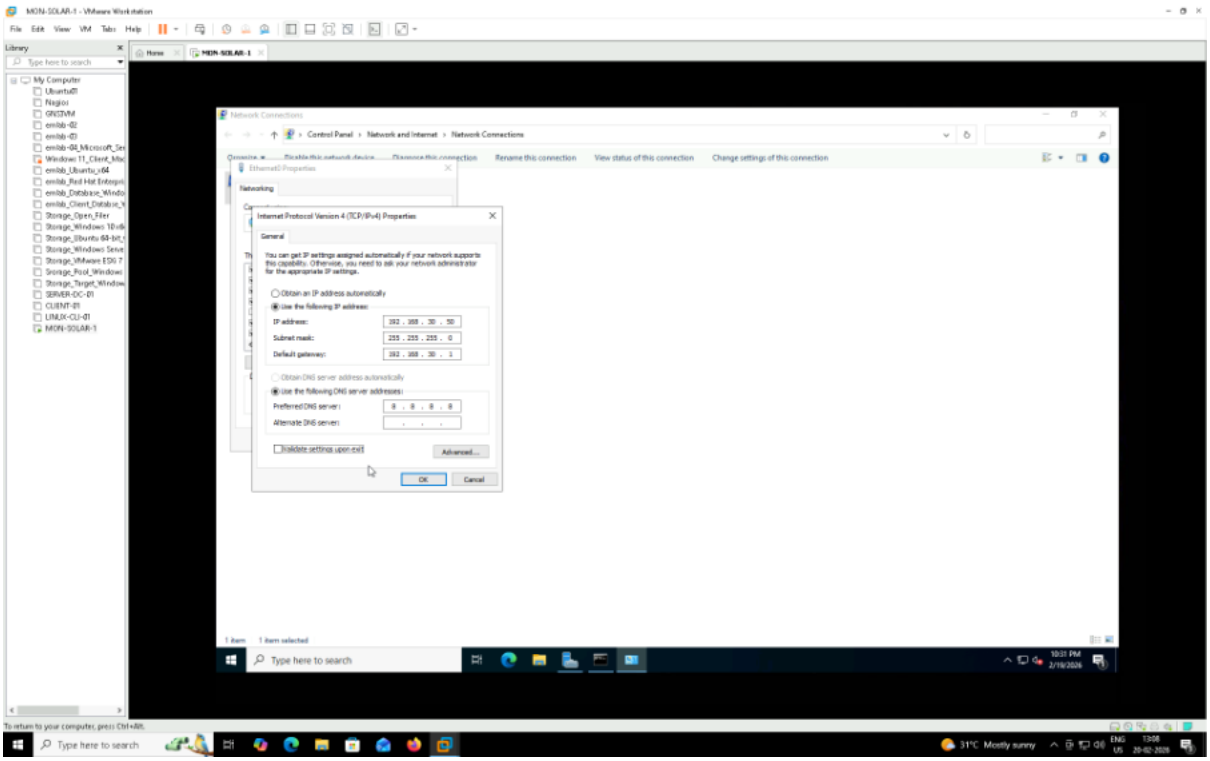
3. Traffic Analysis:
   - Use SolarWinds Network Traffic Analyzer (NTA) to analyse network traffic.

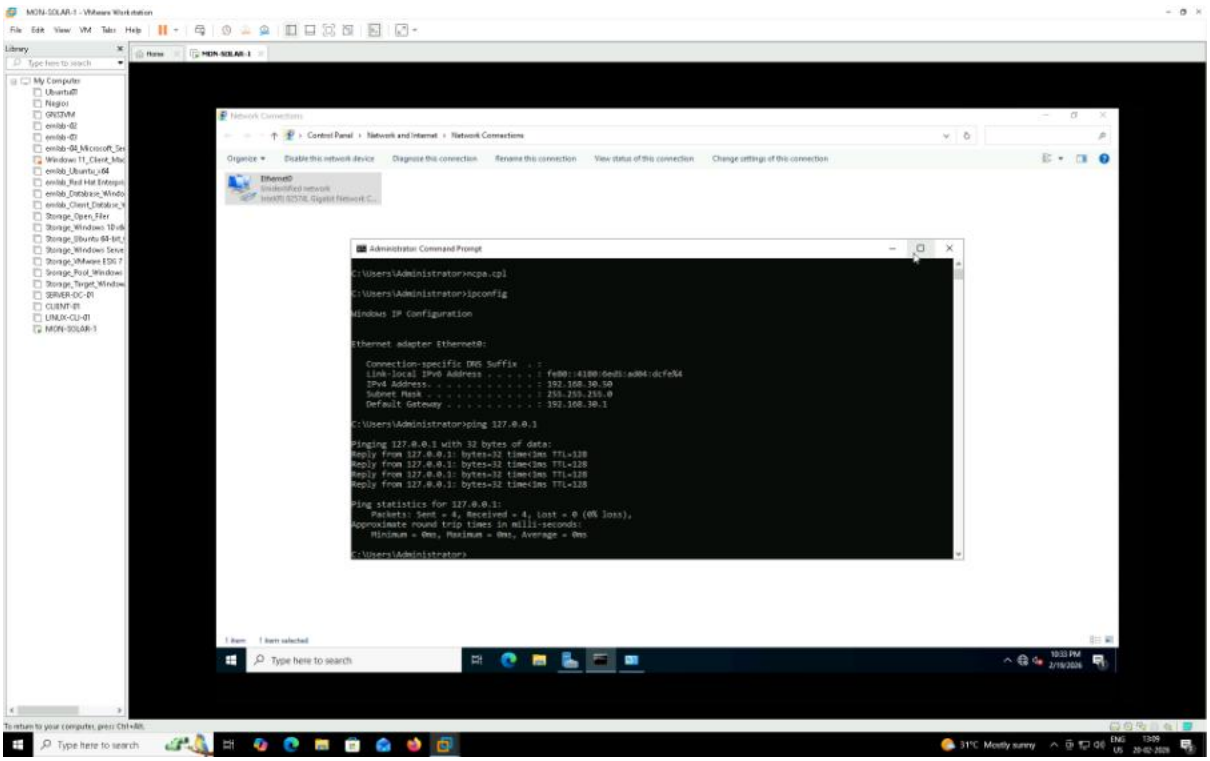


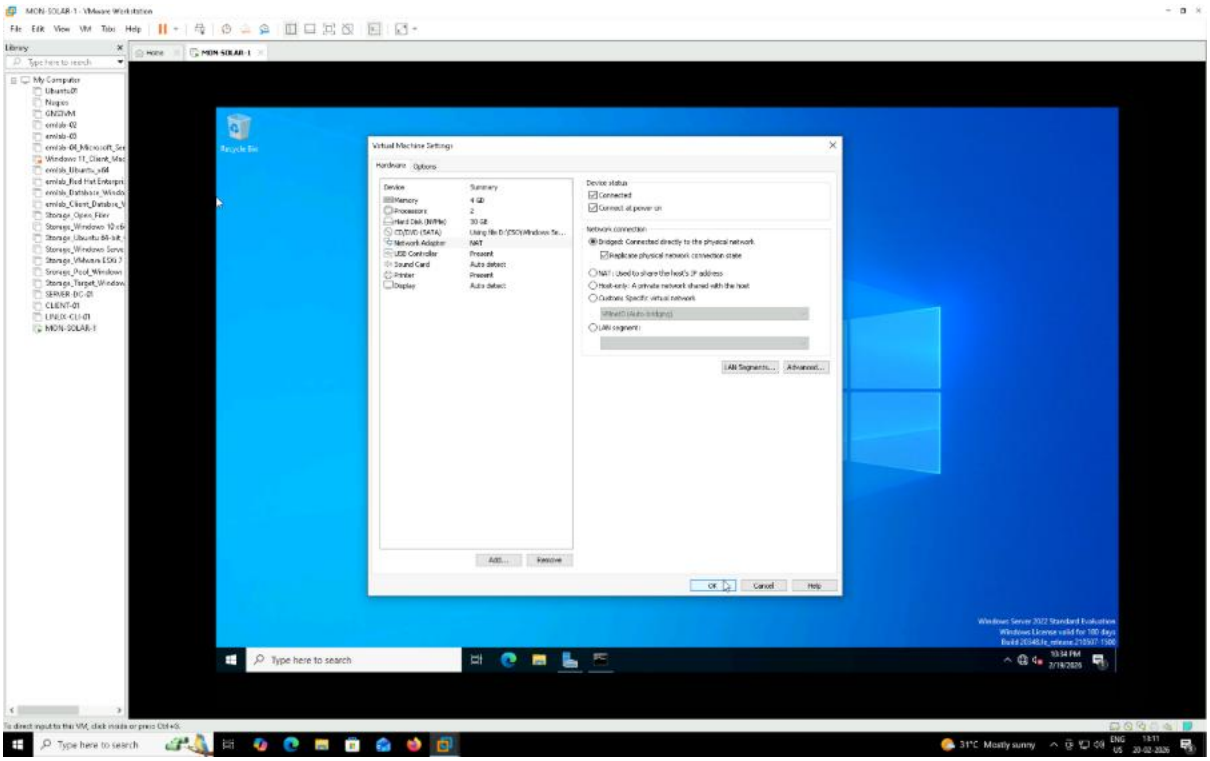

MON-SOLAR-1 - VMware Workstation
MON-SOLAR-1
Devices
Description
Virtual Machine Details
Type here to search

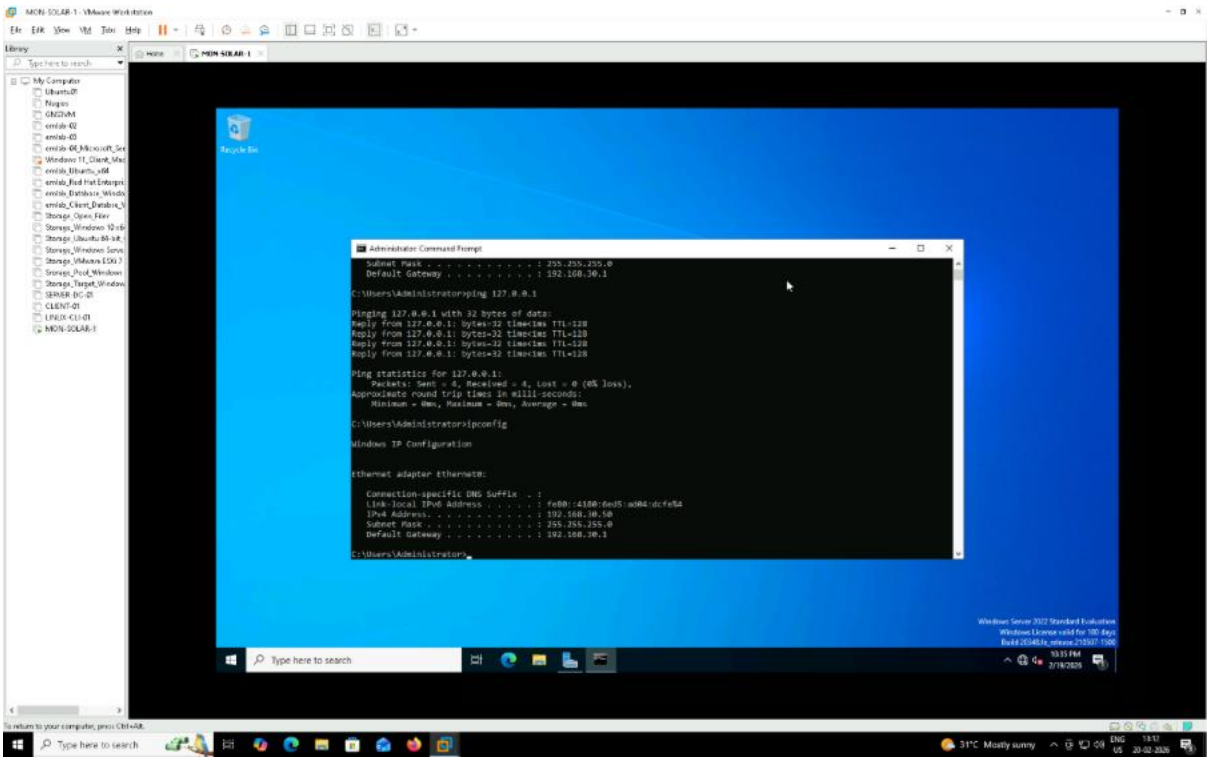

MON-SOLAR-1 - VMware Workstation
Customize settings
Type a password for the built-in administrator account that you can use to sign in to this computer.
User name
Administrator
Password
Reenter password
Finish
Type here to search


MON-SOLAR-1 - VMware Workstation
Server Manager • Local Server
Computer Name/Domain Changes
MON-SOLAR-1
WORKGROUP
OK
Cancel
Type here to search


MON-SOLAR-1 - VMware Workstation
Network Connections
Internet Protocol Version 4 (TCP/IPv4) Properties
General
Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
Type here to search

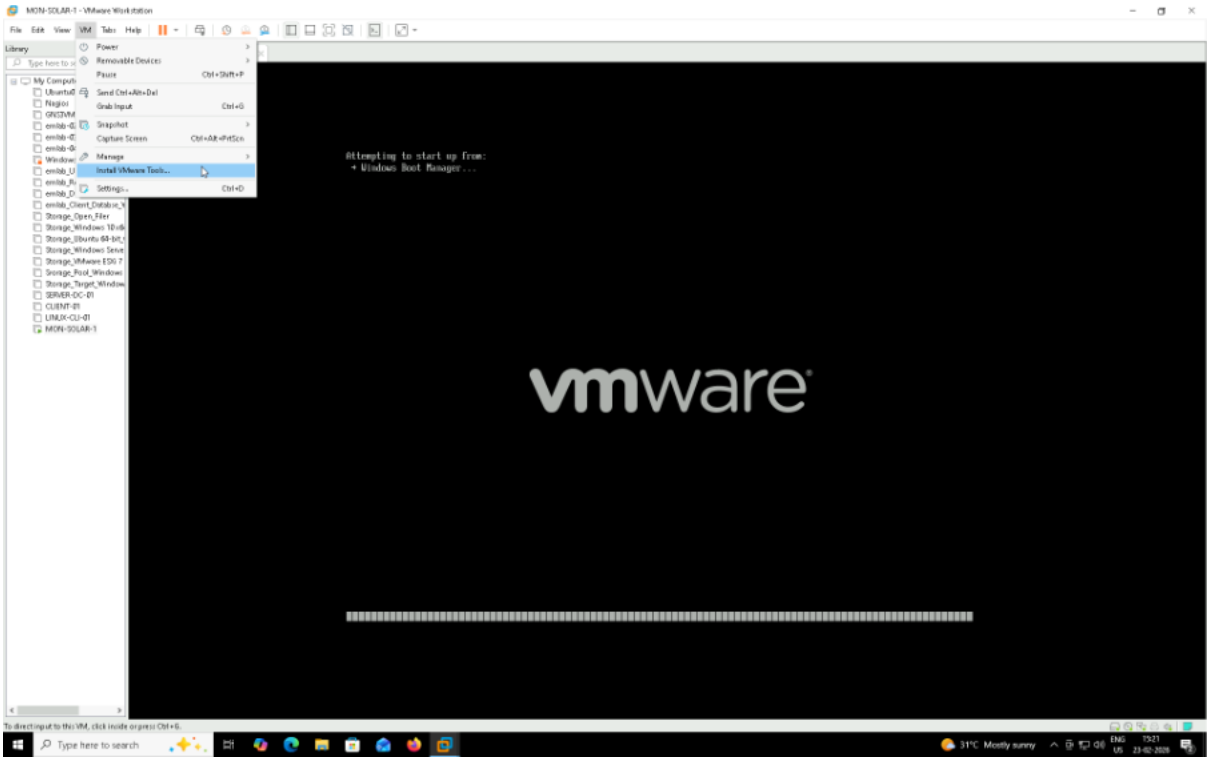

MON-SOLAR-1 - VMware Workstation
Network Connections
Control Panel > Network and Internet > Network Connections
Administrator: Command Prompt
C:\Users\Administrator>ncpa.cpl
C:\Users\Administrator>ipconfig
Windows IP Configuration
Ethernet adapter Ethernet0:
Connection-specific DNS Suffix . :
Link-local IPv6 Address . . . . . : fe80::4180:6ed5:ad04:dcfe%4
IPv4 Address. . . . . : 192.168.30.50
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.30.1
C:\Users\Administrator>ping 127.0.0.1
Pinging 127.0.0.1 with 32 bytes of data:
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Reply from 127.0.0.1: bytes=32 time<1ms TTL=128
Ping statistics for 127.0.0.1:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 0ms, Average = 0ms
C:\Users\Administrator>
Type here to search

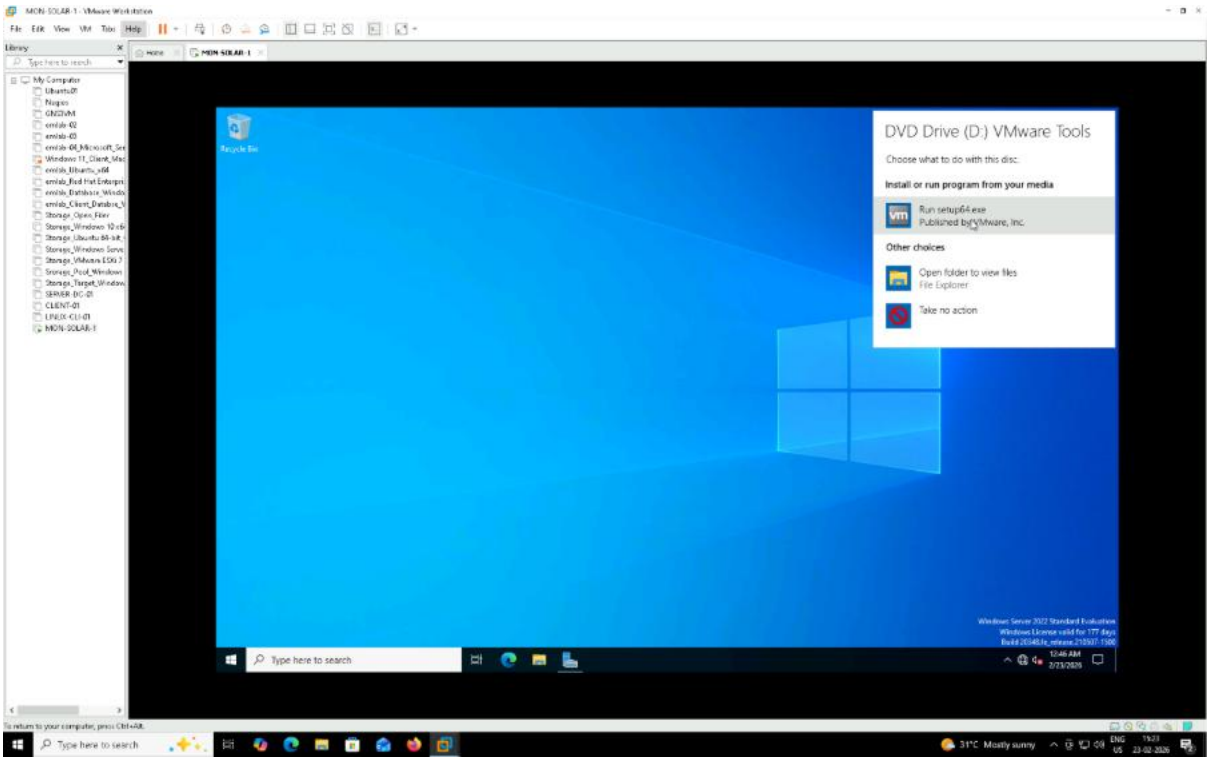

MON-SOLAR-1 - VMware Workstation
Virtual Machine Settings
Hardware
Options
Device
Summary
Memory
4 GB
Processors
2
Hard Disk (NVMe)
30 GB
CD/DVD (SATA)
Network Adapter
NAT
USB Controller
Present
Sound Card
Auto detect
Printer
Present
Display
Auto detect
Add...
Remove
Device status
Connected
Connect at power on
Network connection
Bridged: Connected directly to the physical network
Replicate physical network connection state
NAT: Used to share the host's IP address
Host-only: A private network shared with the host
Custom: Specific virtual network
LAN segment:
LAN Segments...
Advanced...
OK
Cancel
Help
Windows Server 2022 Standard Evaluation
Type here to search



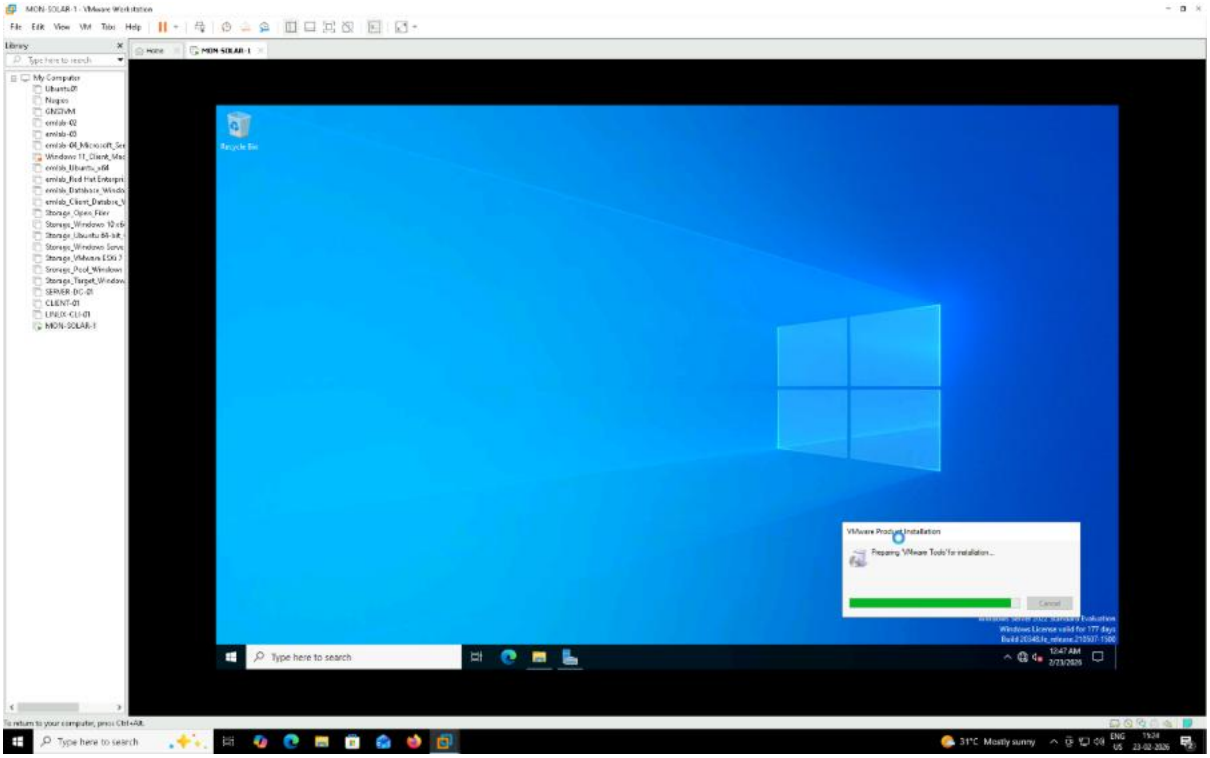
A Windows Server virtual machine MON-SOL-1 was configured with a static IP Address to server as the monitoring server. Necessary network settings were verified to ensure connectivity.

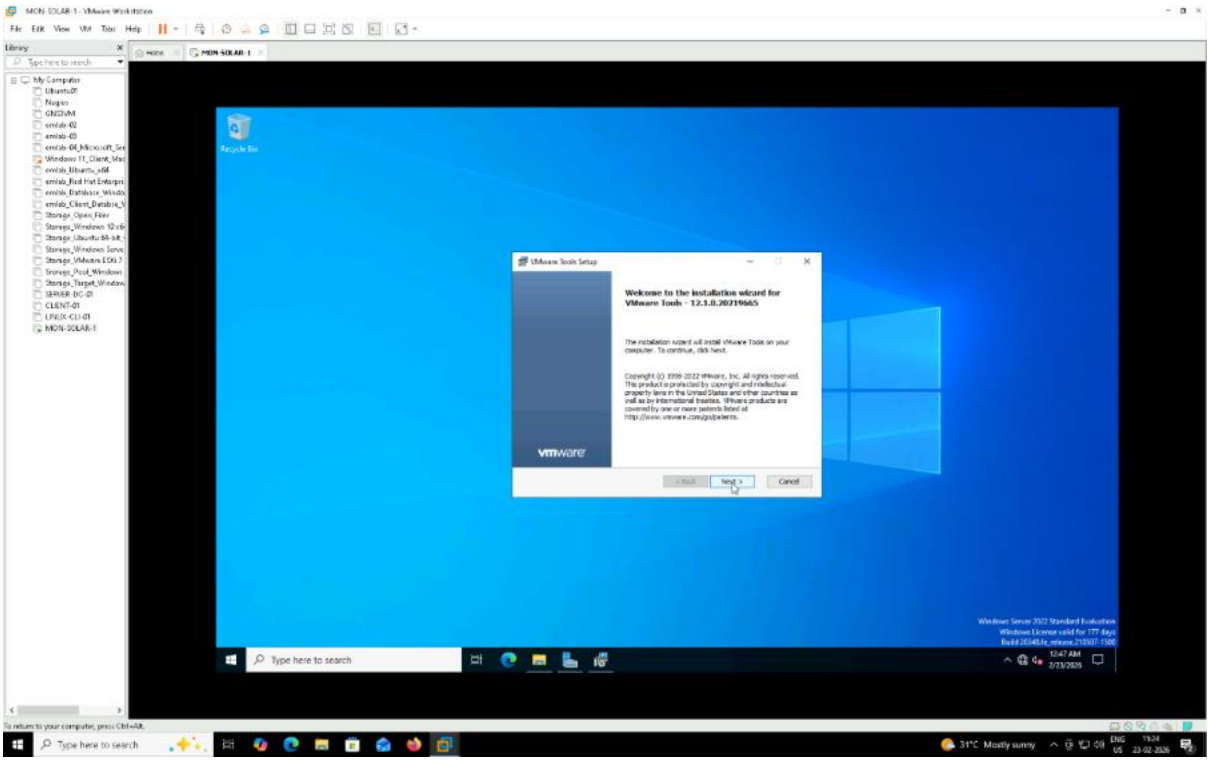
Now we will install VM tools across the VMs so that we can send the setup files in - between them.

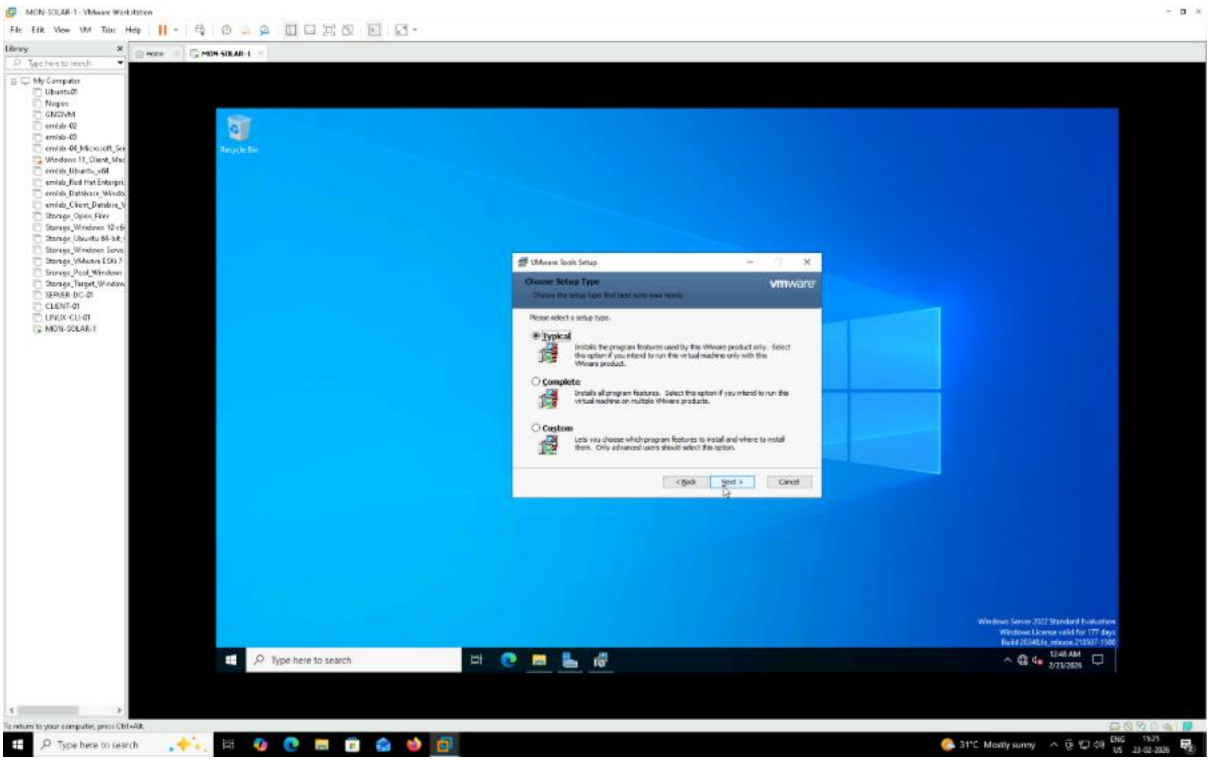




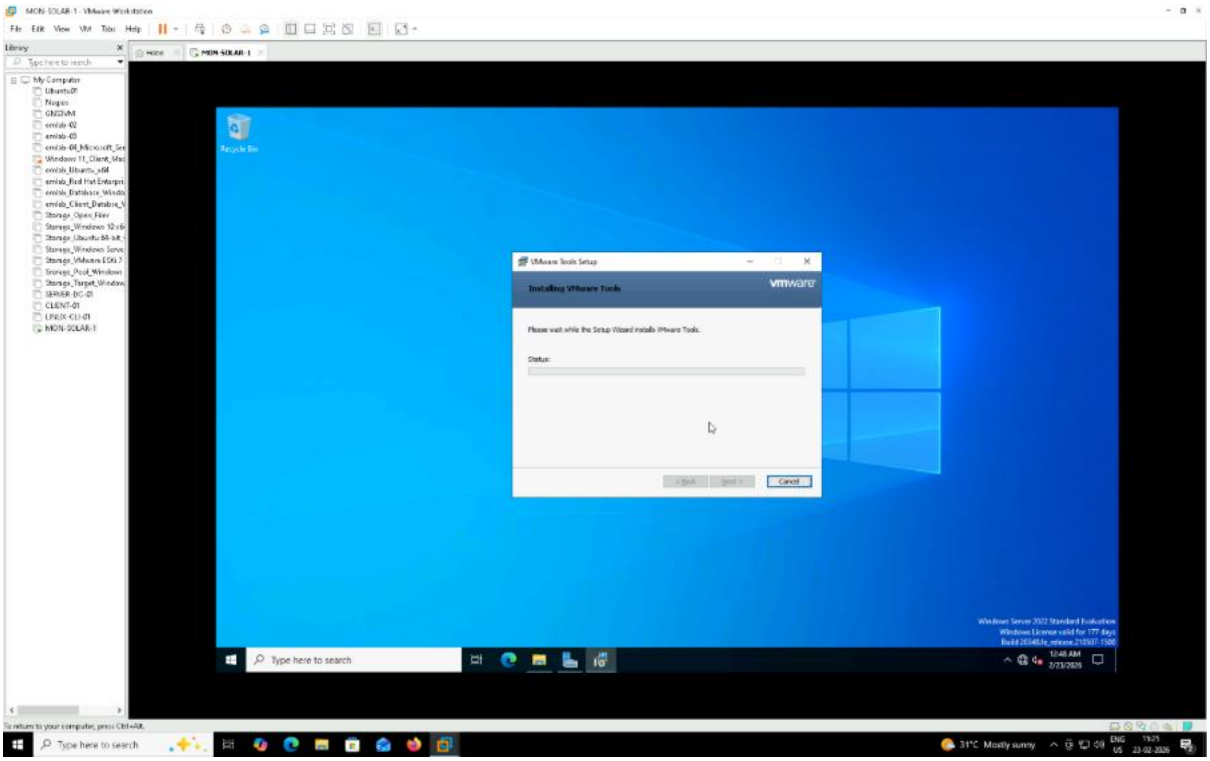
Run the setup64.exe file

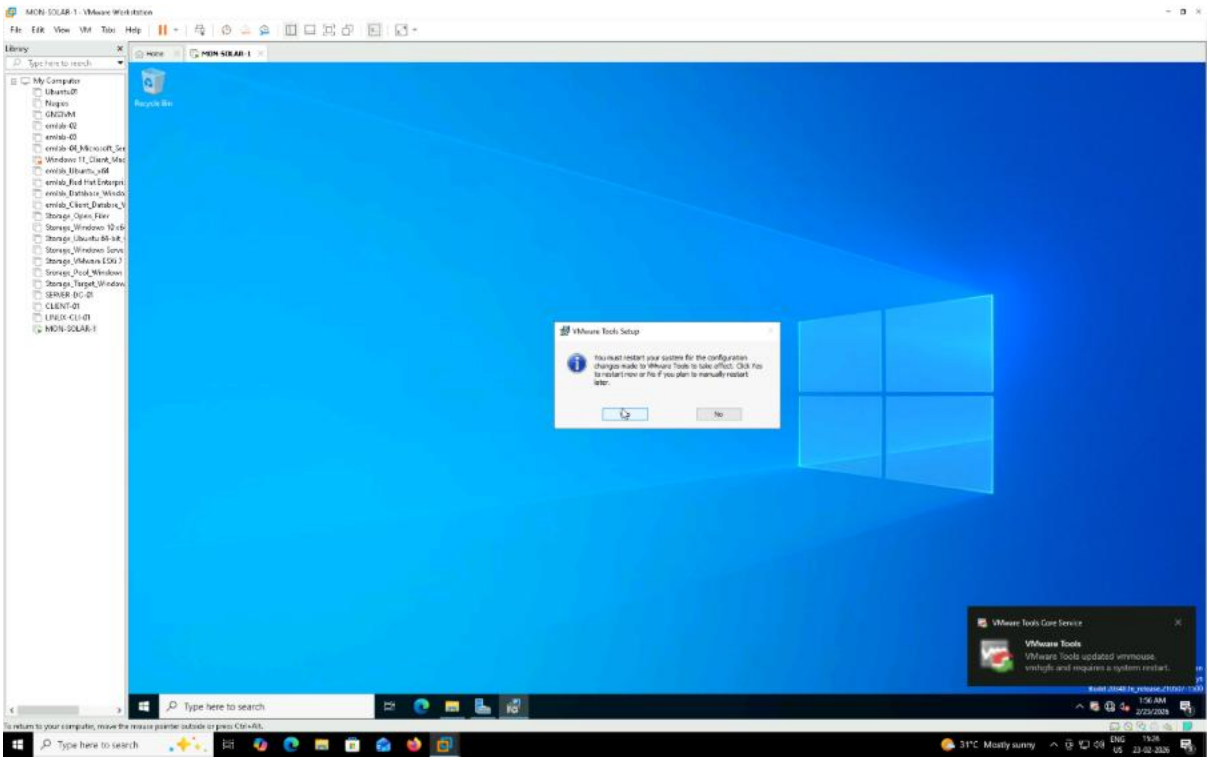

MON-SOLAR-1 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu20
Nagios
GNSVM
emlab-02
emlab-03
emlab-04_Microsoft_Server
Windows 11_Client_Machine
emlab_Ubuntu_x64
emlab_Red Hat Enterprise
emlab_Database_Windows
emlab_Client_Database_V
Storage_Open_Filer
Storage_Windows 10 x64
Storage_Ubuntu 64-bit_1
Storage_Windows Server
Storage_VMware ESXi 7
Storage_Pool_Windows
Storage_Target_Windows
SERVER-DC-01
CLIENT-01
LINUX-CLI-01
MON-SOLAR-1
Home
MON-SOLAR-1
Recycle Bin
VMware Product Installation
Preparing VMware Tools for installation...
Cancel
Type here to search
12:47 AM
2/23/2026
To return to your computer, press Ctrl+Alt.
Type here to search
31°C Mostly sunny
ENG
US
19:24
23-02-2026


MON-SOLAR-1 - VMware Workstation
File Edit View VM Tabs Help
Library
Type here to search
My Computer
Ubuntu20
Nagios
GNSVM
emlab-02
emlab-03
emlab-04_Microsoft_Server
Windows 11_Client_Machine
emlab_Ubuntu_x64
emlab_Red Hat Enterprise
emlab_Database_Windows
emlab_Client_Database_V
Storage_Open_Filer
Storage_Windows 10 x64
Storage_Ubuntu 64-bit_1
Storage_Windows Server
Storage_VMware ESXi 7
Storage_Pool_Windows
Storage_Target_Windows
SERVER-DC-01
CLIENT-01
LINUX-CLI-01
MON-SOLAR-1
Home
MON-SOLAR-1
Recycle Bin
VMware Tools Setup
Welcome to the installation wizard for VMware Tools - 12.1.0.20219665
The installation wizard will install VMware Tools on your computer. To continue, click Next.
Copyright (c) 1998-2022 VMware, Inc. All rights reserved. This product is protected by copyright and intellectual property laws in the United States and other countries as well as by international treaties. VMware products are covered by one or more patents listed at http://www.vmware.com/go/patents.
vmware
< Back
Next >
Cancel
Type here to search
12:47 AM
2/23/2026
To return to your computer, press Ctrl+Alt.
Type here to search
31°C Mostly sunny
ENG
US
19:24
23-02-2026

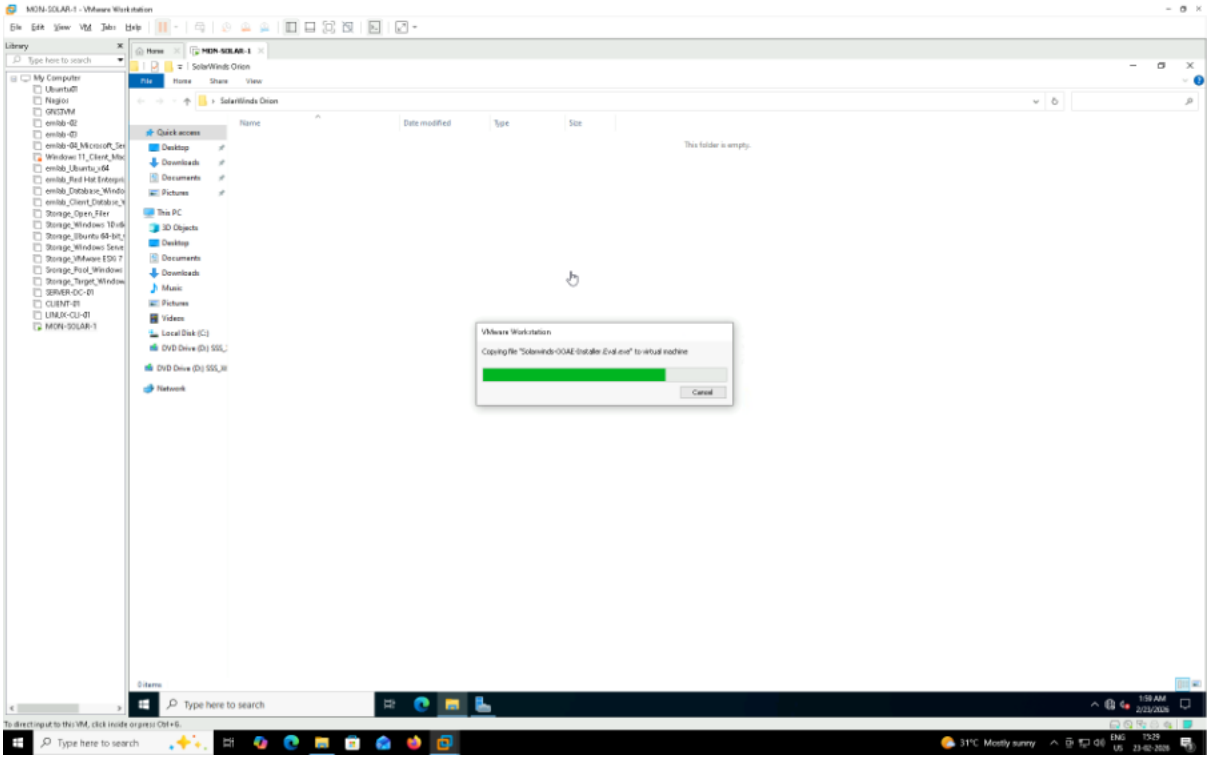


Choose typical mode and click on next.



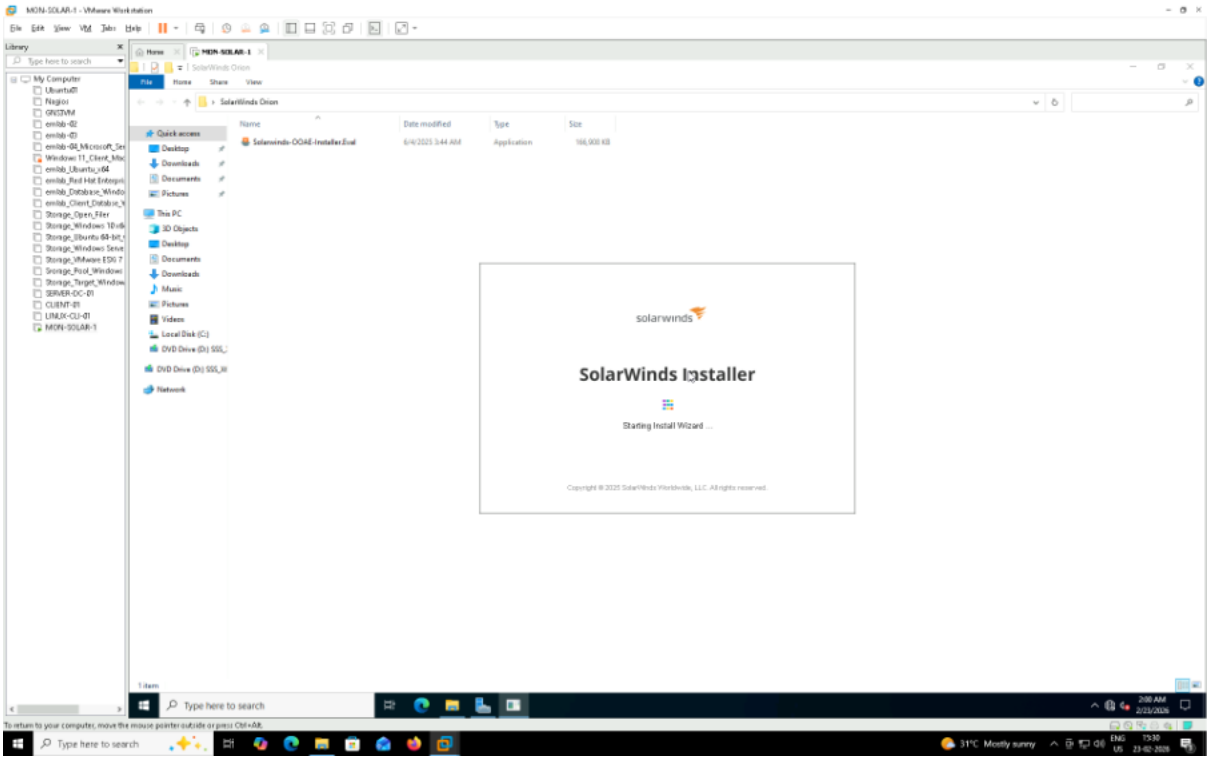


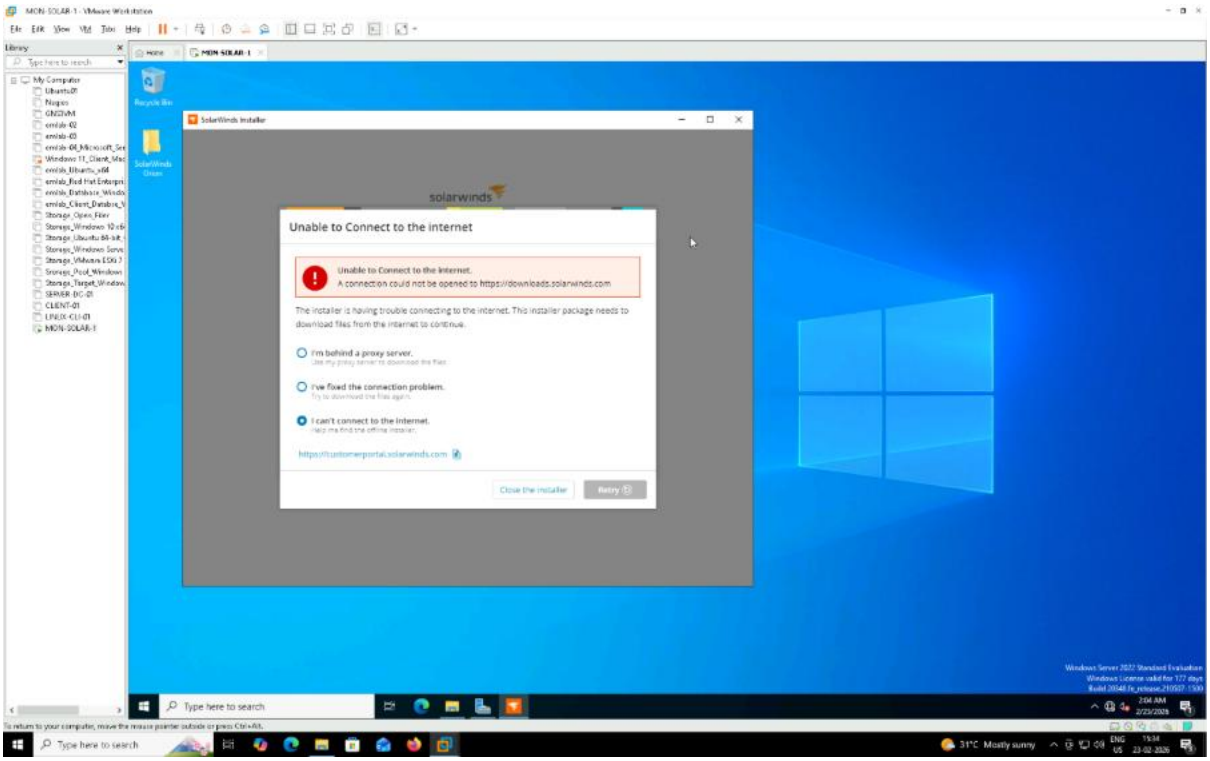
Restart the VM to install the VM tools successfully.



After the successful installation of VM tools, we can copy from one VM to another VM.

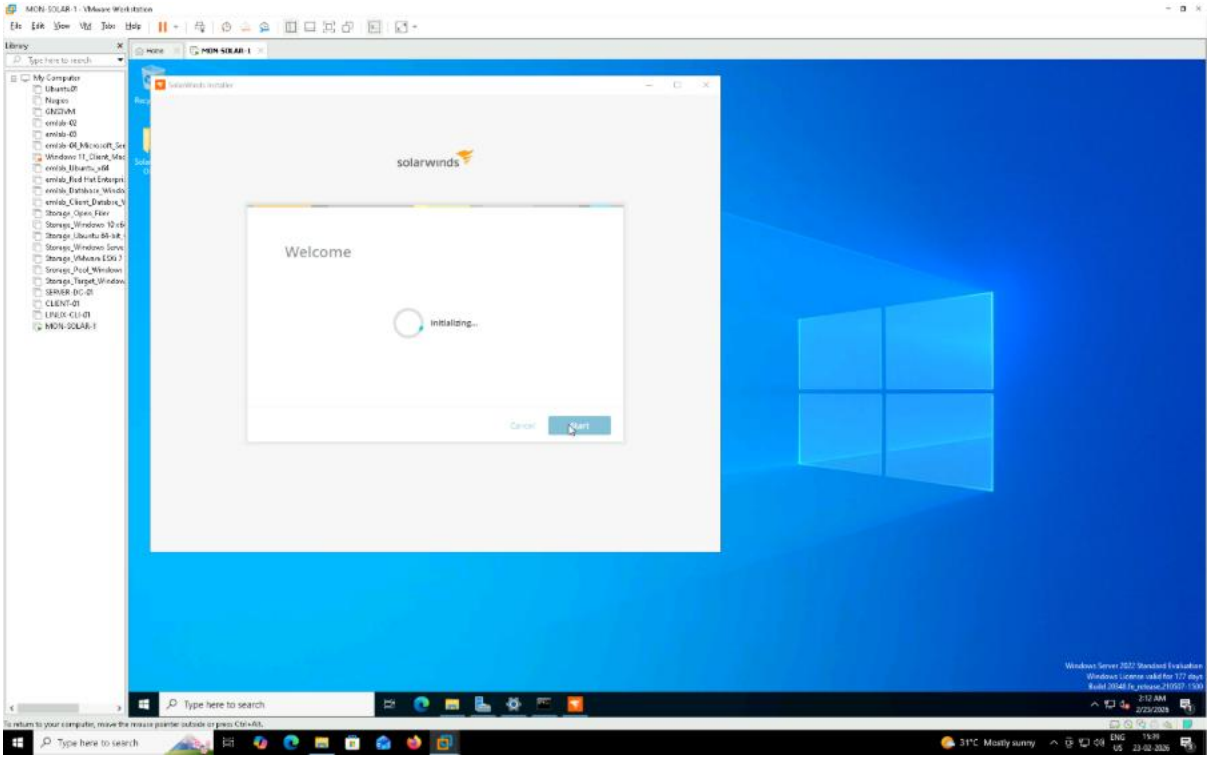
Now install the SolarWinds Orion installer

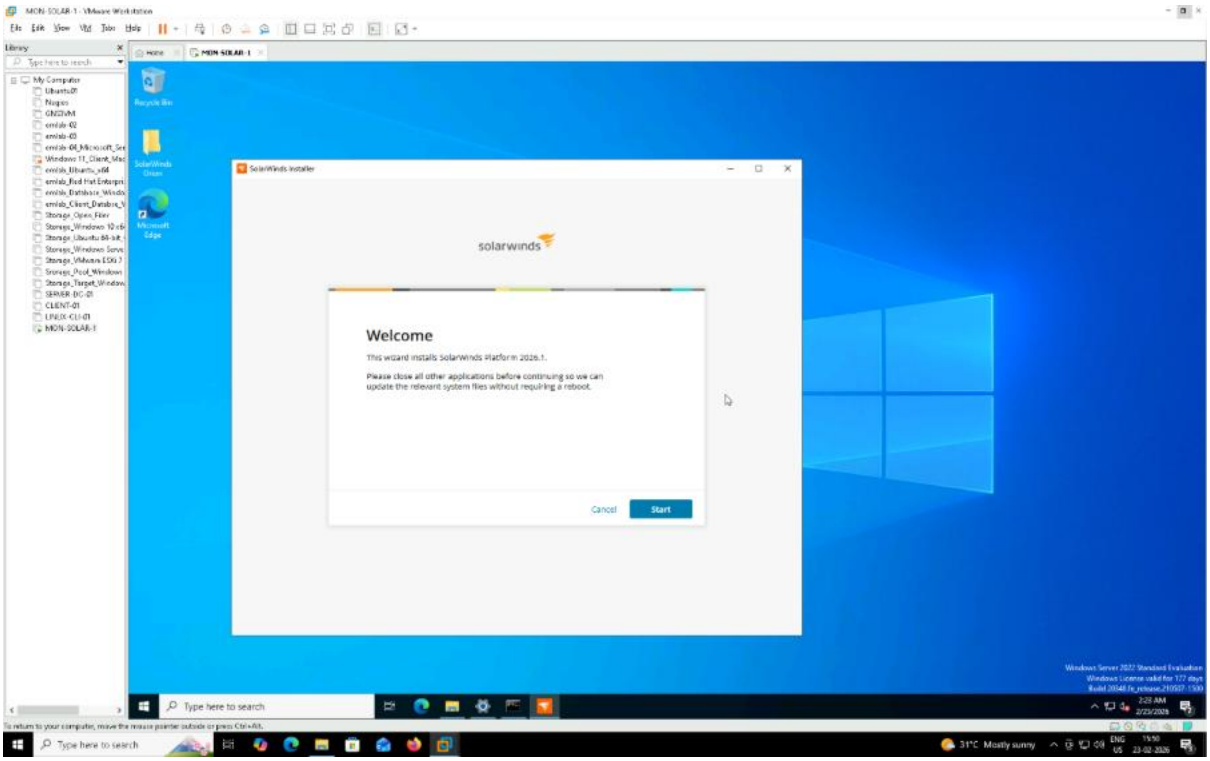




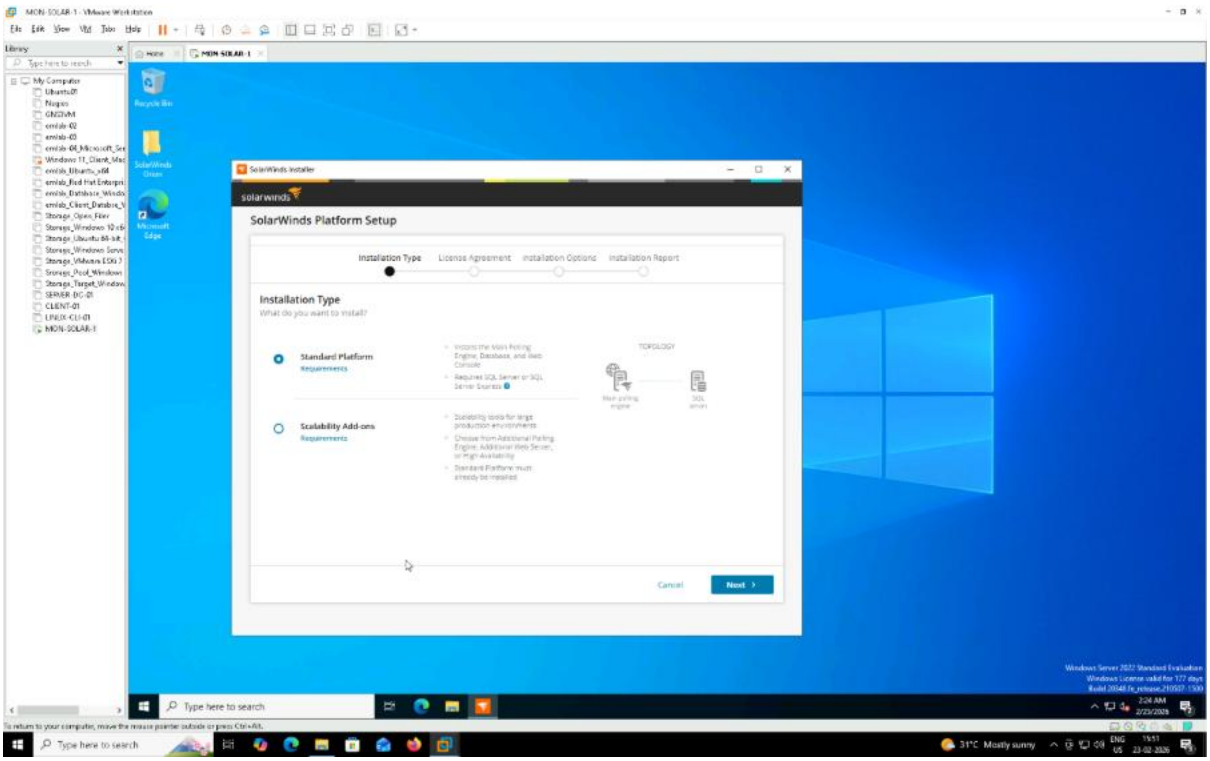
We have faced this issue because we have assigned a Static IP address which isn't connected to the Internet.

For now, we can change the IP configuration to Dynamic and then after installation we can change it back.

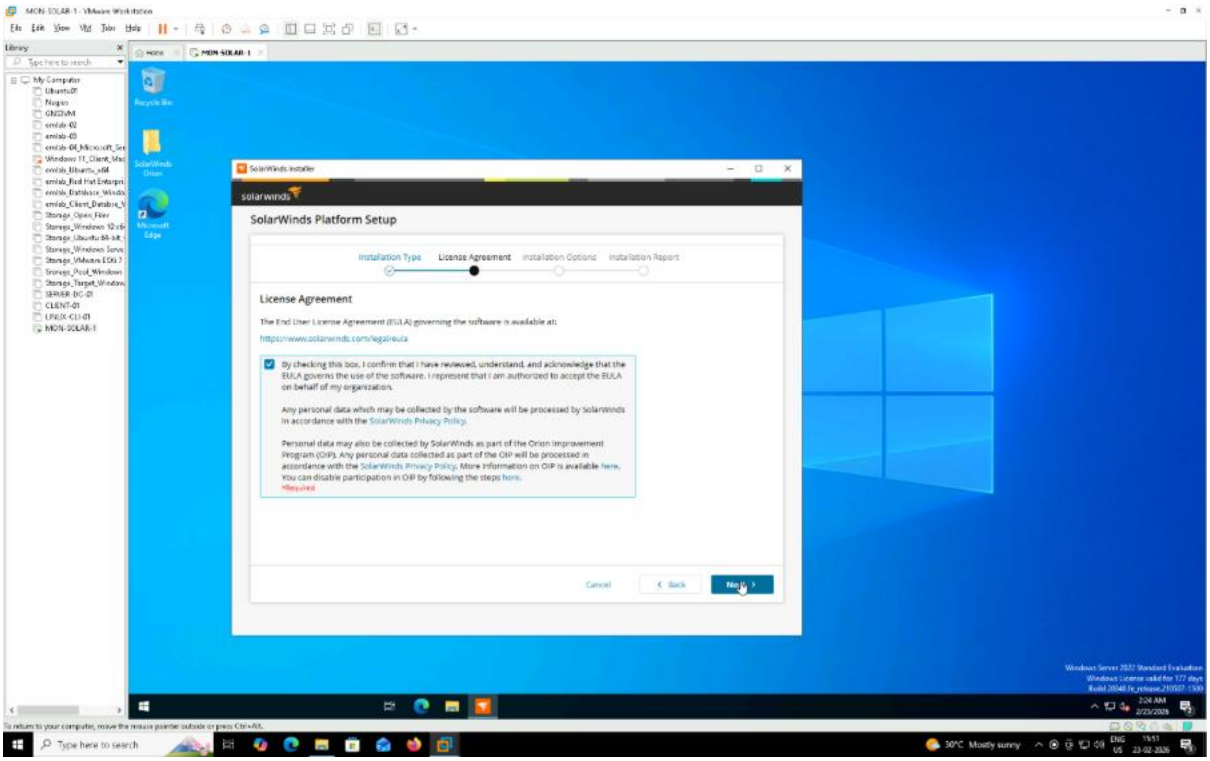




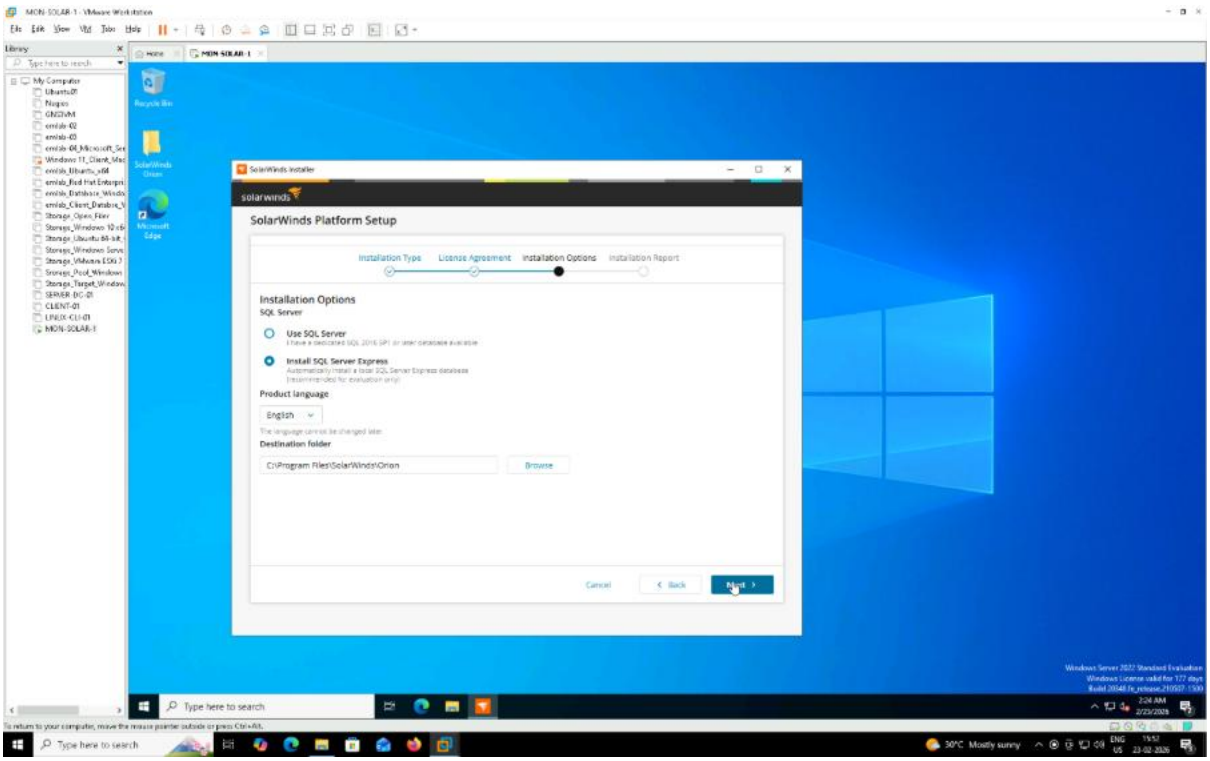
Let’s proceed with the installation of the SolarWinds.



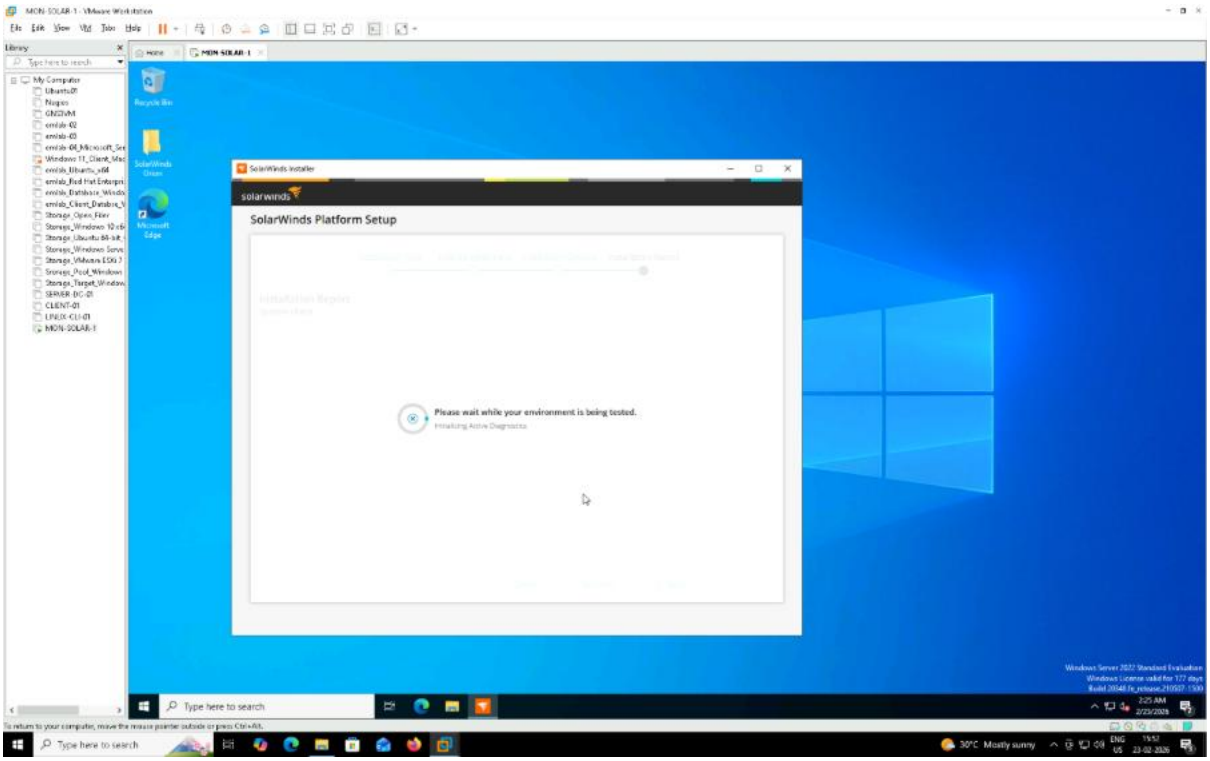
We will choose the Standard Platform option.



Accepting the User end agreement.



Here we will choose the option to Download the SQL Server Express.

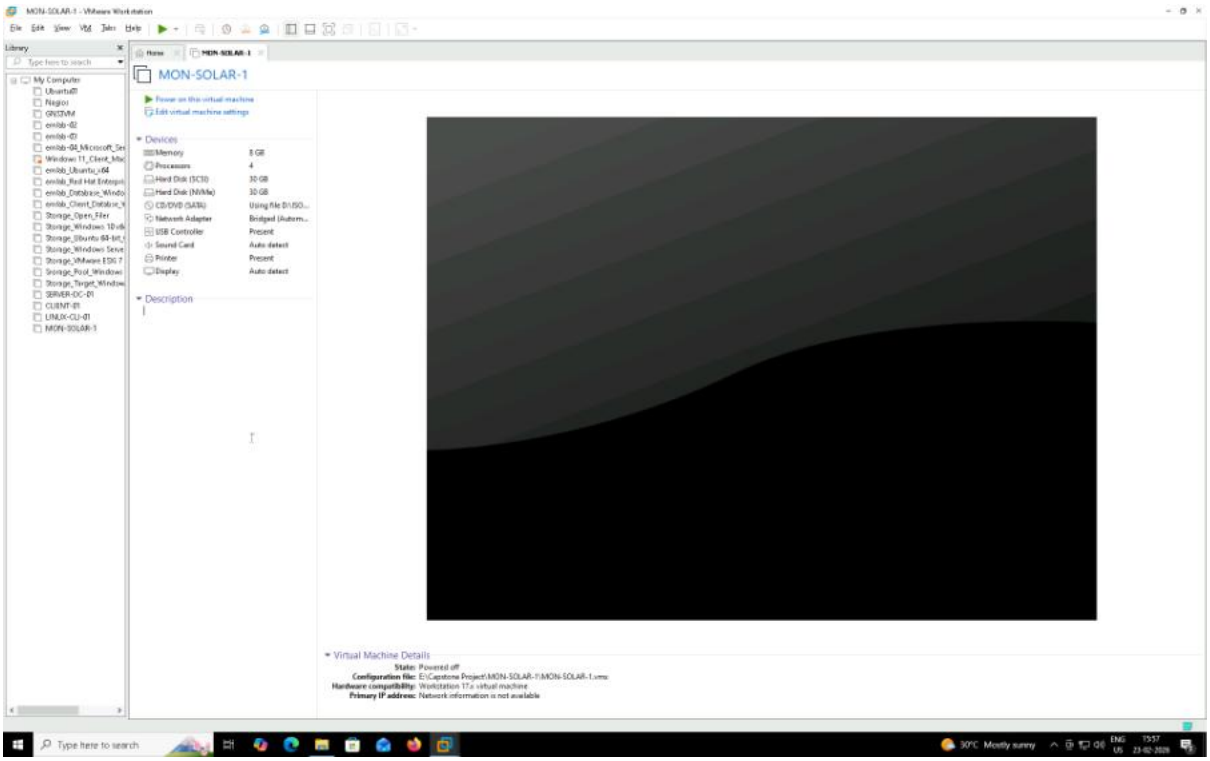


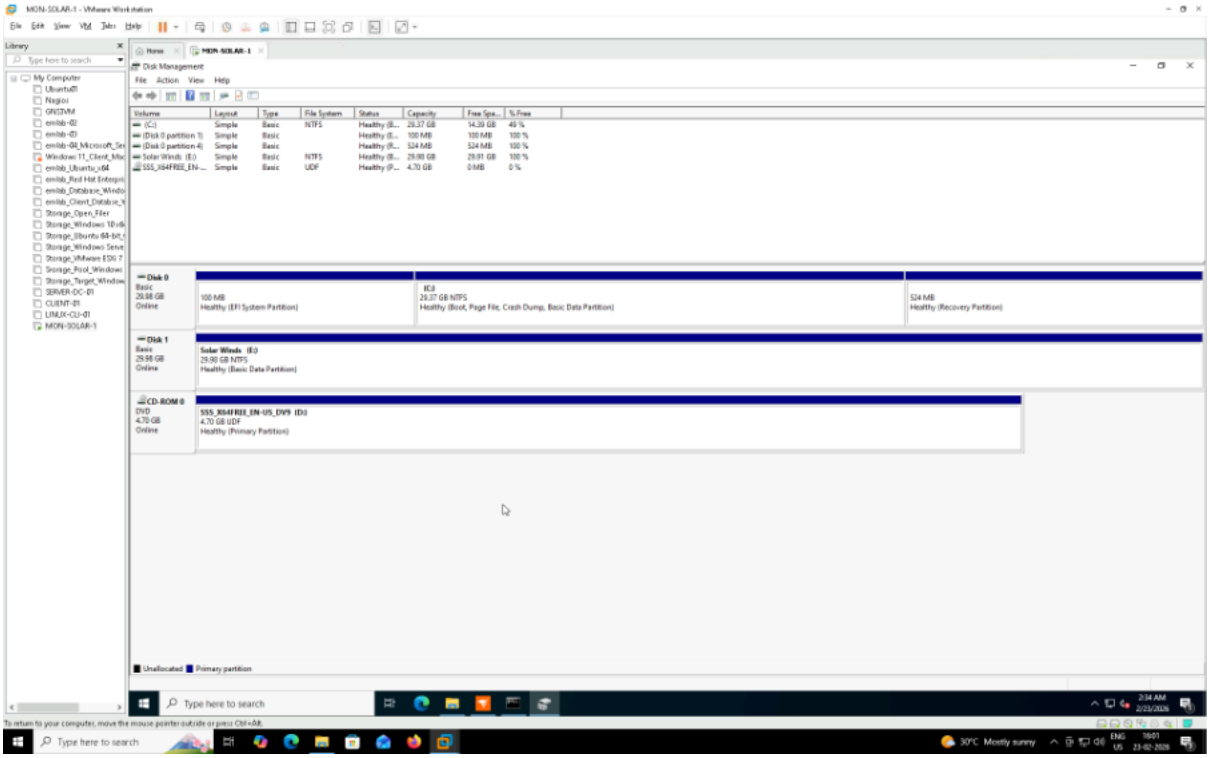
Installation has been initiated.

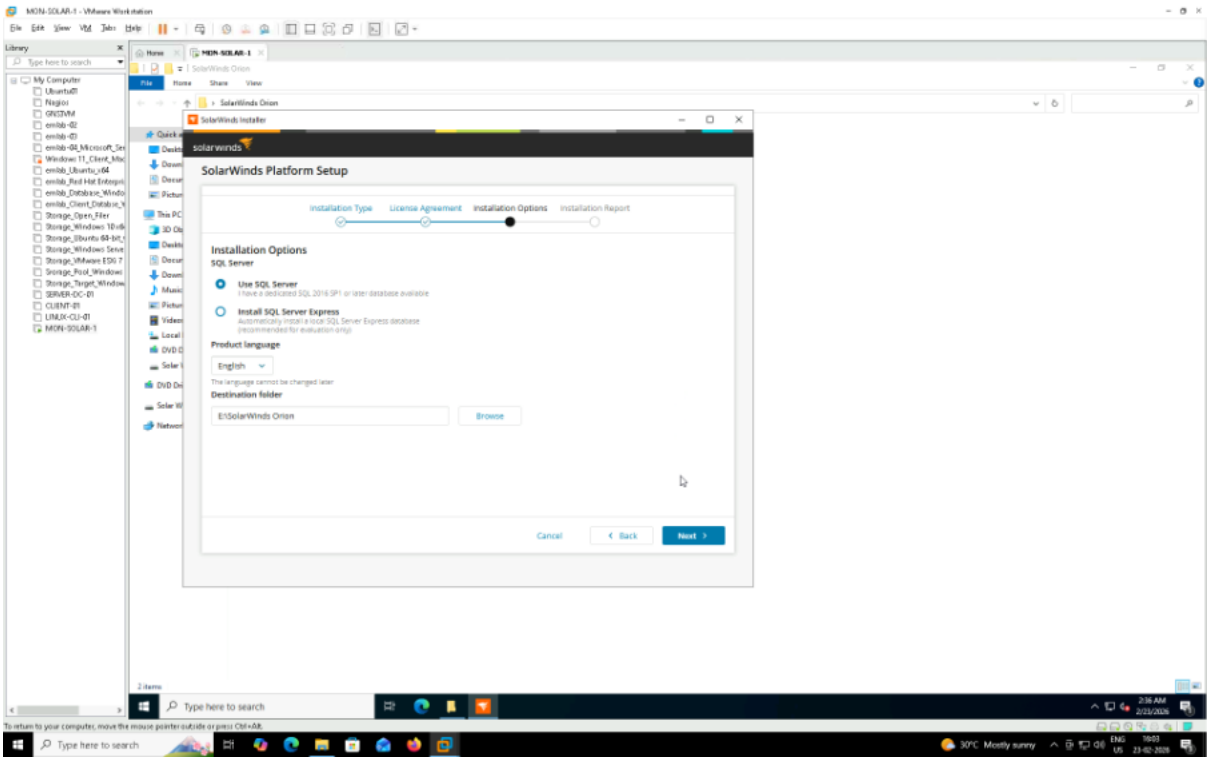
We have got a few issues while doing so,

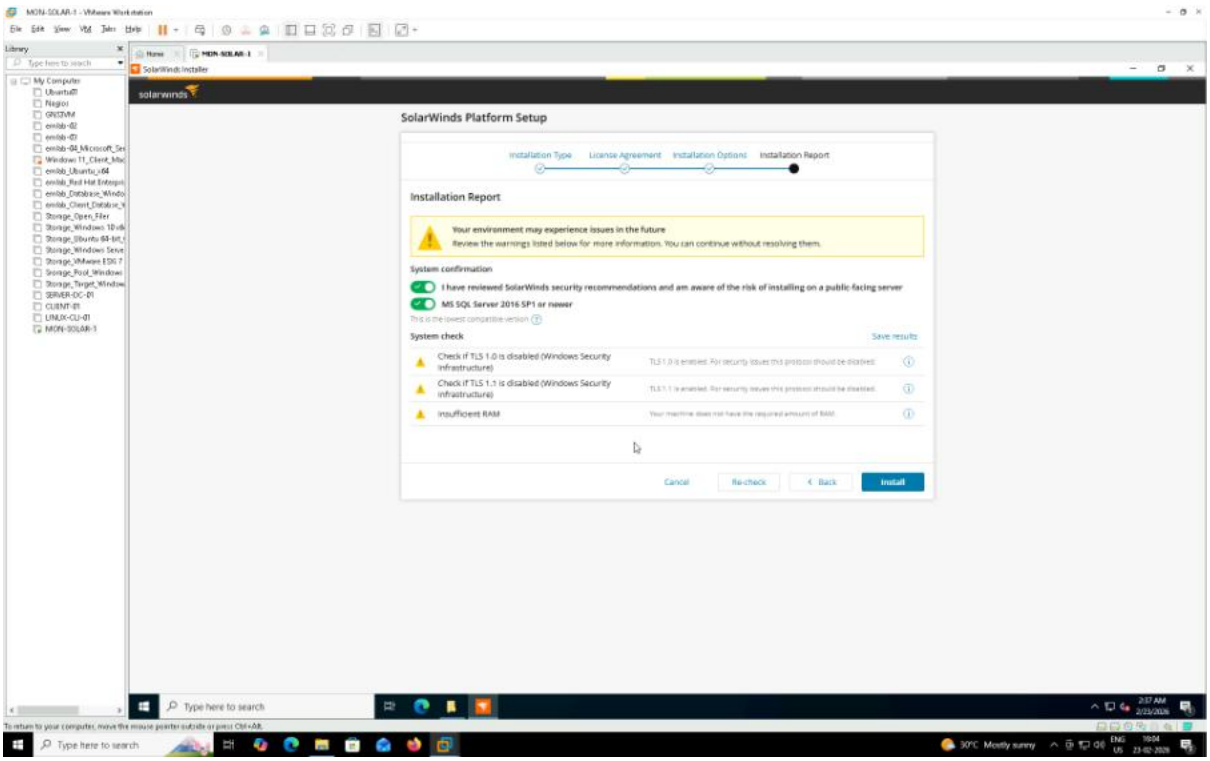
- RAM – 8 GB (Required)
- Cores – 4 (Required)
- Storage Space – 17 GB (Required)

We will now edit the VM settings and match these and proceed further.

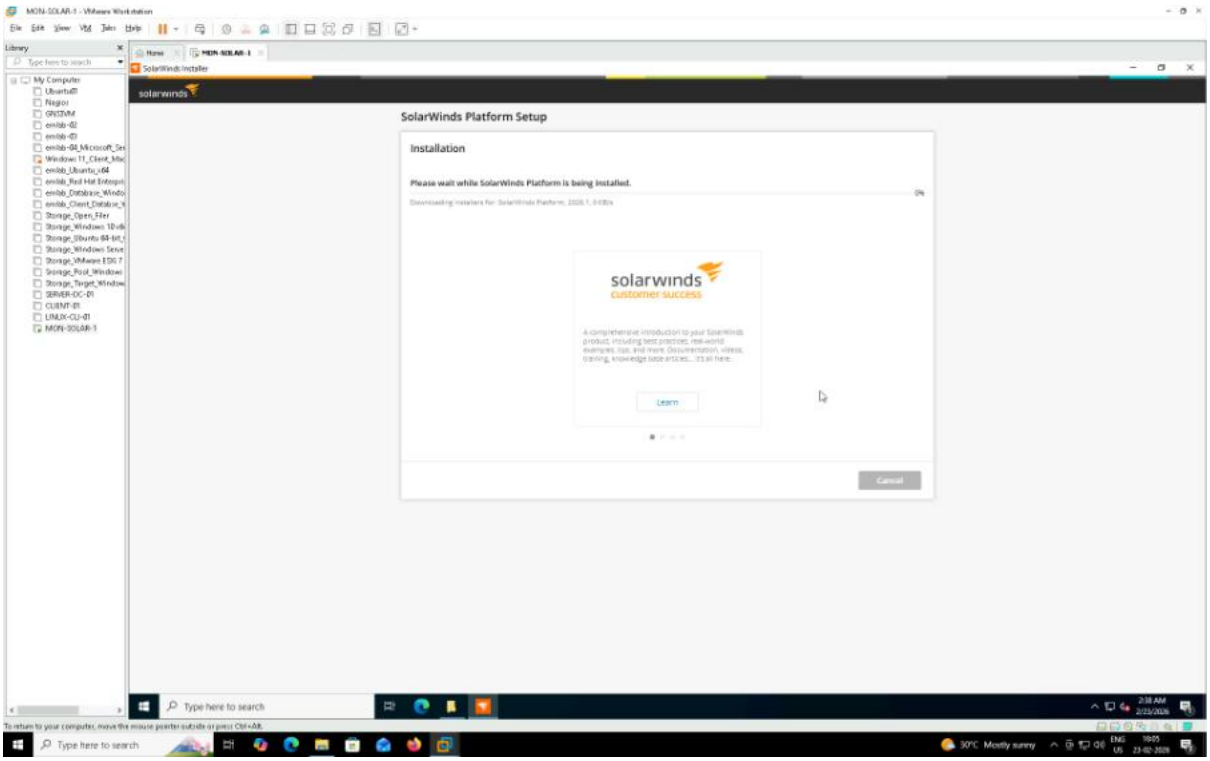



MON-SOLAR-1 - VMware Workstation
Disk Management
Disk 0
Disk 1
CD-ROM 0
Solar Winds (E:)
Unallocated
Primary partition

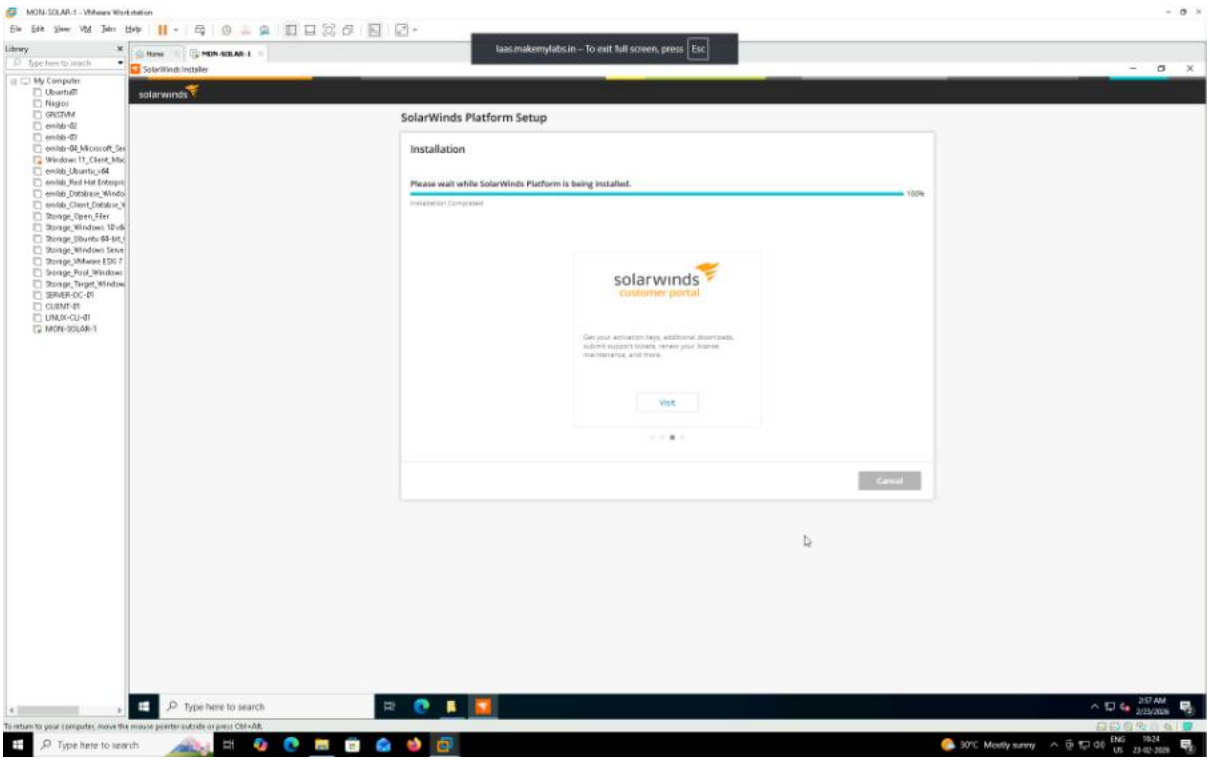

MON-SOLAR-1 - VMware Workstation
SolarWinds Installer
SolarWinds Platform Setup
Installation Type
License Agreement
Installation Options
Installation Report
Installation Options
SQL Server
Use SQL Server
Install SQL Server Express
Product language
English
Destination folder
E:\SolarWinds Orion
Browse
Cancel
Back
Next



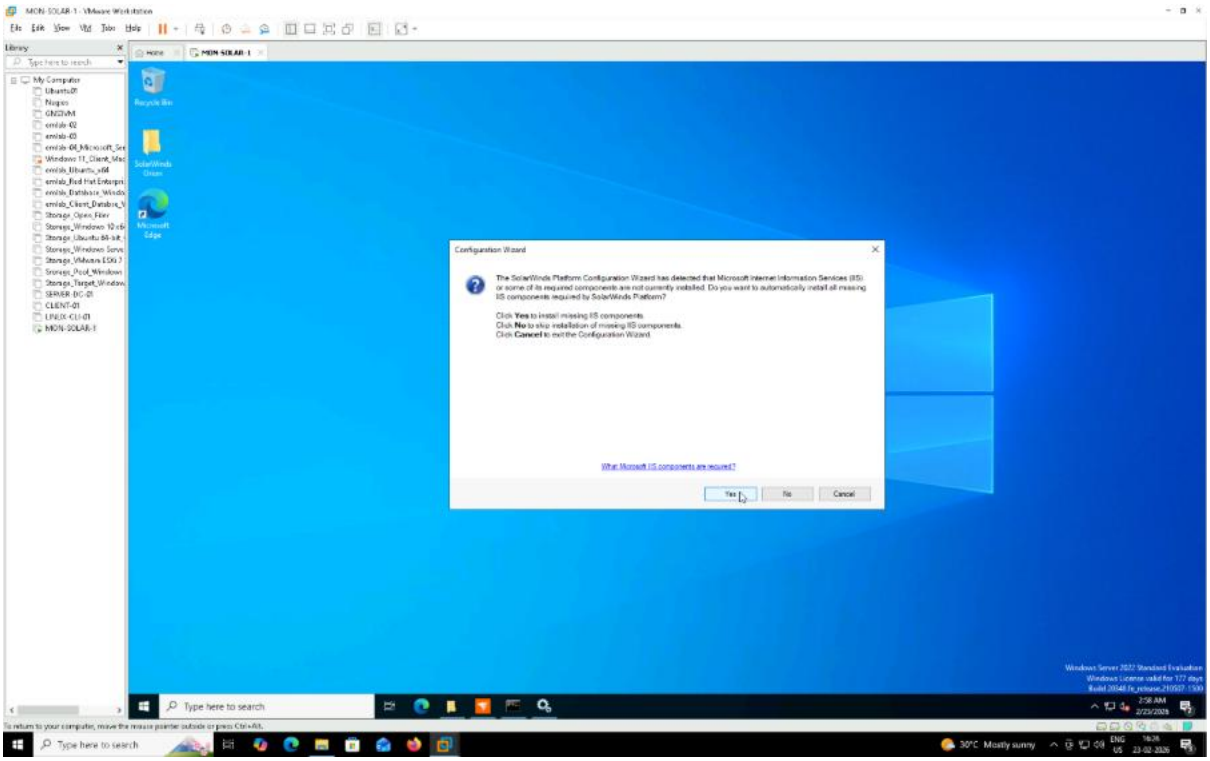
The monitoring server was provisioned with 8 GB RAM, which is sufficient for small-scale lab deployment of SolarWinds Orion platform using SQL Express.



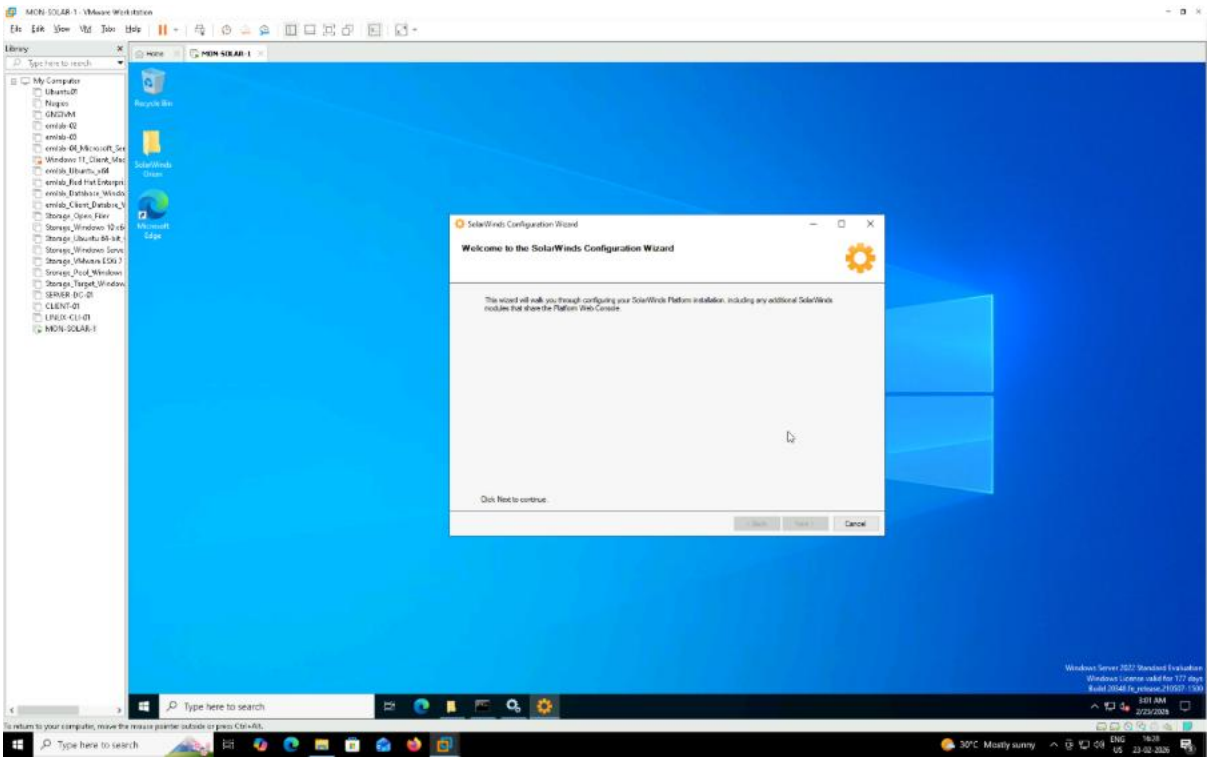
Installation has started successfully.

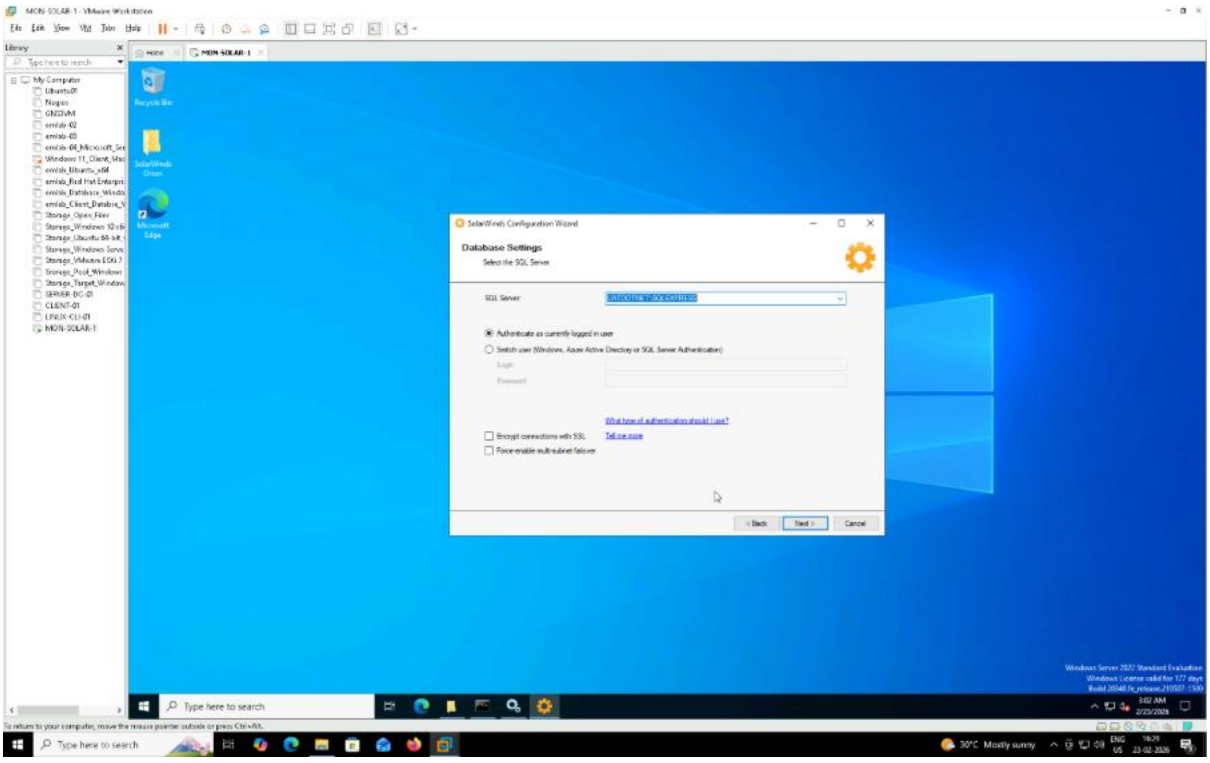


Installation has been completed successfully.

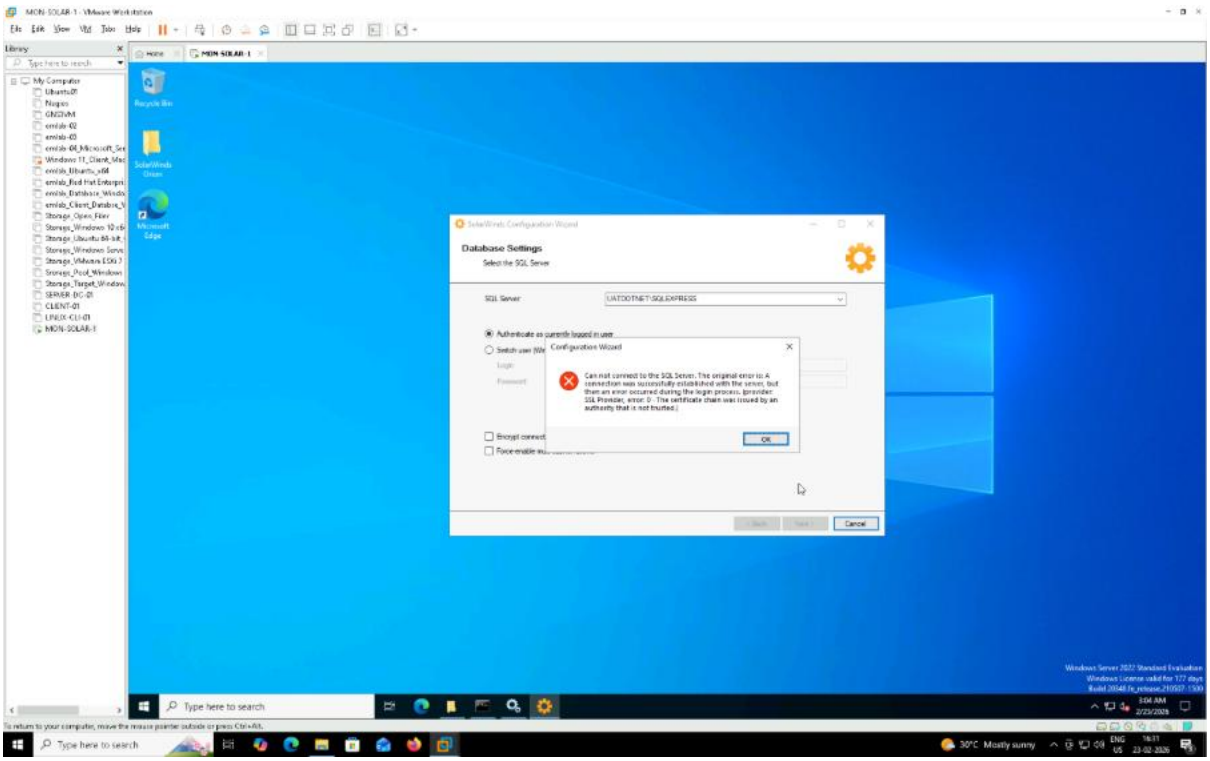


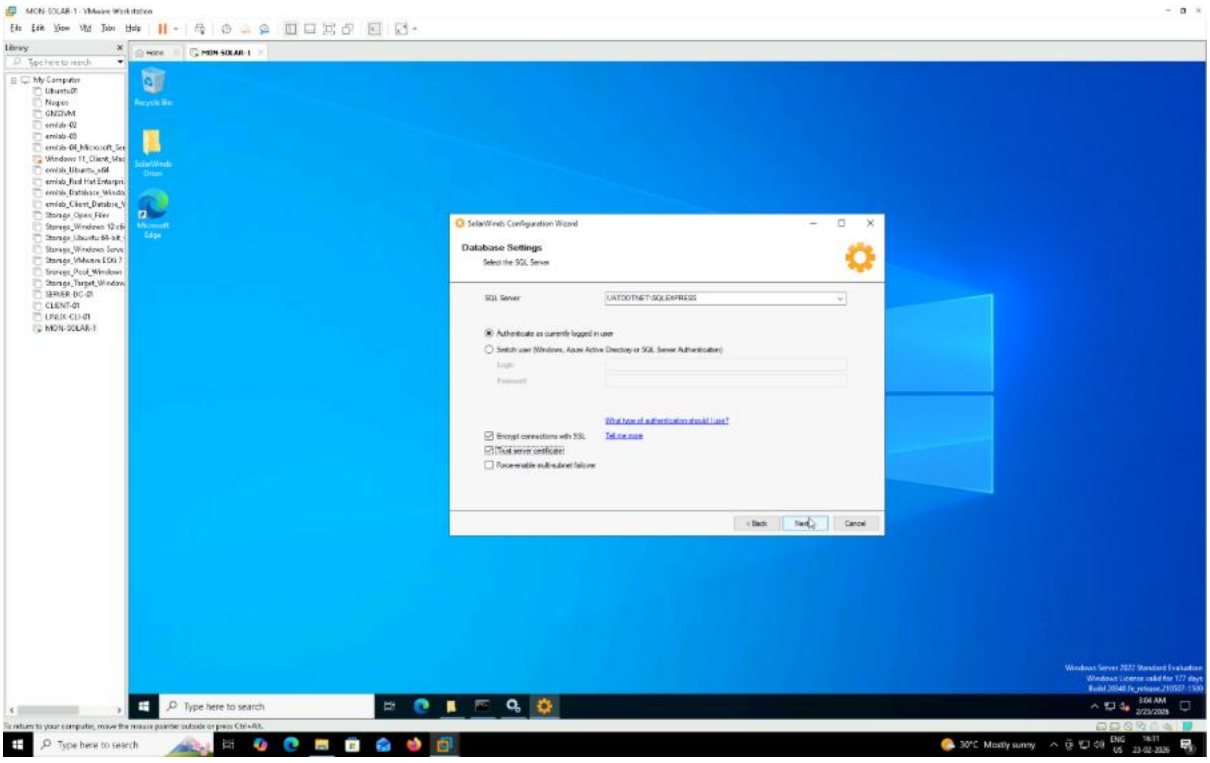
These tools are required by SolarWinds to function.



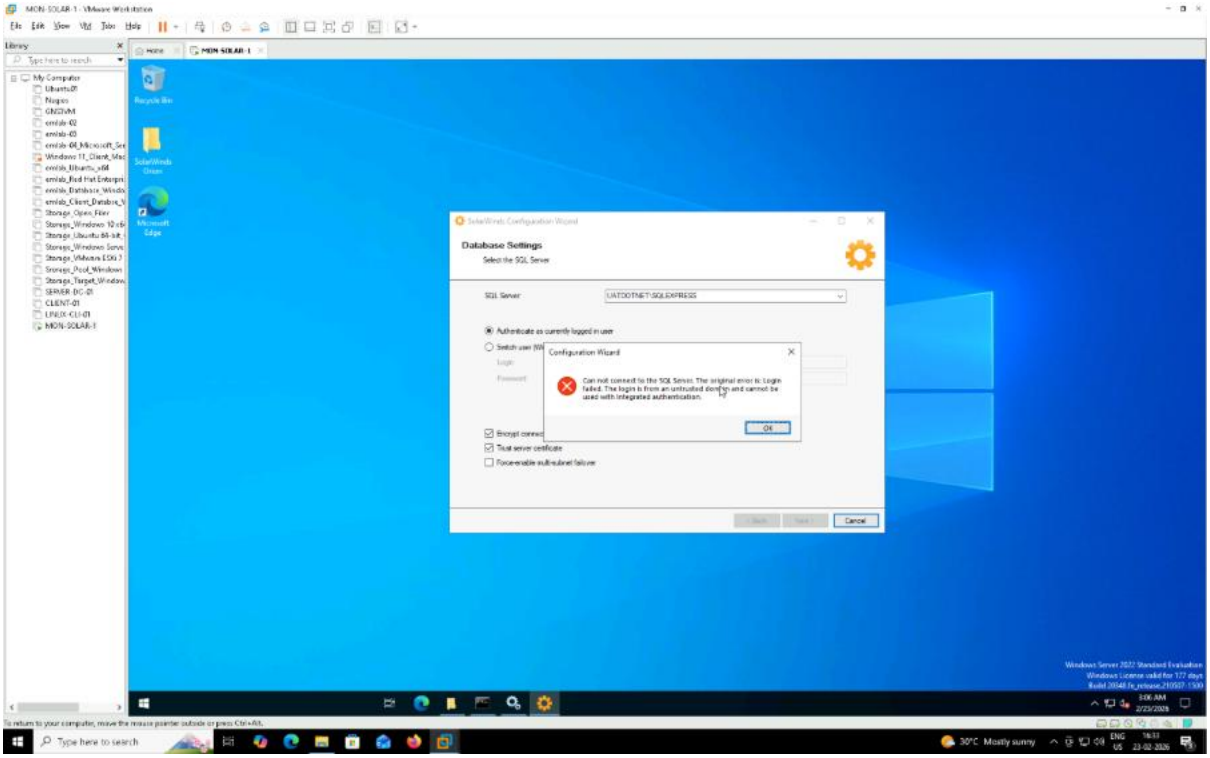


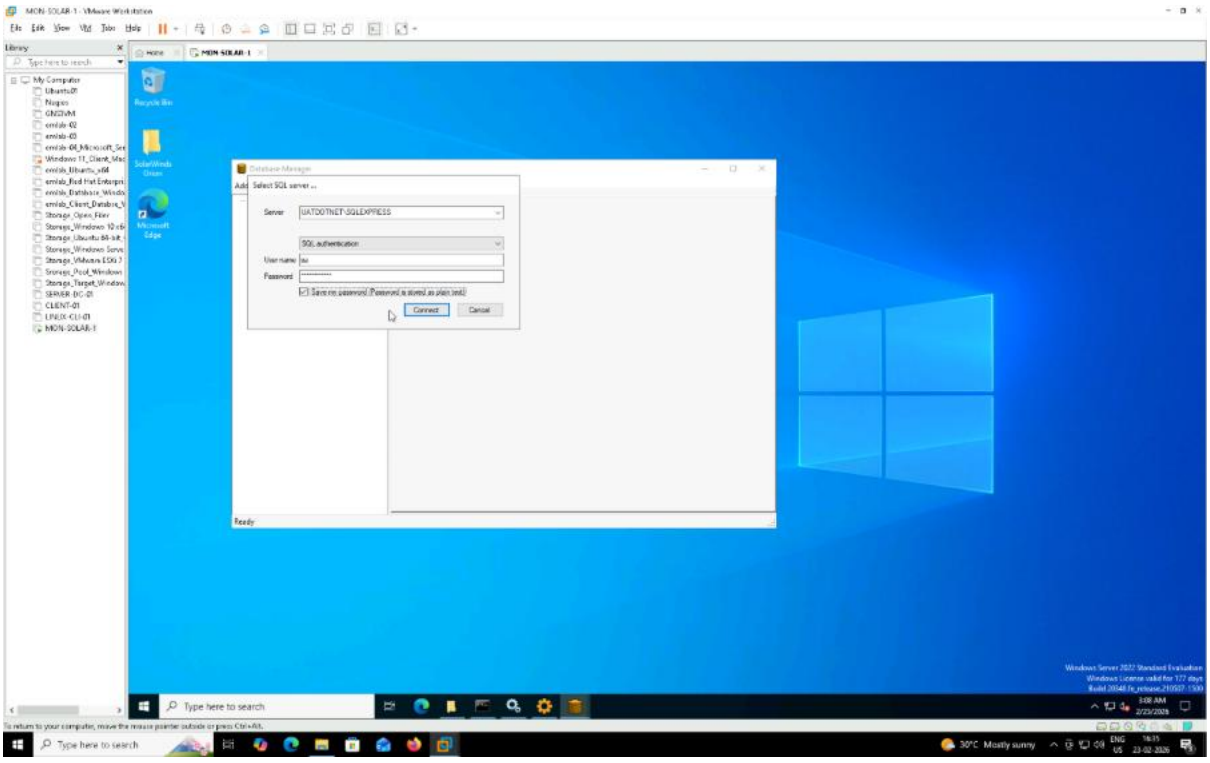
The SolarWinds Orion Platform was configured to use the SQL Server Express instance (SQLEXPRESS) installed locally on the monitoring server. Windows Authentication was used for secure database connectivity.



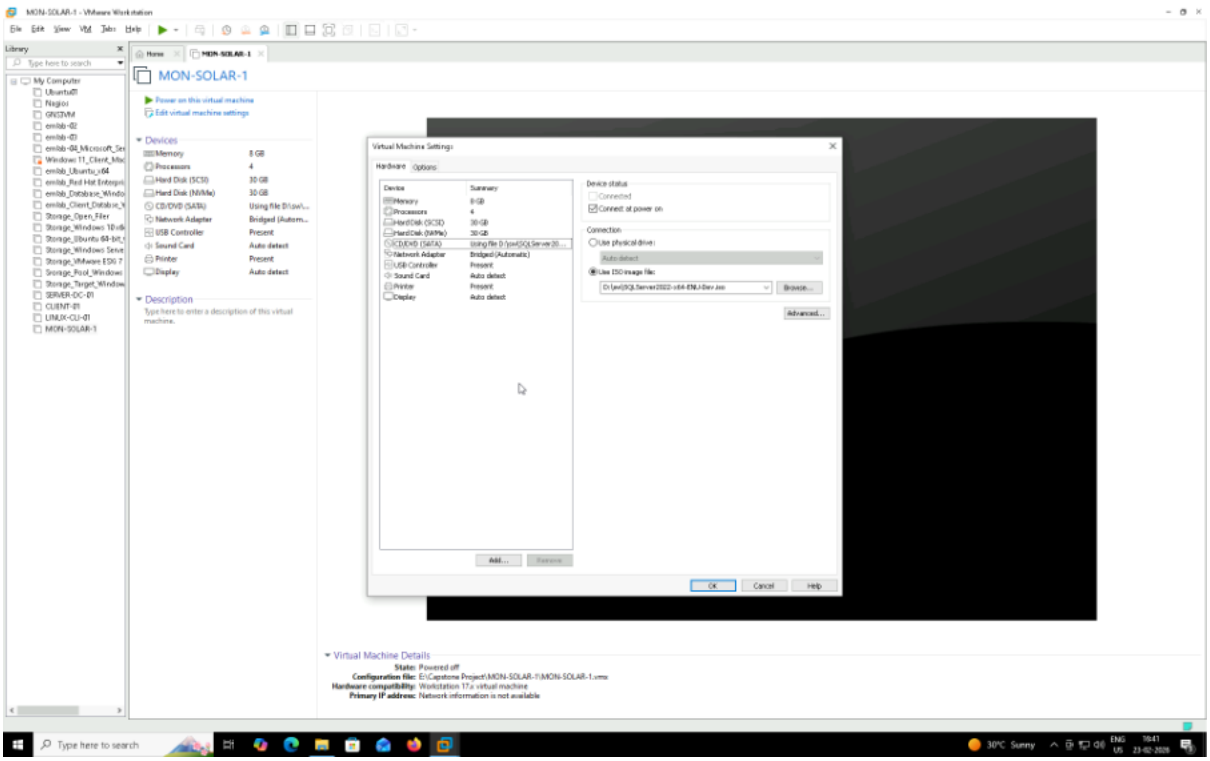


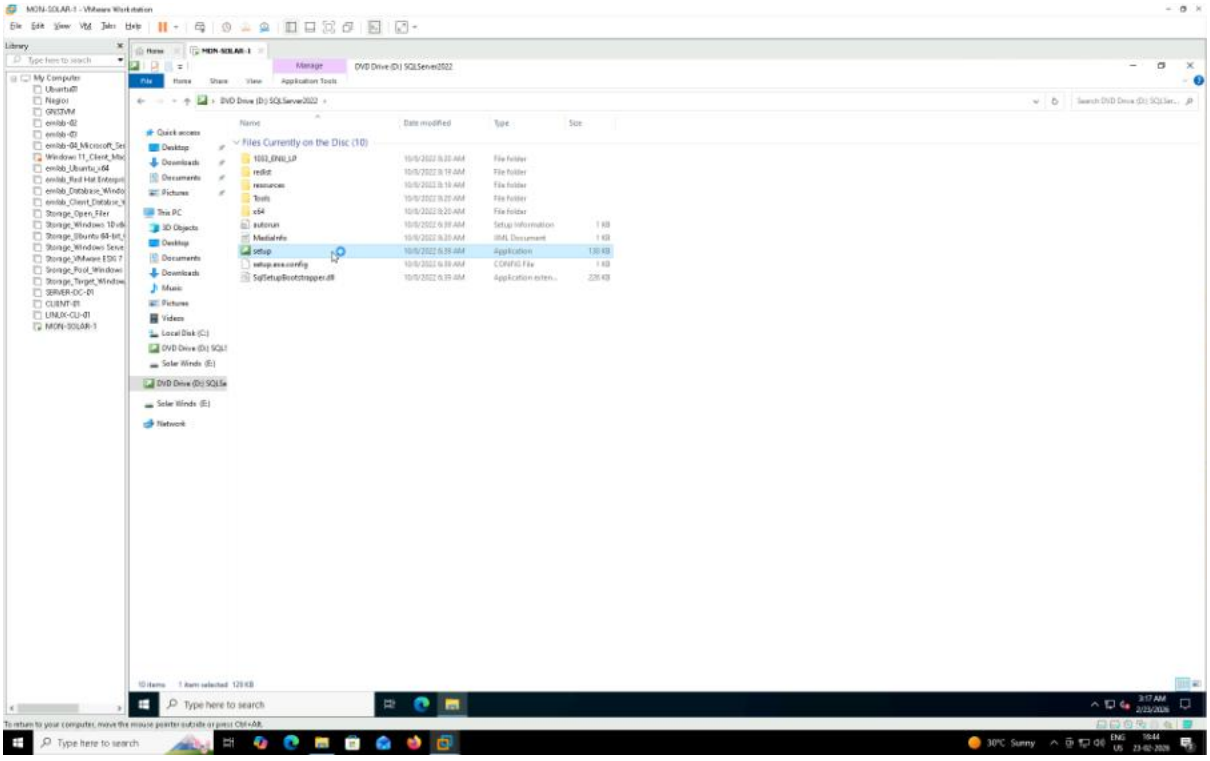
During configuration, an SSL certificate validation issue occurred with SQL Server Express. Encryption was enabled and the server certificate was trusted to establish secure database connectivity in the lab environment.

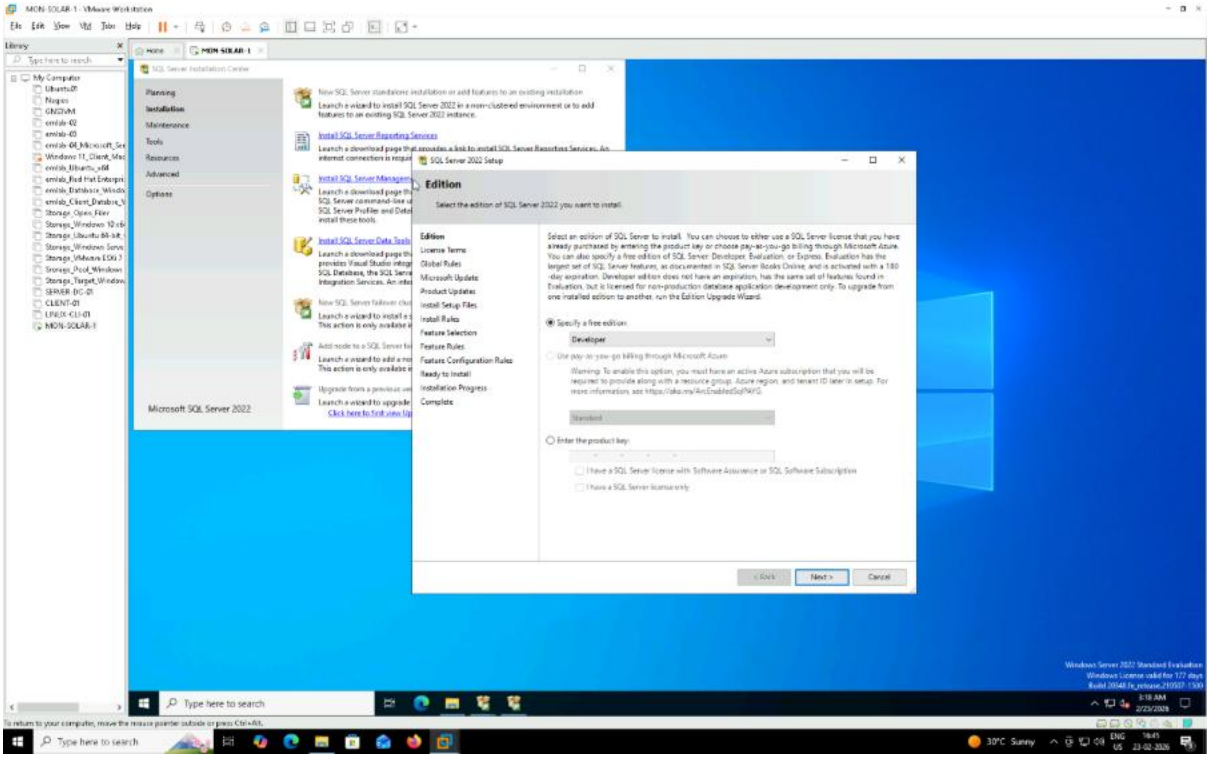


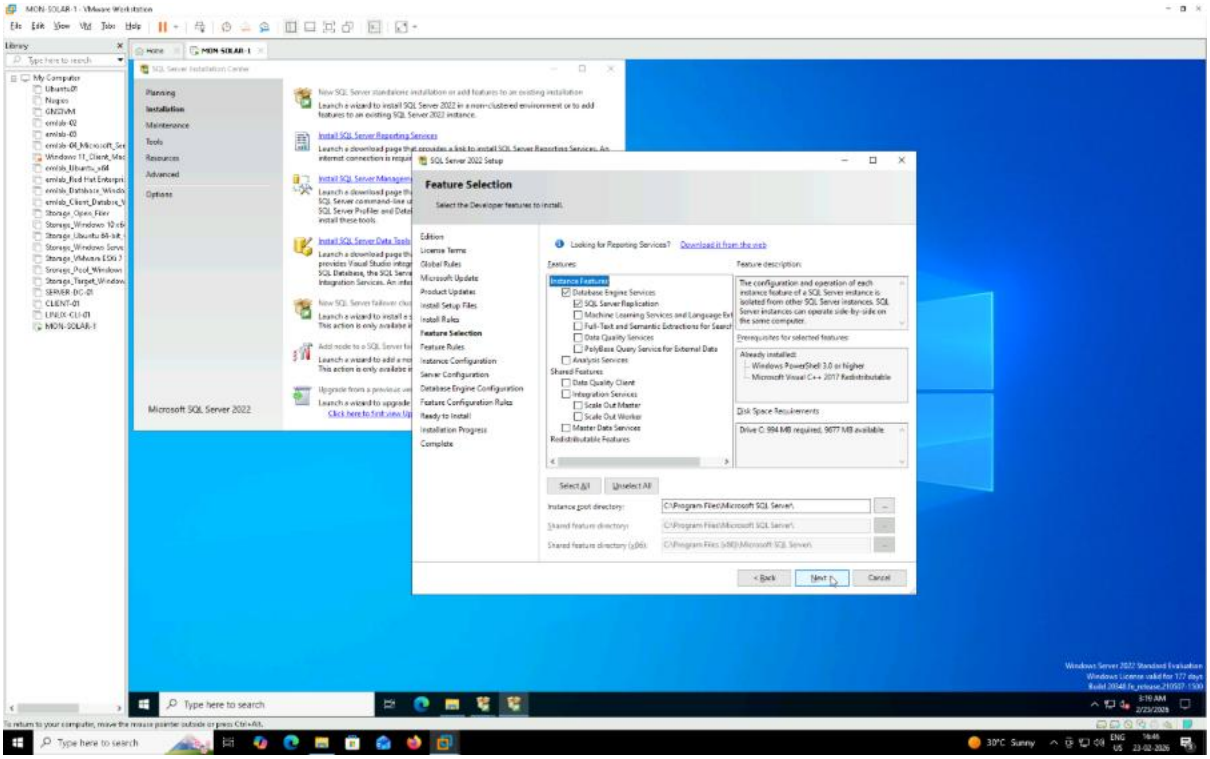


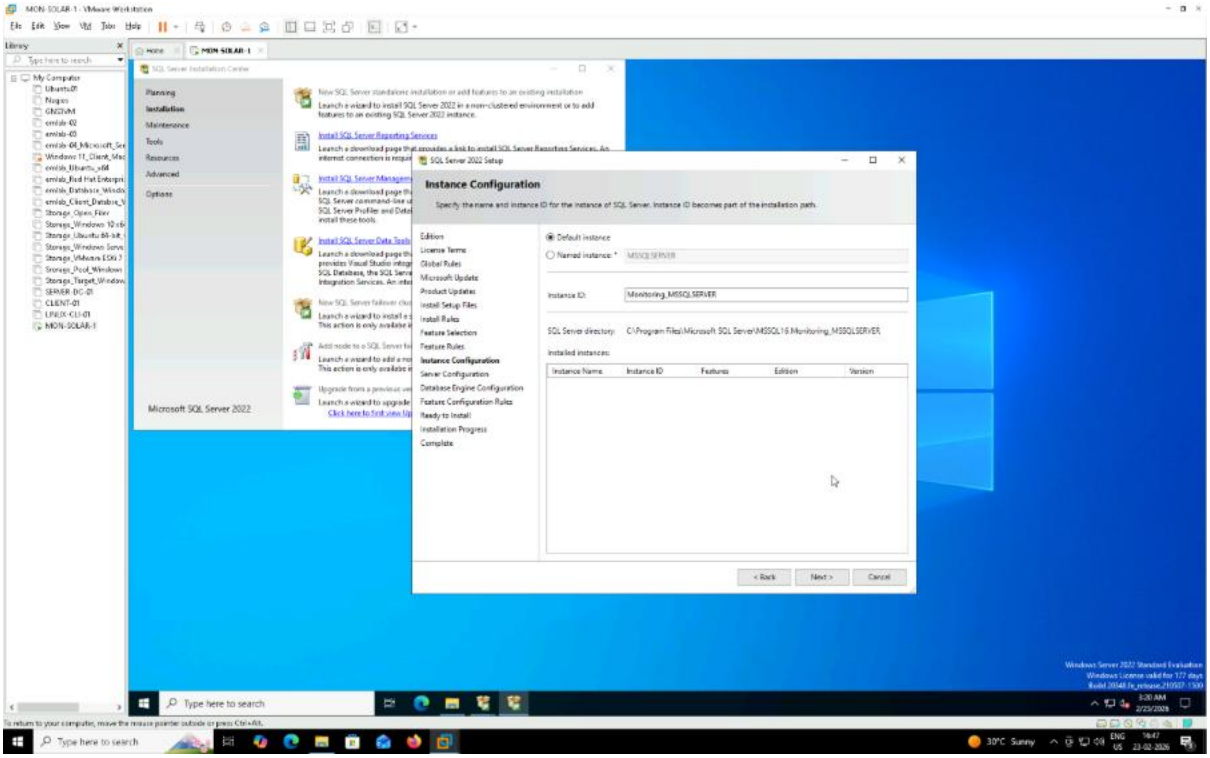
This means that SQL Server hasn't been installed successfully. We will now mount the SQL Server ISO Image and install the SQL Server.

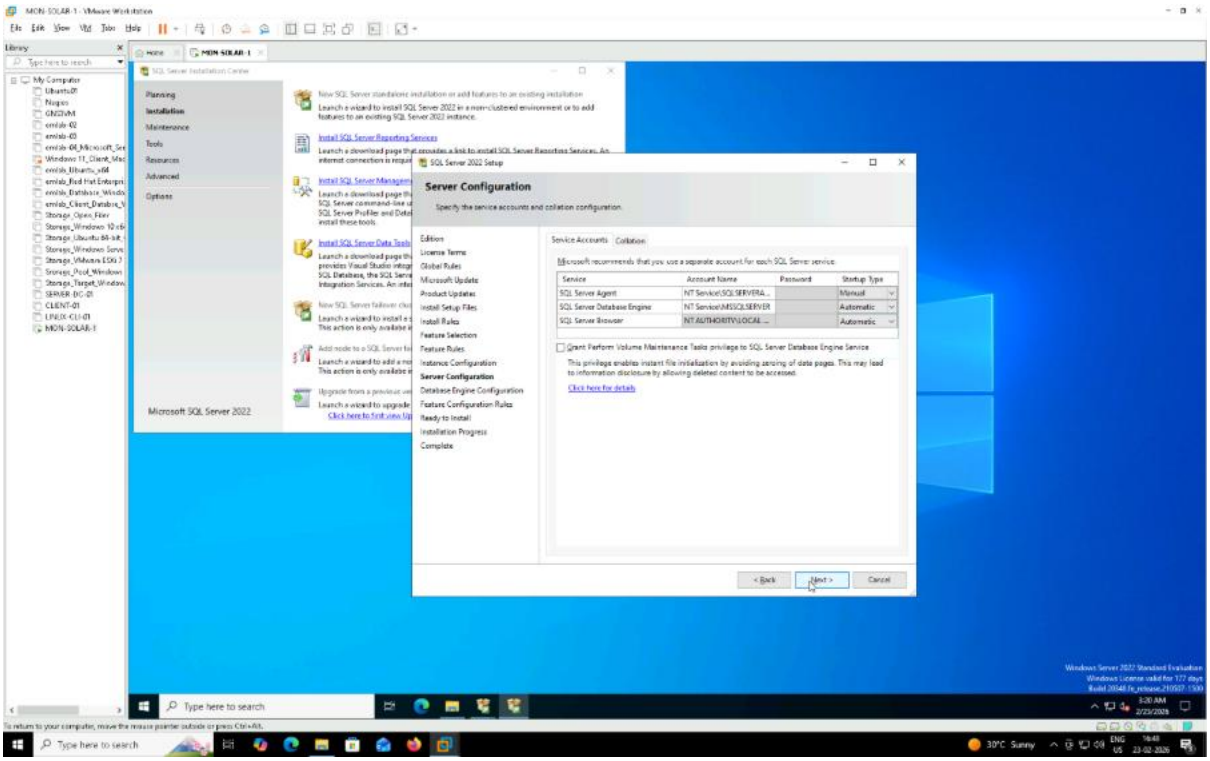

MON-SOLAR-1 - VMware Workstation
MON-SOLAR-1
Devices
Description
Virtual Machine Settings
Hardware
Options
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\MON-SOLAR-1\MON-SOLAR-1.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available

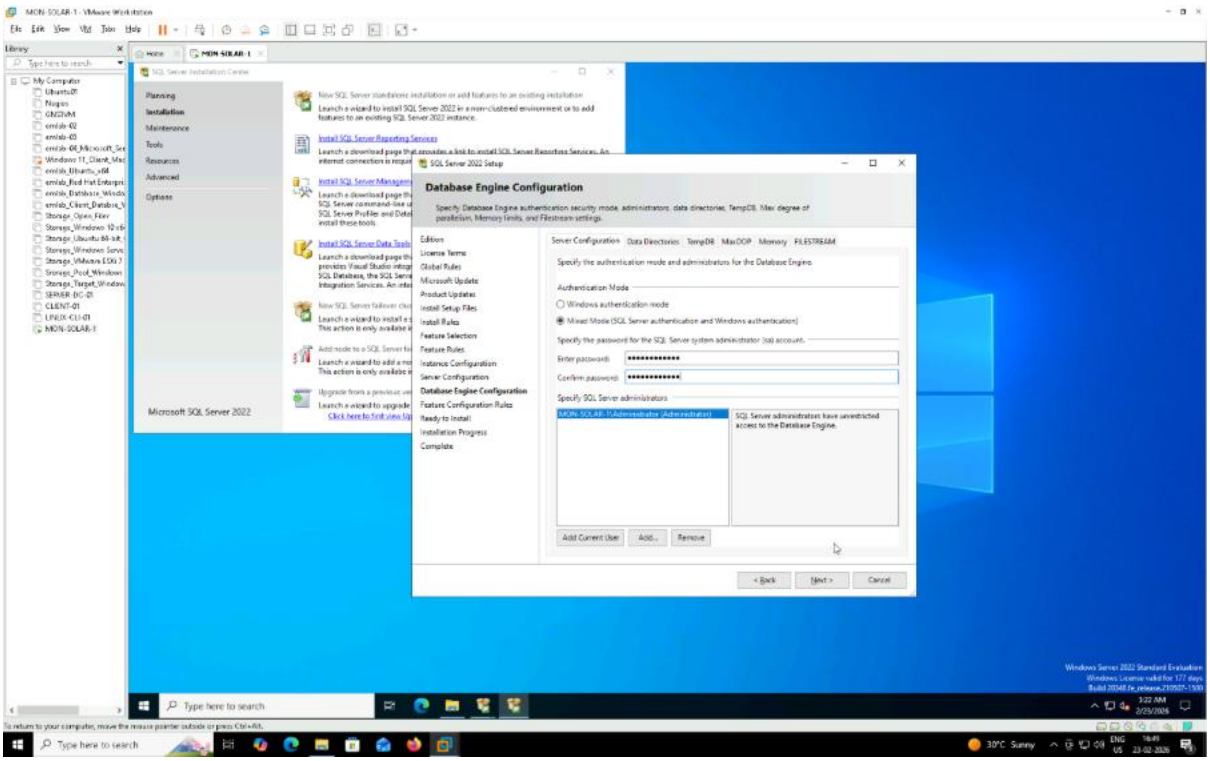

MON-SOLAR-1 - VMware Workstation
DVD Drive (D:) SQLServer2022
Files Currently on the Disc (10)
setup
autorun
MediaInfo
Type here to search

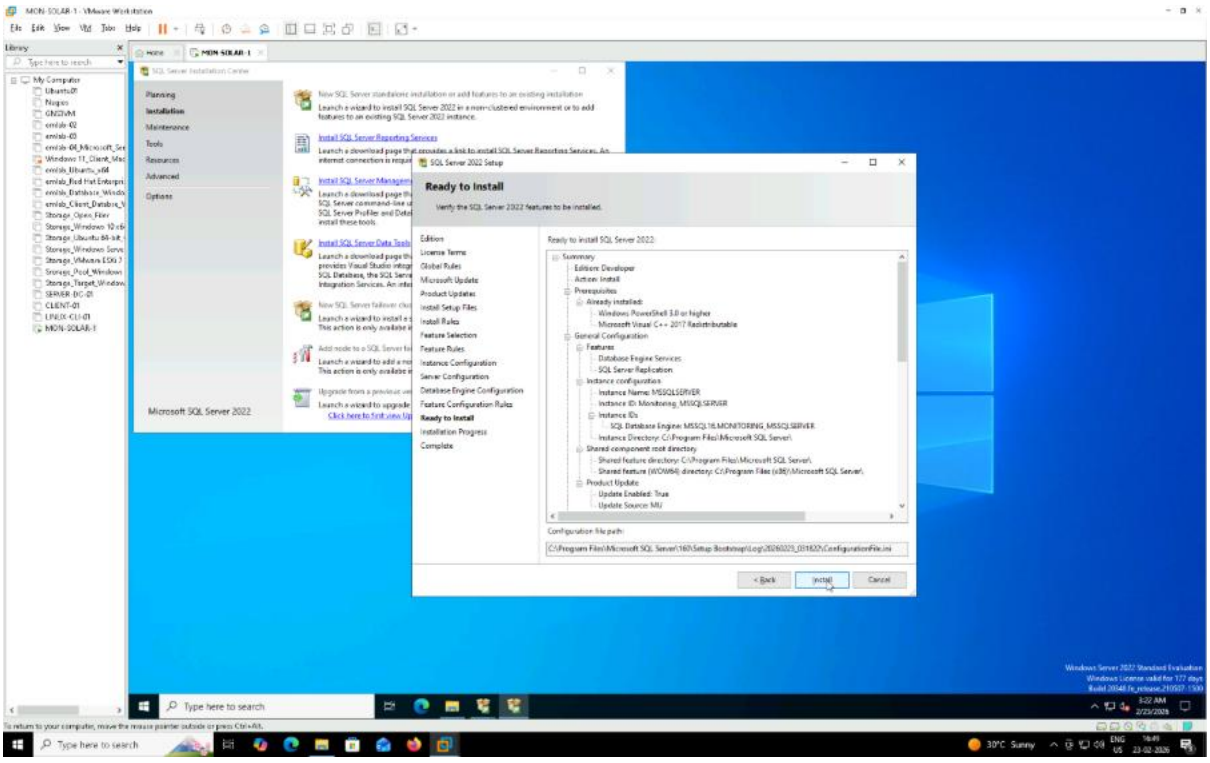

SQL Server 2022 Setup
Edition
Select the edition of SQL Server 2022 you want to install.
Edition
License Terms
Global Rules
Microsoft Update
Product Updates
Install Setup Files
Install Rules
Feature Selection
Feature Rules
Feature Configuration Rules
Ready to Install
Installation Progress
Complete
Specify a free edition
Developer
Enter the product key
< Back
Next >
Cancel
Microsoft SQL Server 2022


SQL Server 2022 Setup
Feature Selection
Select the Developer features to install.
Edition
License Terms
Global Rules
Microsoft Update
Product Updates
Install Setup Files
Install Rules
Feature Selection
Feature Rules
Instance Configuration
Server Configuration
Database Engine Configuration
Feature Configuration Rules
Ready to Install
Installation Progress
Complete
Looking for Reporting Services?
Features:
Instance Features
Database Engine Services
SQL Server Replication
Data Quality Services
Analysis Services
Shared Features
Data Quality Client
Integration Services
Master Data Services
Redistributable Features
Feature description:
Prerequisites for selected features:
Already installed:
Windows PowerShell 3.0 or higher
Microsoft Visual C++ 2017 Redistributable
Disk Space Requirements
Drive C: 994 MB required, 9077 MB available
Select All
Unselect All
Instance root directory:
C:\Program Files\Microsoft SQL Server\
< Back
Next >
Cancel
Microsoft SQL Server 2022

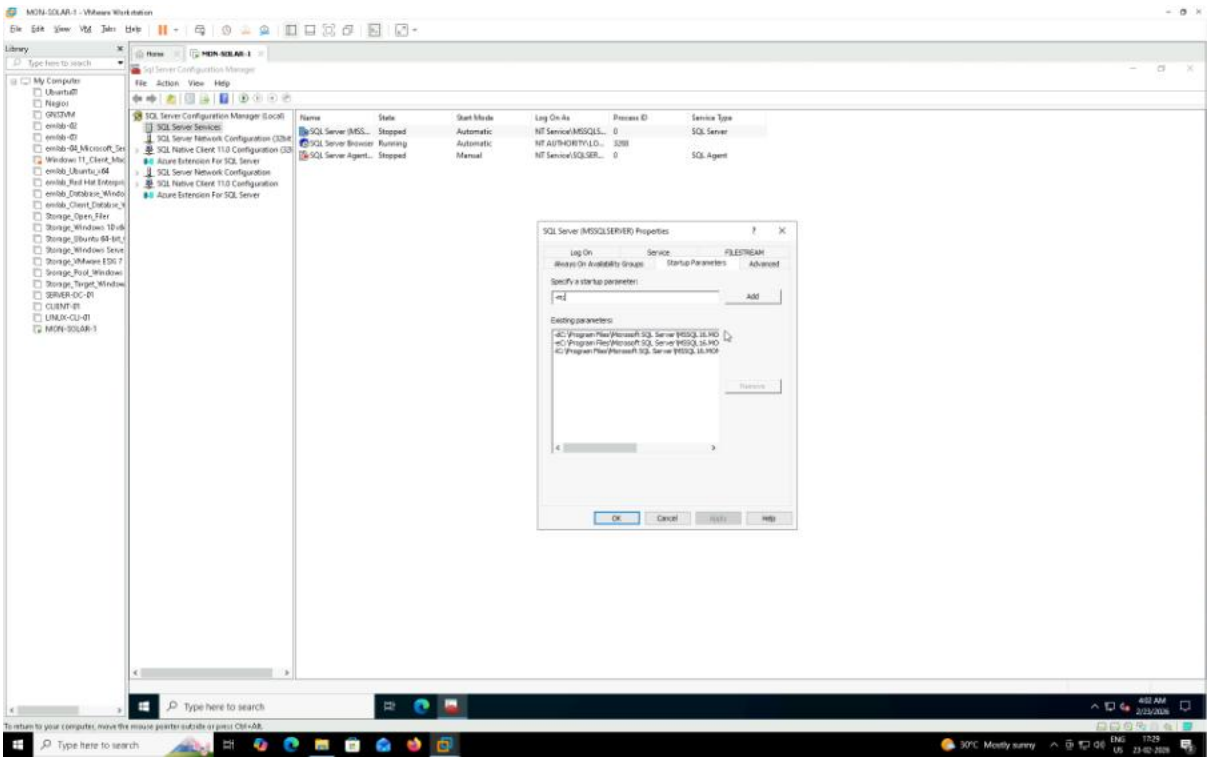

SQL Server 2022 Setup
Instance Configuration
Specify the name and instance ID for the instance of SQL Server. Instance ID becomes part of the installation path.
Default instance
Named instance:
Instance ID:
Monitoring_MSSQLSERVER
SQL Server directory:
C:\Program Files\Microsoft SQL Server\MSSQL16.Monitoring_MSSQLSERVER
Installed instances:
Instance Name
Instance ID
Features
Edition
Version
< Back
Next >
Cancel
Microsoft SQL Server 2022

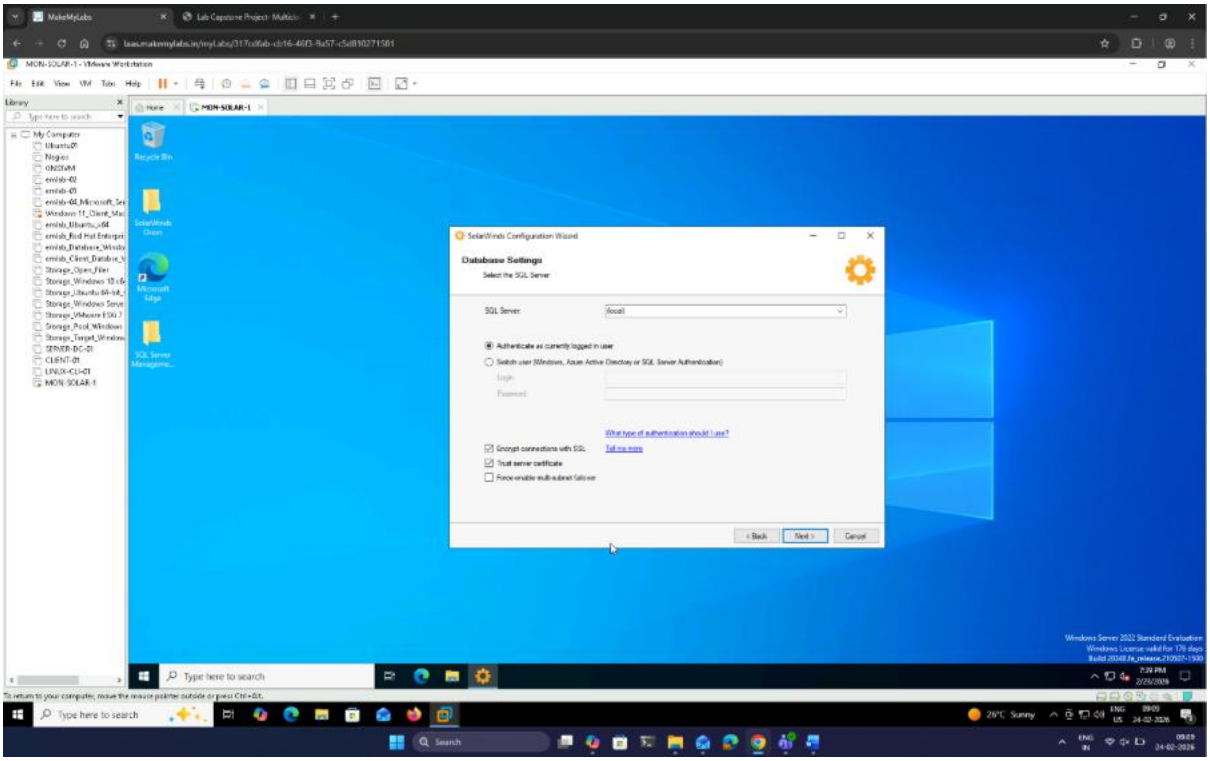

SQL Server 2022 Setup
Server Configuration
Specify the service accounts and collation configuration.
Service Accounts
Collation
Service
Account Name
Password
Startup Type
SQL Server Agent
SQL Server Database Engine
SQL Server Browser
Automatic
Manual
Grant Perform Volume Maintenance Tasks privilege to SQL Server Database Engine Service
Click here for details
< Back
Next >
Cancel
Microsoft SQL Server 2022



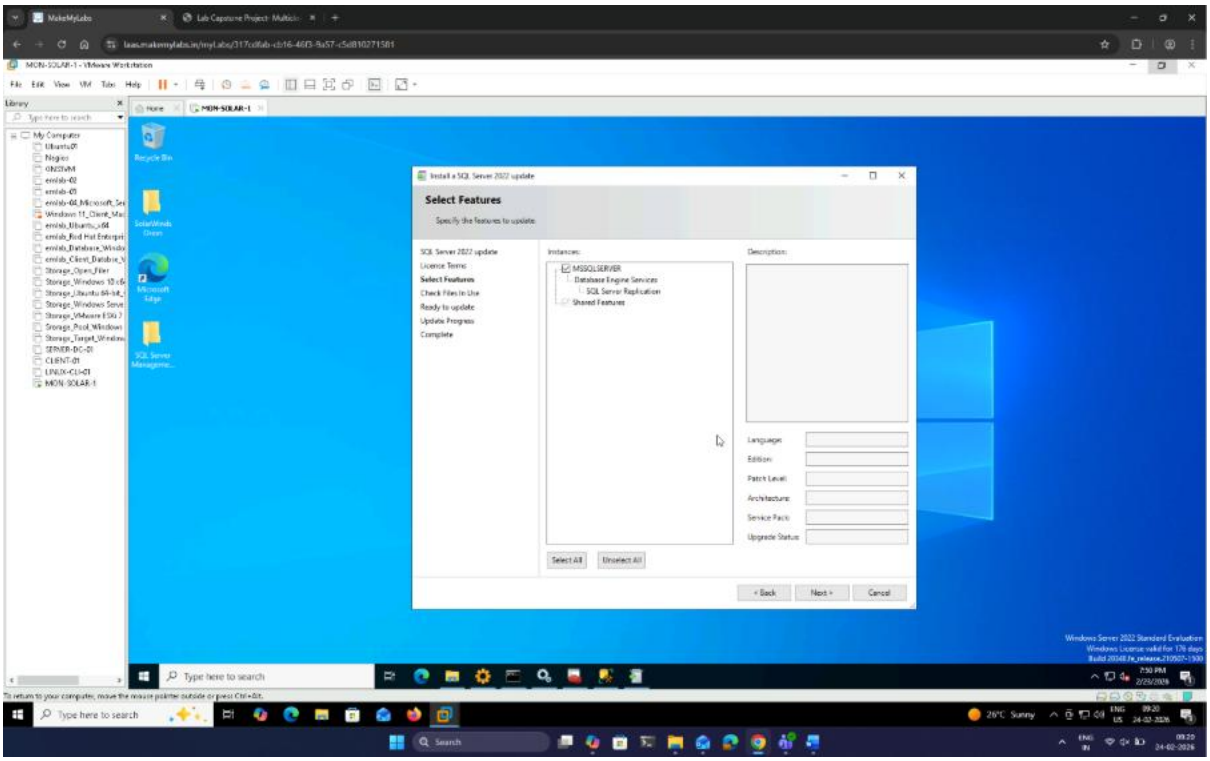


SQL Server was mounted from ISO and installed in Mixed Mode Authentication. Database Engine Services were configured with SQL Authentication enabled to ensure compatibility with SolarWinds Orion Installer.

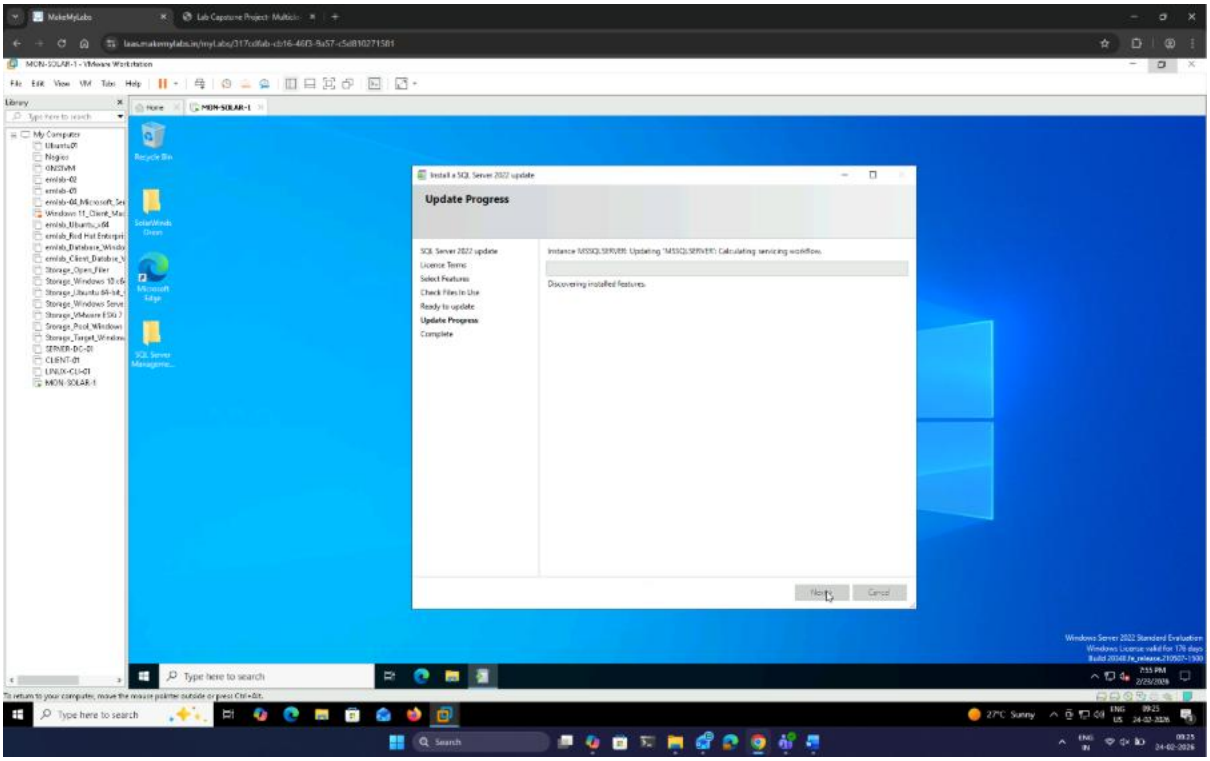




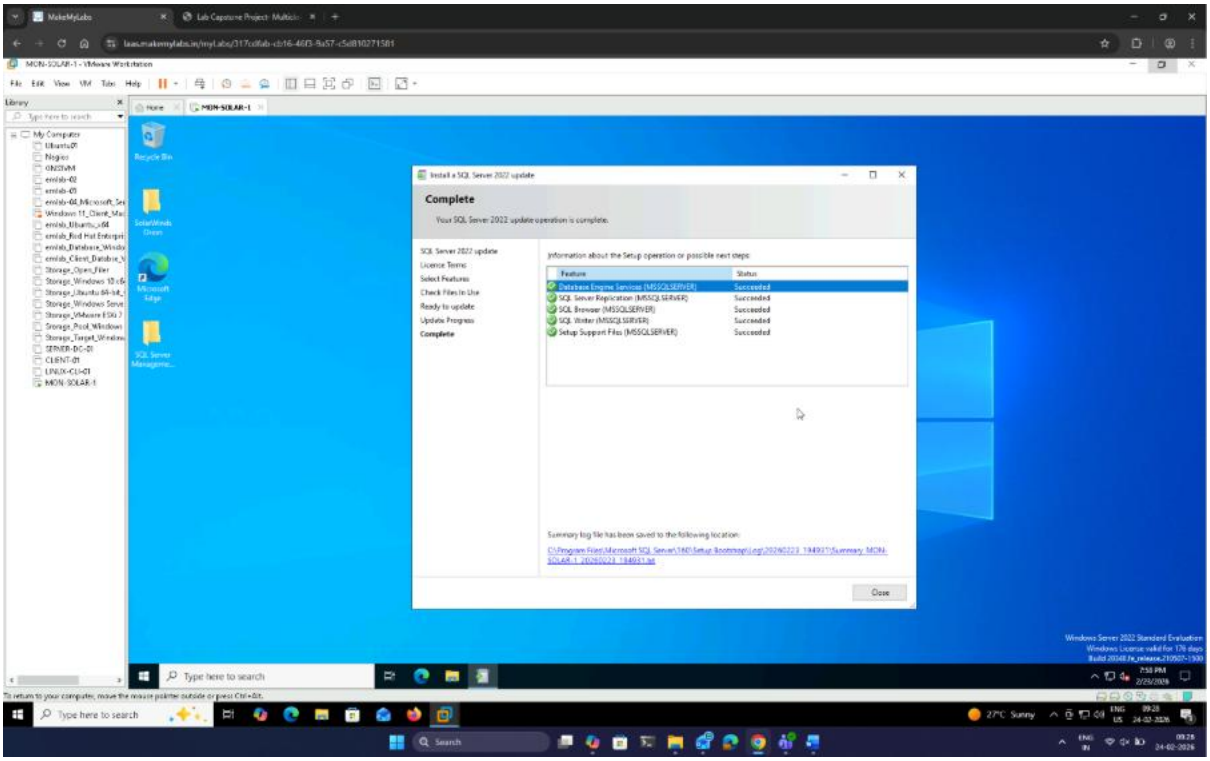
Trying to connect once again and got an error.

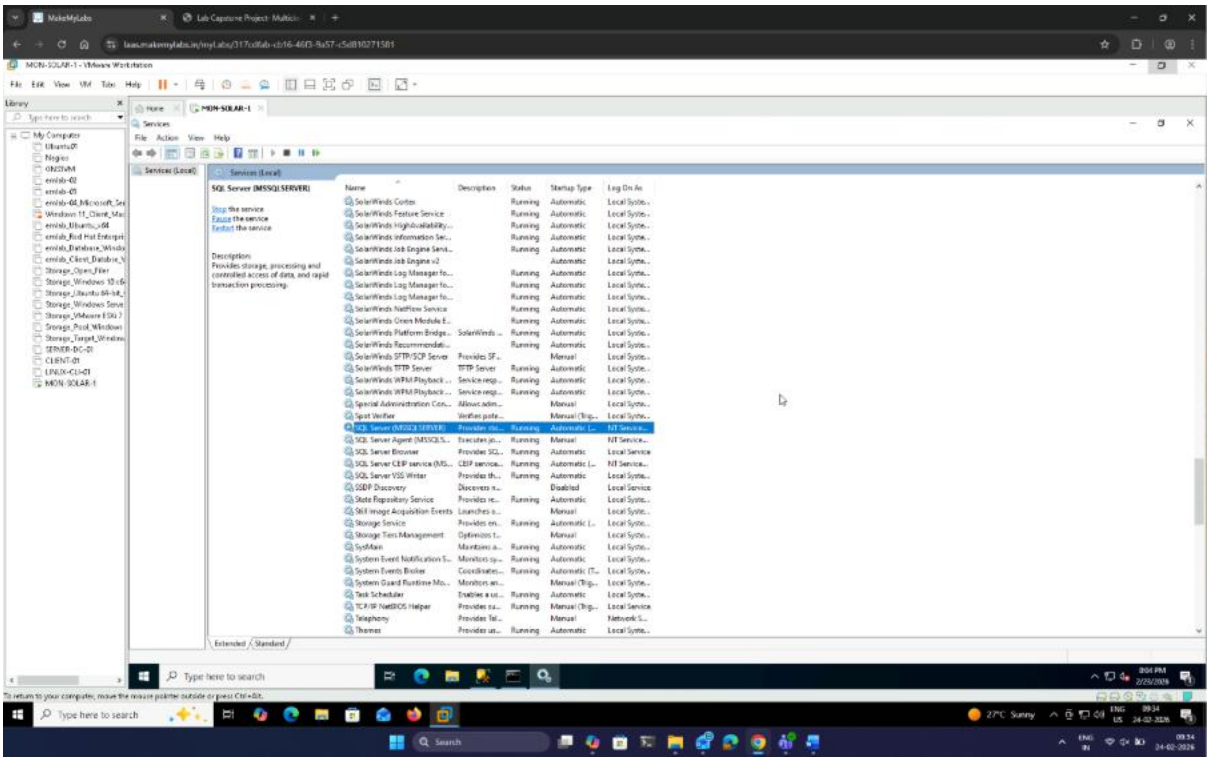


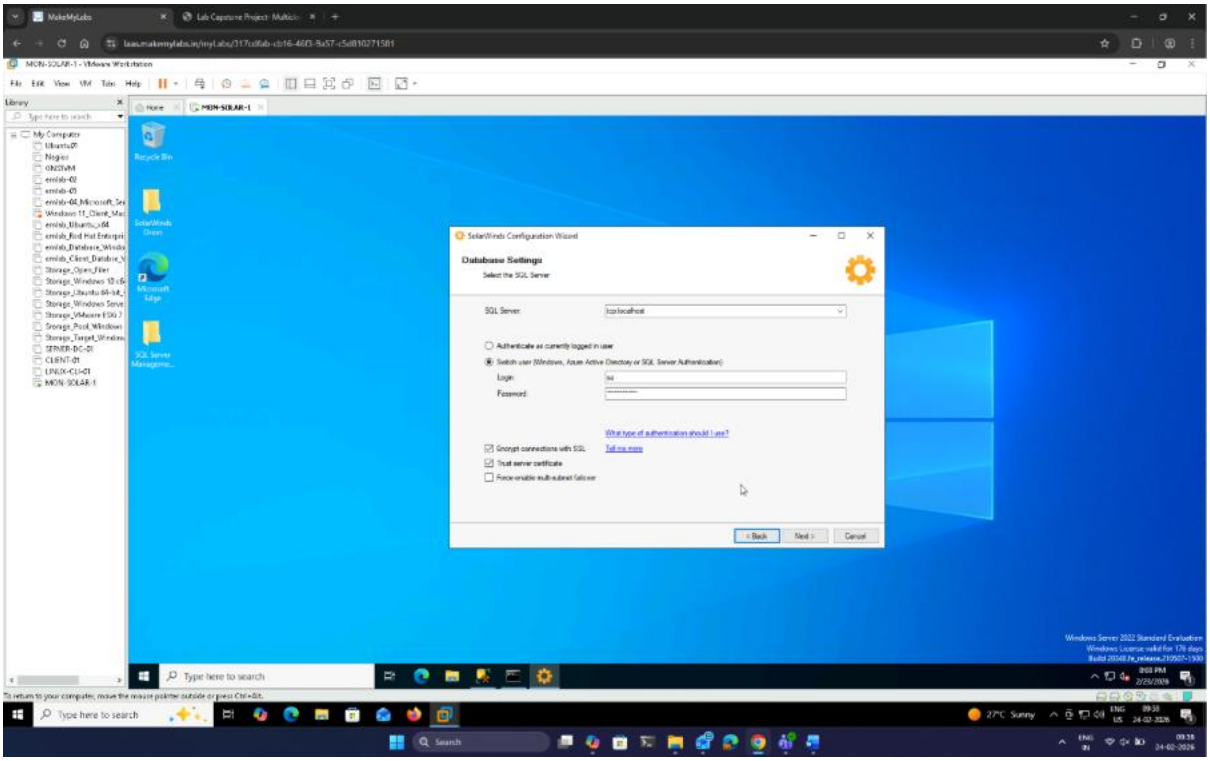
SolarWinds required SQL Server 2022 Cumulative Update 1 or later. The latest cumulative update was installed to ensure compatibility, stability, and security compliance with SolarWinds Orion platform requirements.

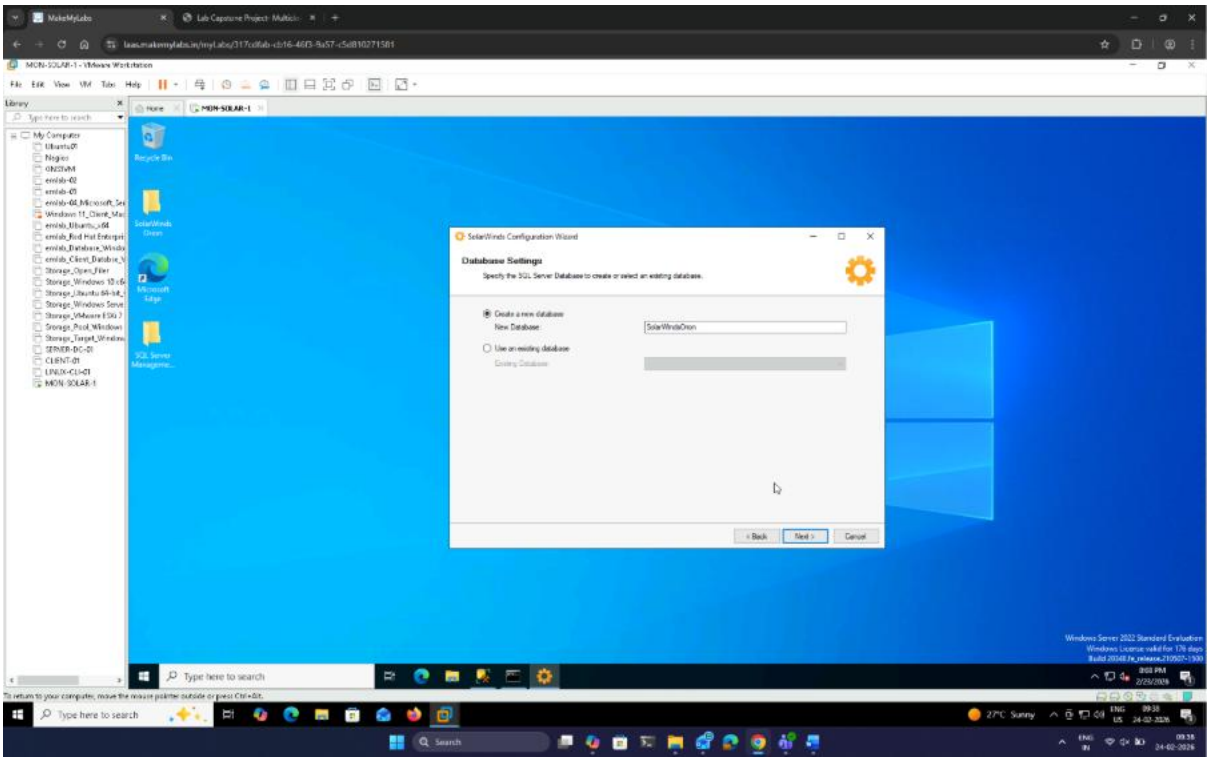


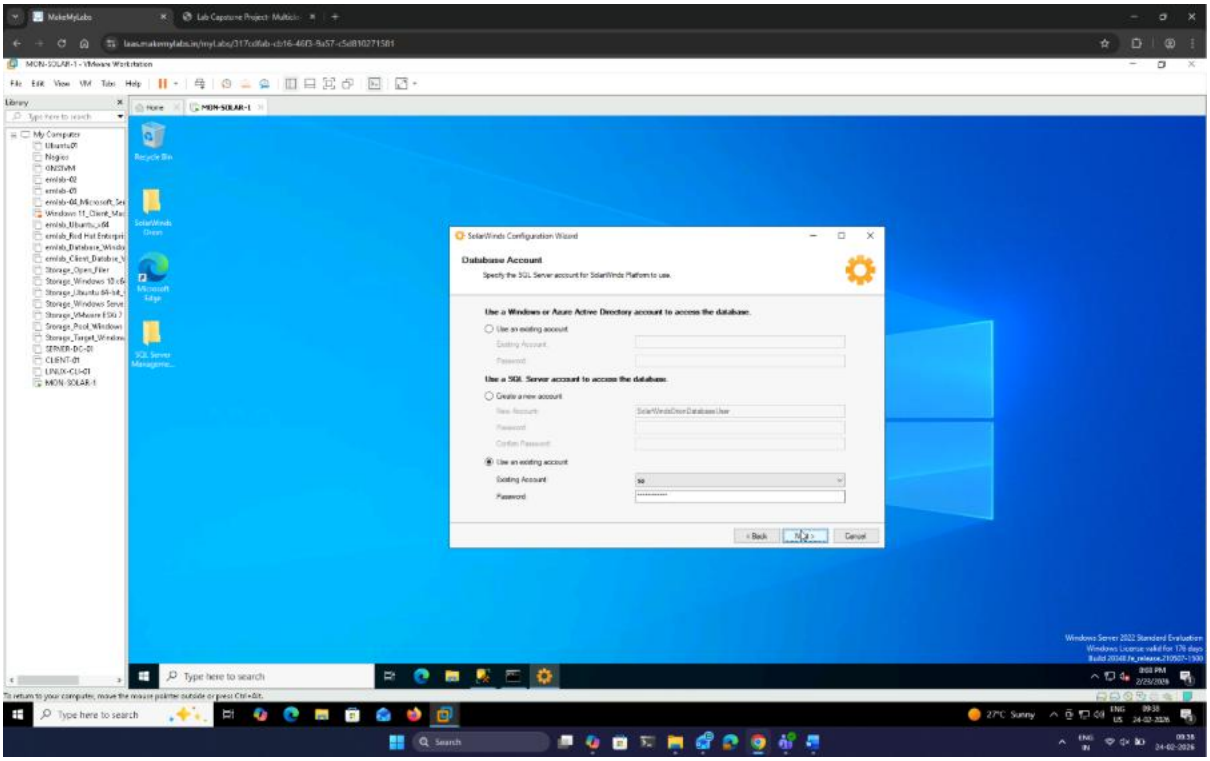
During the cumulative installation, the Windows Management Instrumentation (WMI) process was detected using SQL-related files. The installation was allowed to proceed after stopping SQL services and restarting the server to ensure a clean update environment.



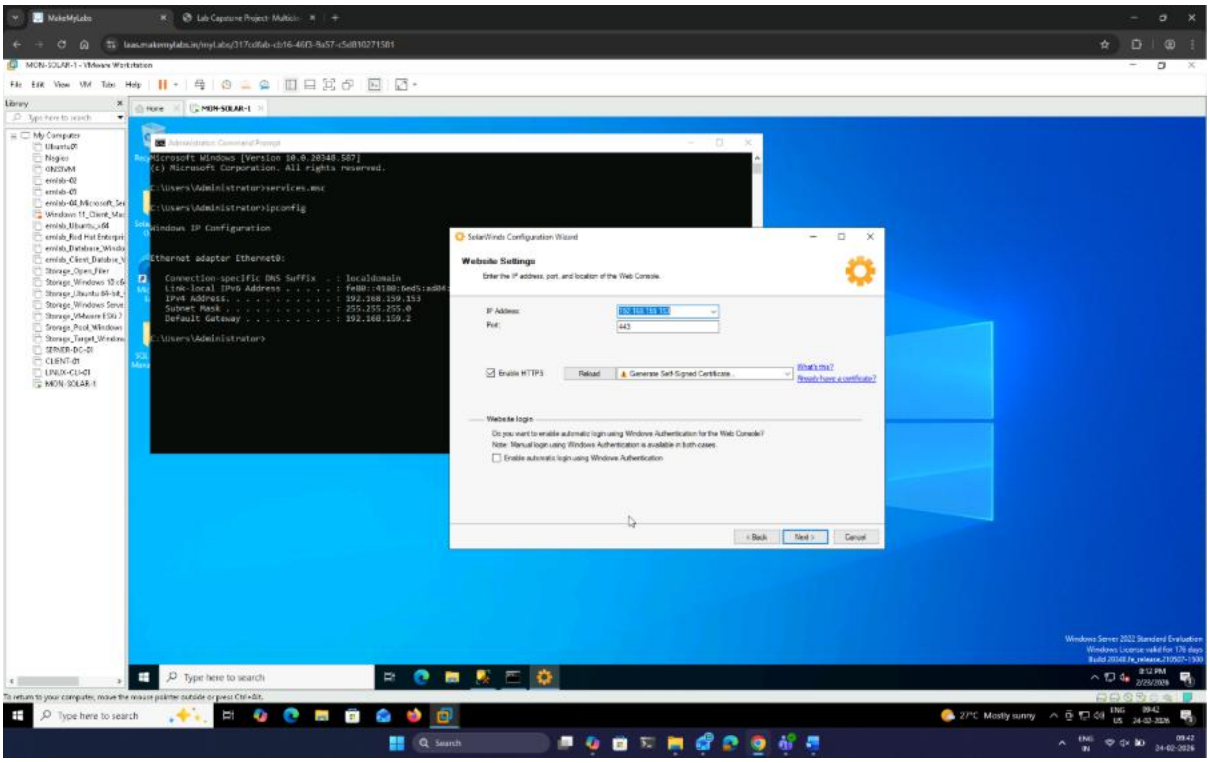



SolarWinds Configuration Wizard
Database Settings
Select the SQL Server
SQL Server:
localhost
Authenticate as currently logged in user
Switch user (Windows, Azure Active Directory or SQL Server Authentication)
Login:
Password:
What type of authentication should I use?
Encrypt connections with SSL
Tell me more
Trust server certificate
Force enable multi-subnet failover
< Back
Next >
Cancel

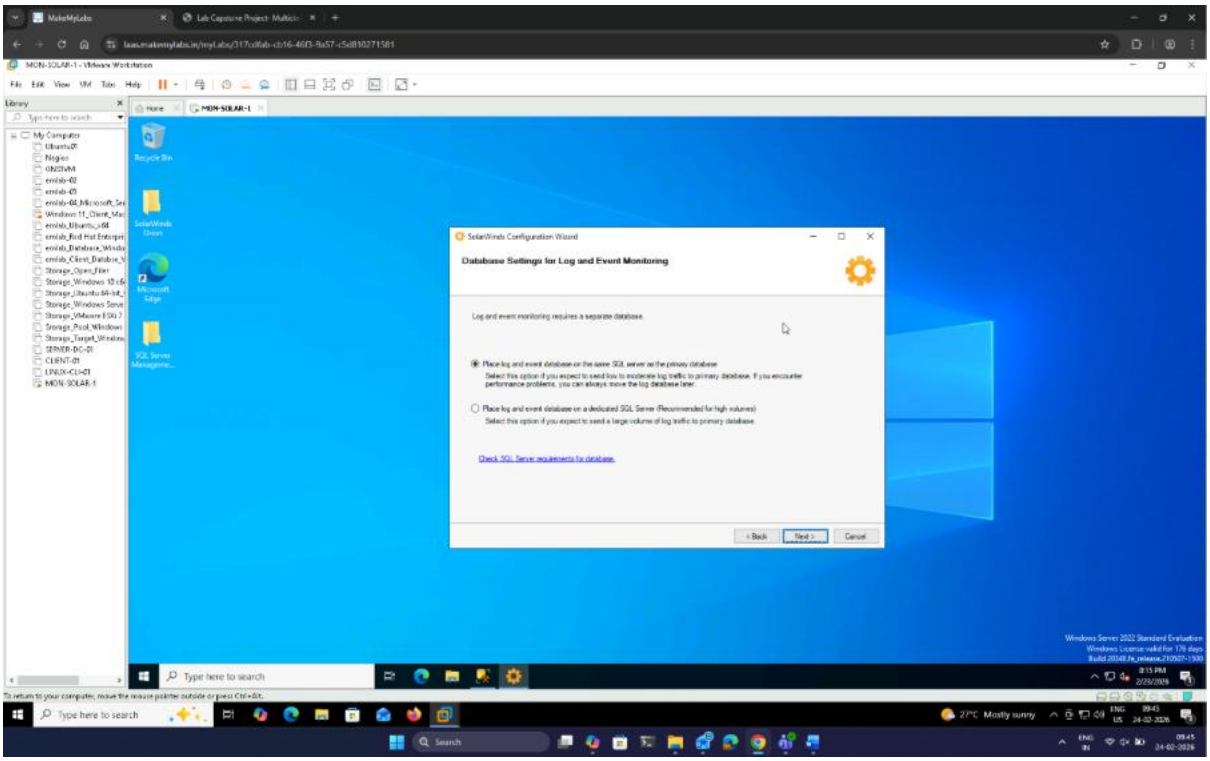

SolarWinds Configuration Wizard
Database Settings
Specify the SQL Server Database to create or select an existing database.
Create a new database
New Database:
SolarWindsOrion
Use an existing database
Existing Database:
< Back
Next >
Cancel

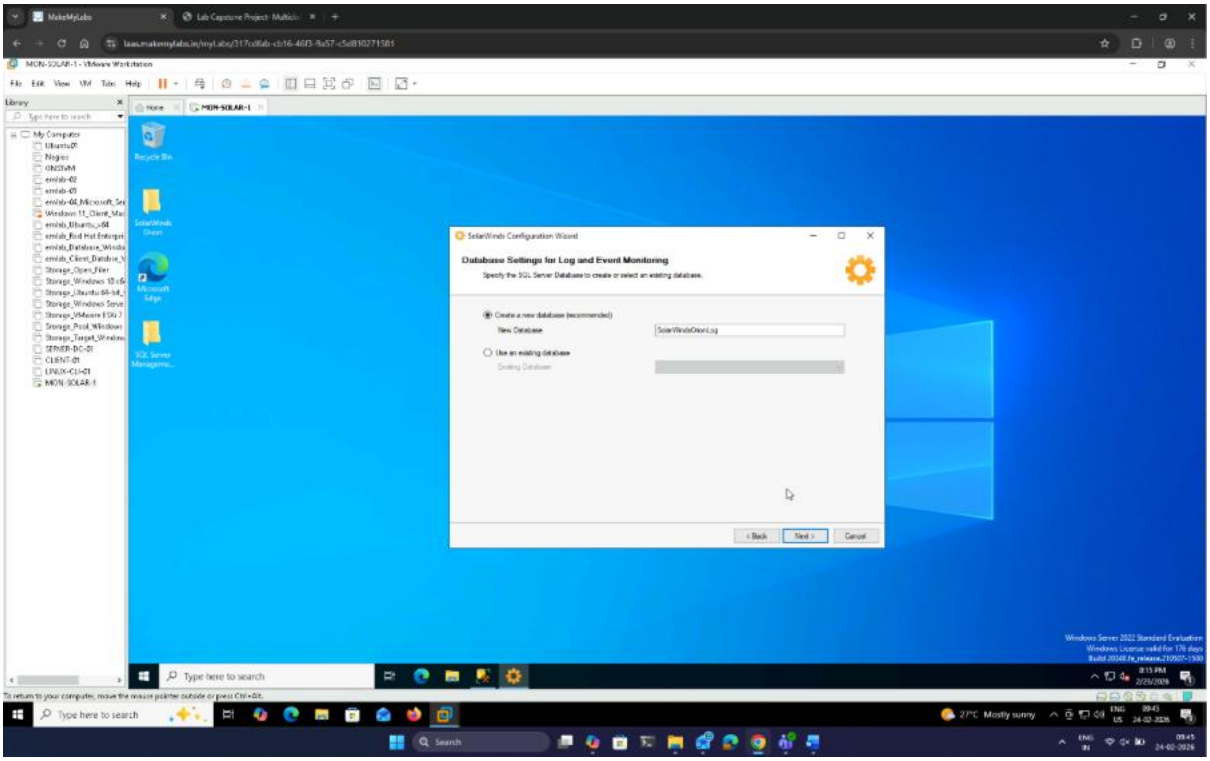

SolarWinds Configuration Wizard
Database Account
Specify the SQL Server account for SolarWinds Platform to use.
Use a Windows or Azure Active Directory account to access the database.
Use an existing account
Existing Account:
Password:
Use a SQL Server account to access the database.
Create a new account
New Account:
SolarWindsOrionDatabaseUser
Password:
Confirm Password:
Use an existing account
Existing Account:
sa
Password:
< Back
Next >
Cancel

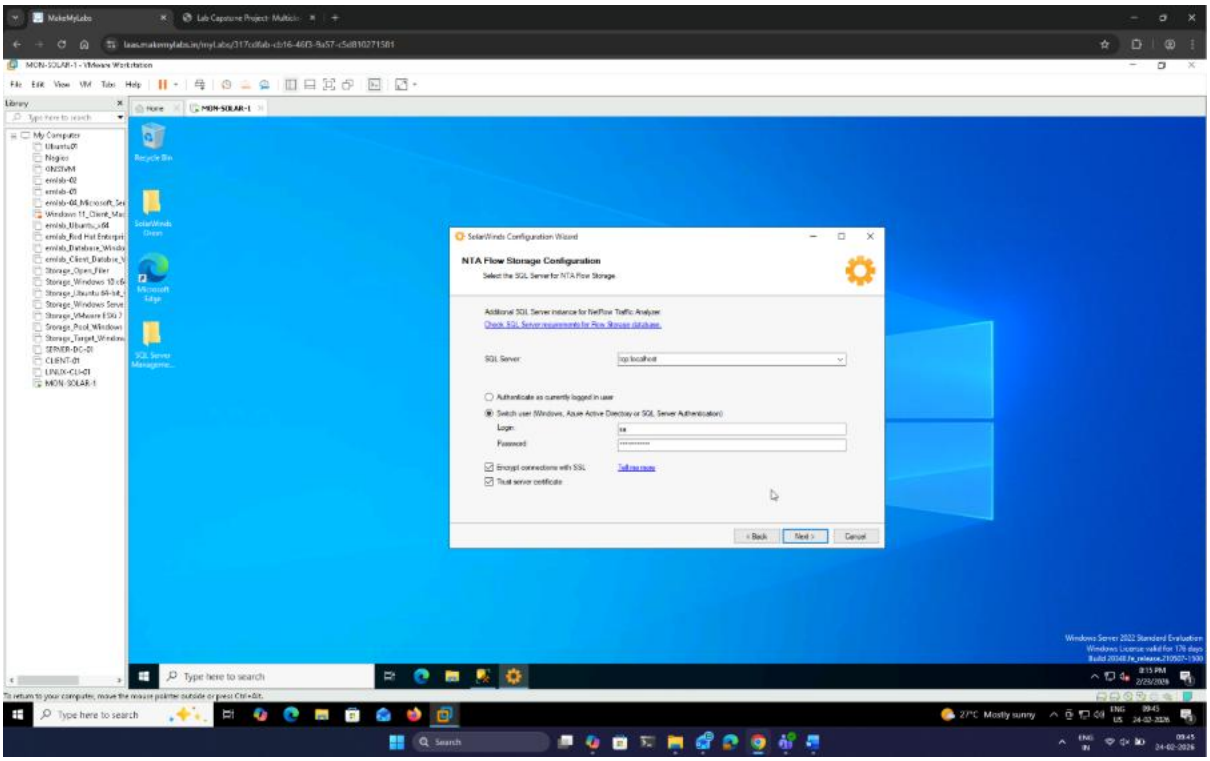
The monitoring server was configured with a virtualized NAT network environment. External internet reachability was not required for SolarWinds configuration, as core components and SQL Services were already installed.

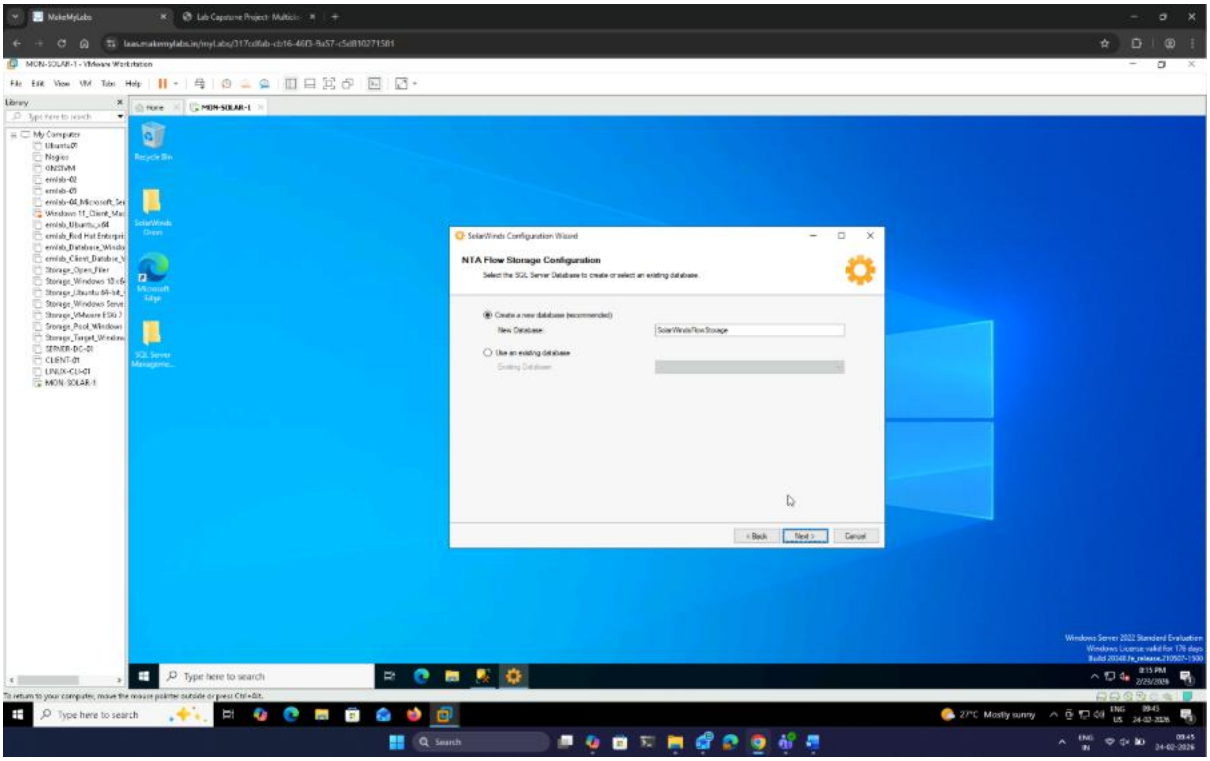


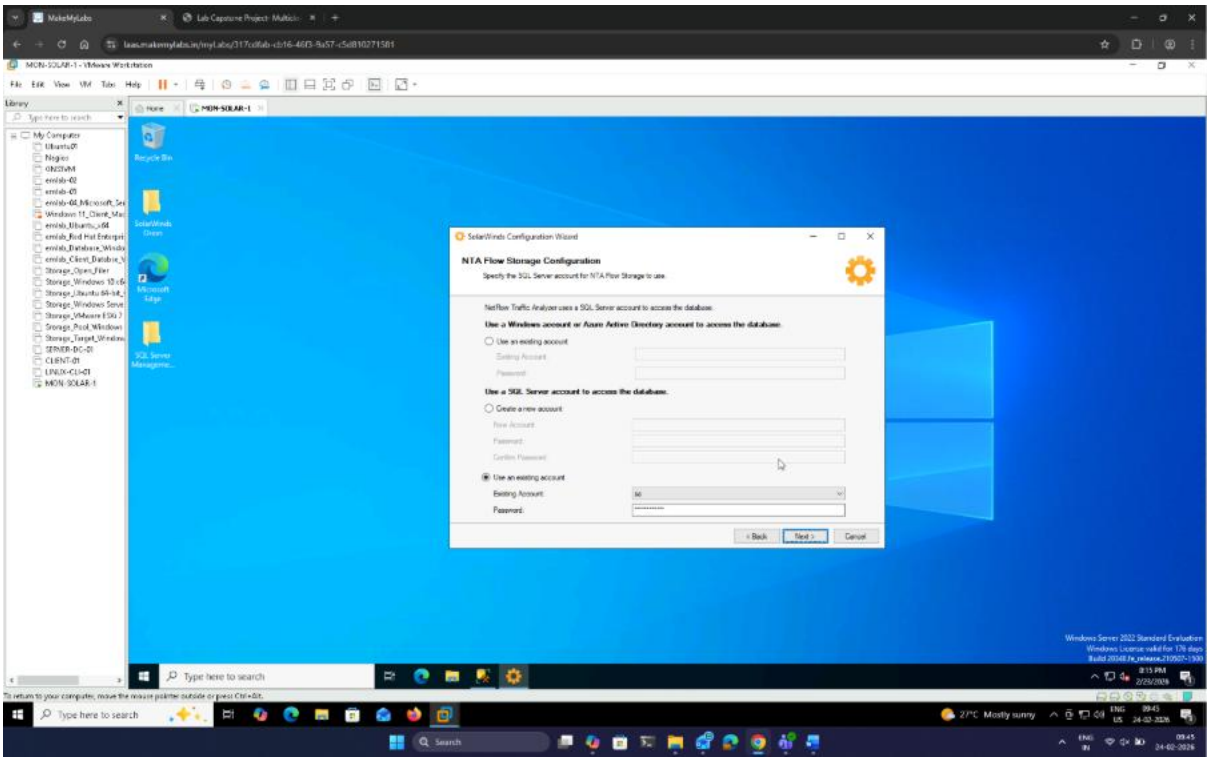
The SolarWinds console was configured to bind to the Server's active IP address (192.168.159.153) using HTTPS on port 443. A self-signed SSL certificate was generated to enable secure web access to the monitoring platform.

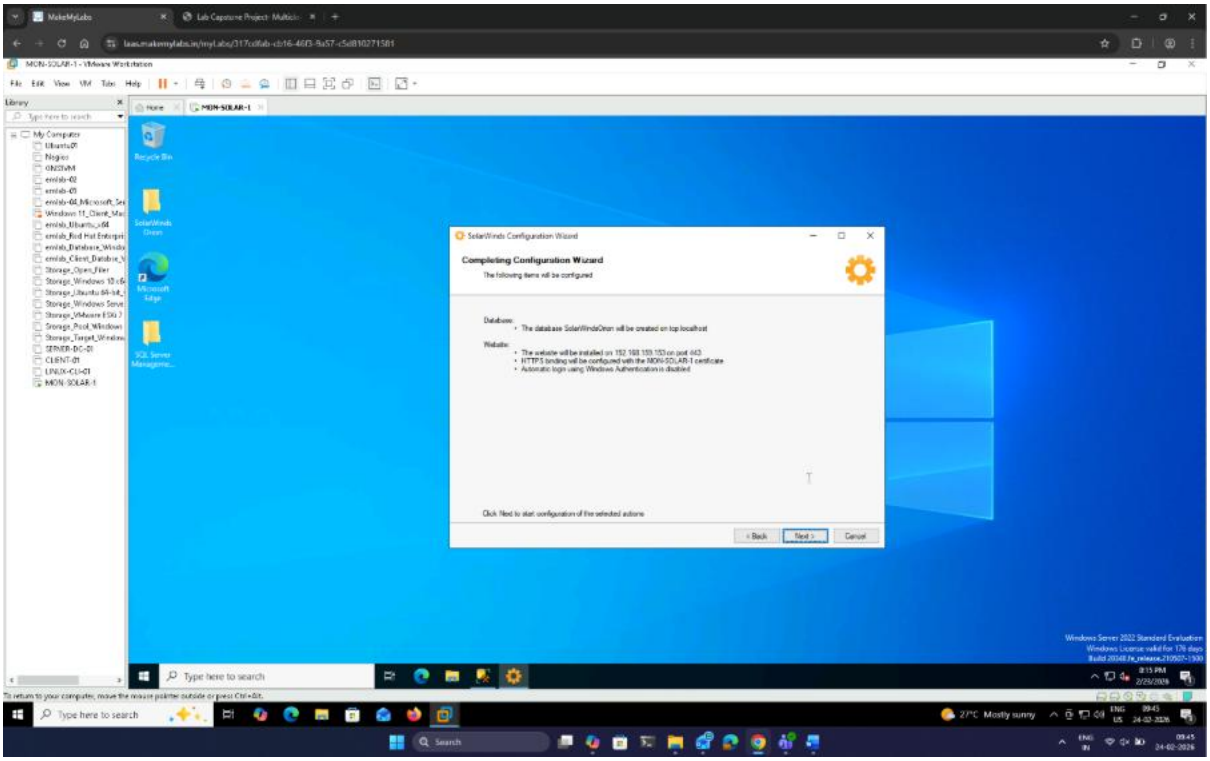

SolarWinds Configuration Wizard
Database Settings for Log and Event Monitoring
Log and event monitoring requires a separate database.
Place log and event database on the same SQL server as the primary database
Select this option if you expect to send low to moderate log traffic to primary database. If you encounter performance problems, you can always move the log database later.
Place log and event database on a dedicated SQL Server (Recommended for high volumes)
Select this option if you expect to send a large volume of log traffic to primary database.
Check SQL Server requirements for database
< Back
Next >
Cancel

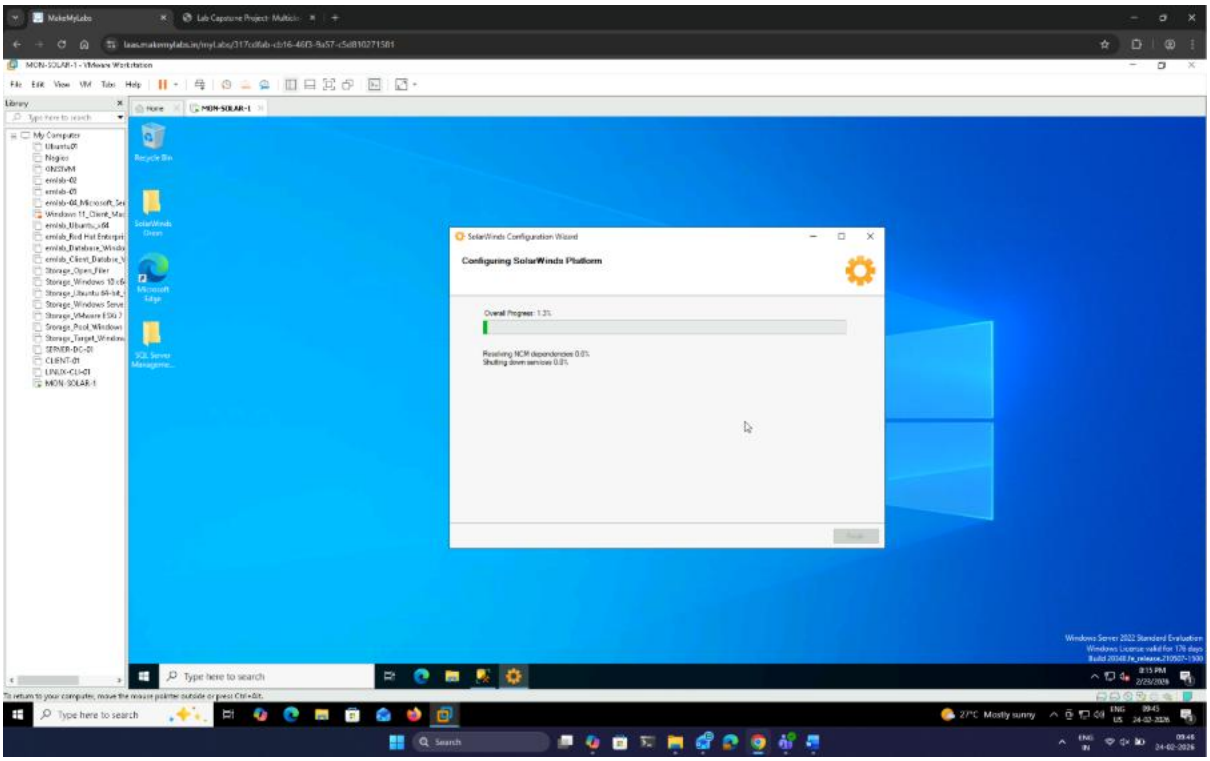

SolarWinds Configuration Wizard
Database Settings for Log and Event Monitoring
Specify the SQL Server Database to create or select an existing database.
Create a new database (recommended)
New Database
SolarWindsOrionLog
Use an existing database
Existing Database
< Back
Next >
Cancel


SolarWinds Configuration Wizard
NTA Flow Storage Configuration
Select the SQL Server for NTA Flow Storage
Additional SQL Server instance for NetFlow Traffic Analyzer
Check SQL Server requirements for Flow Storage database
SQL Server
tcp:localhost
Authenticate as currently logged in user
Switch user (Windows, Azure Active Directory or SQL Server Authentication)
Login:
sa
Password:
Encrypt connections with SSL
Tell me more
Trust server certificate
< Back
Next >
Cancel
Type here to search
Windows Server 2022 Standard Evaluation

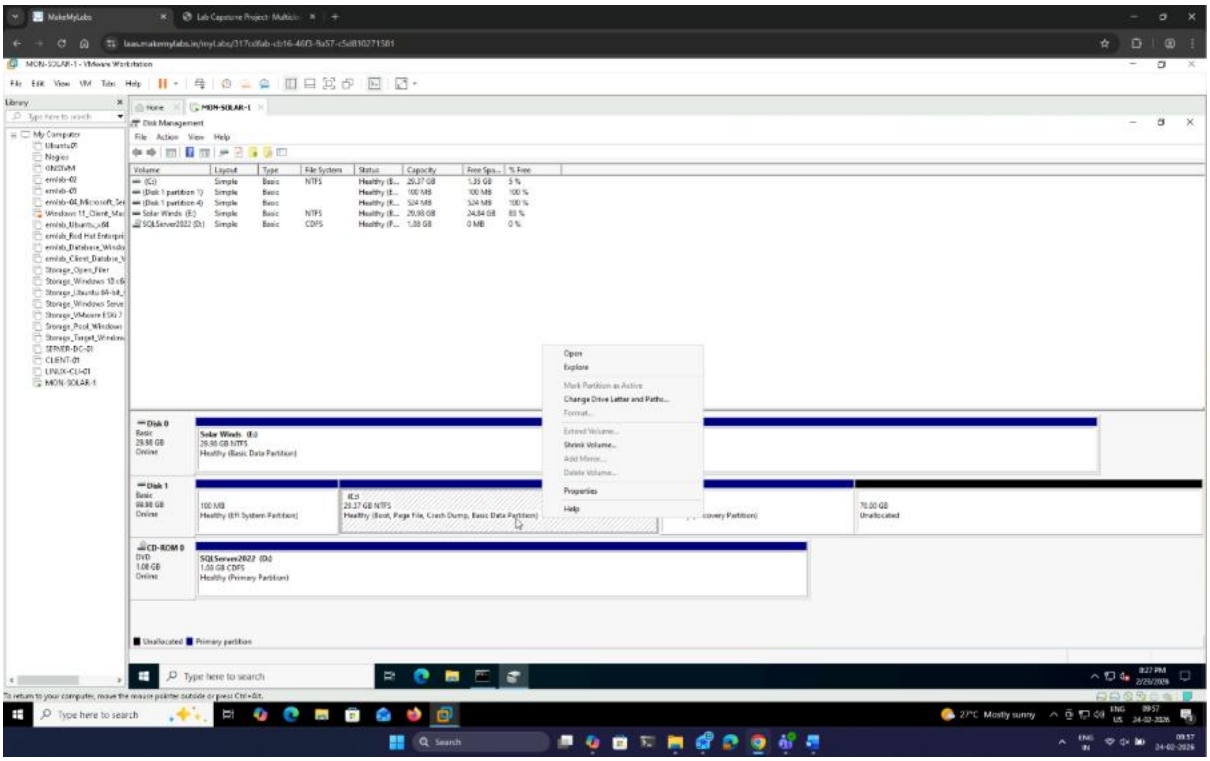

SolarWinds Configuration Wizard
NTA Flow Storage Configuration
Select the SQL Server Database to create or select an existing database.
Create a new database (recommended)
New Database:
SolarWindsFlowStorage
Use an existing database
< Back
Next >
Cancel
Type here to search
Windows Server 2022 Standard Evaluation


SolarWinds Configuration Wizard
NTA Flow Storage Configuration
Specify the SQL Server account for NTA Flow Storage to use.
NetFlow Traffic Analyzer uses a SQL Server account to access the database.
Use a Windows account or Azure Active Directory account to access the database.
Use an existing account
Use a SQL Server account to access the database.
Create a new account
Use an existing account
Existing Account:
Password:
< Back
Next >
Cancel

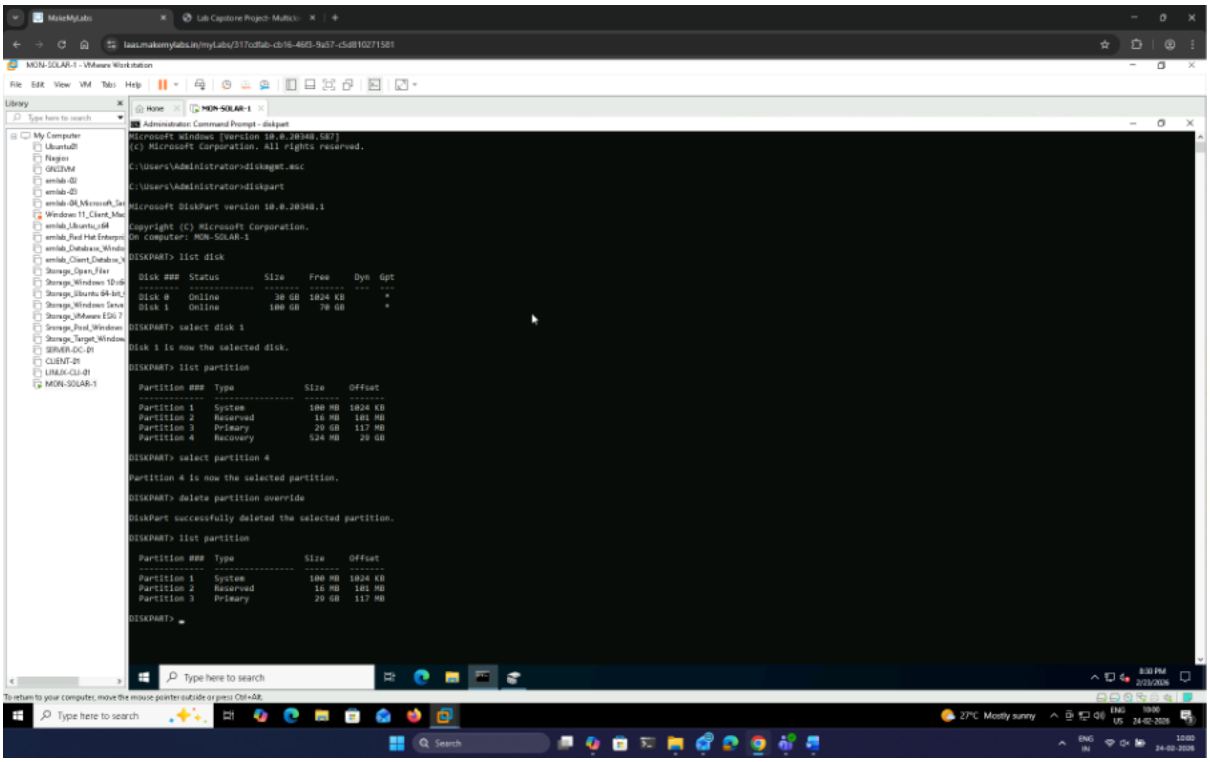

SolarWinds Configuration Wizard
Completing Configuration Wizard
The following items will be configured
Database:
Website:
Click Next to start configuration of the selected actions
< Back
Next >
Cancel

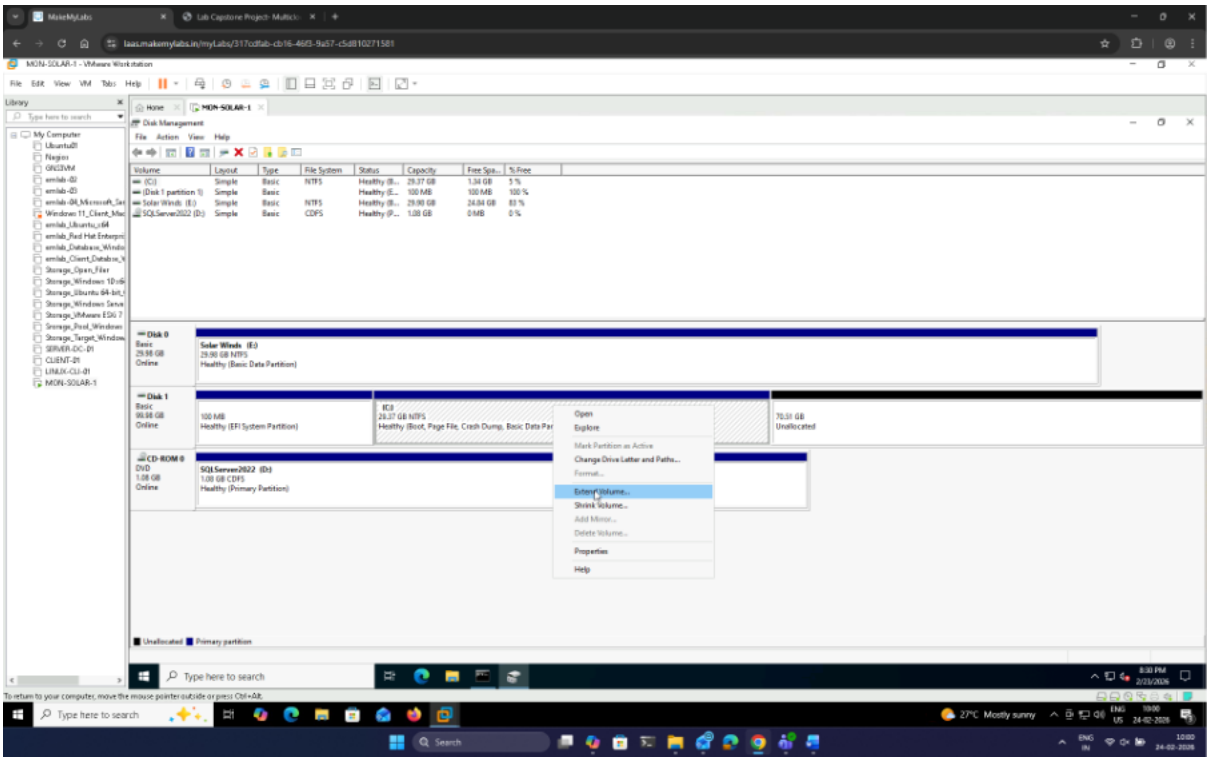


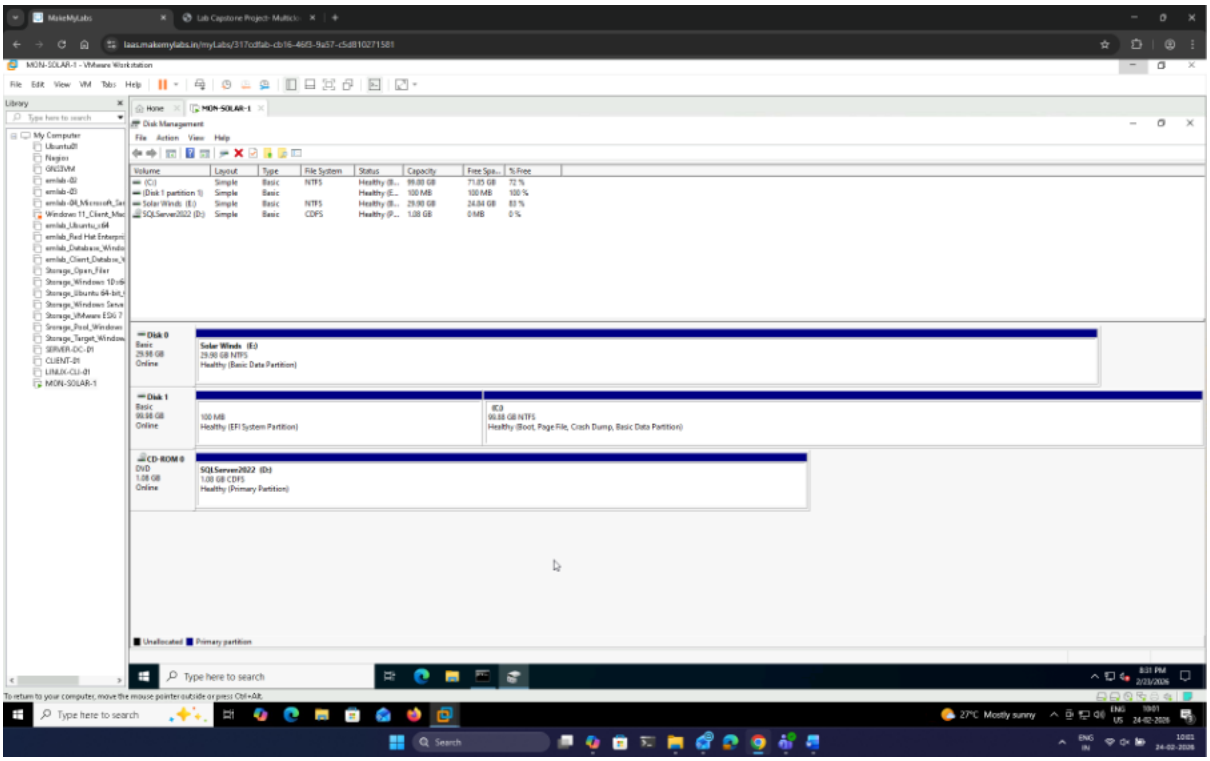
After all the configurations is completed, the configuration of the SolarWinds platform is initiated.



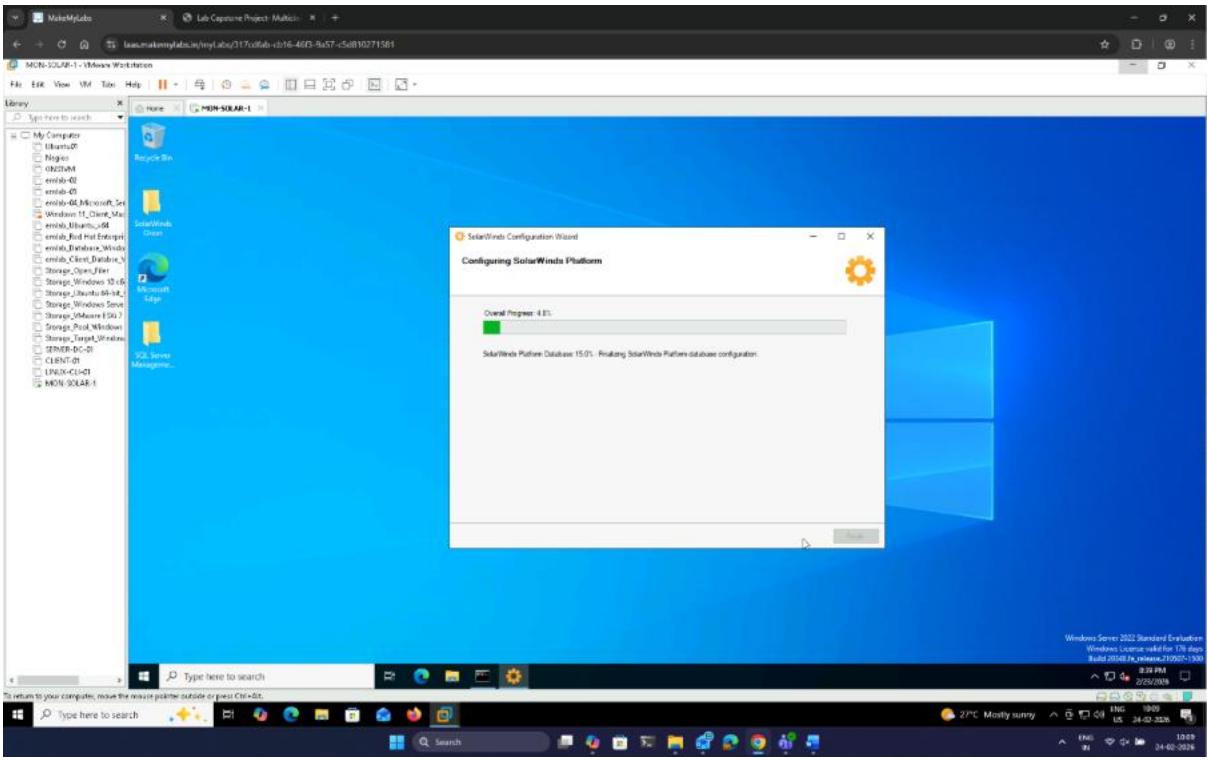
The Recovery Partition is blocking the extension, since this is a Lab VM, the clean and safe solution is to remove the Recovery partition.



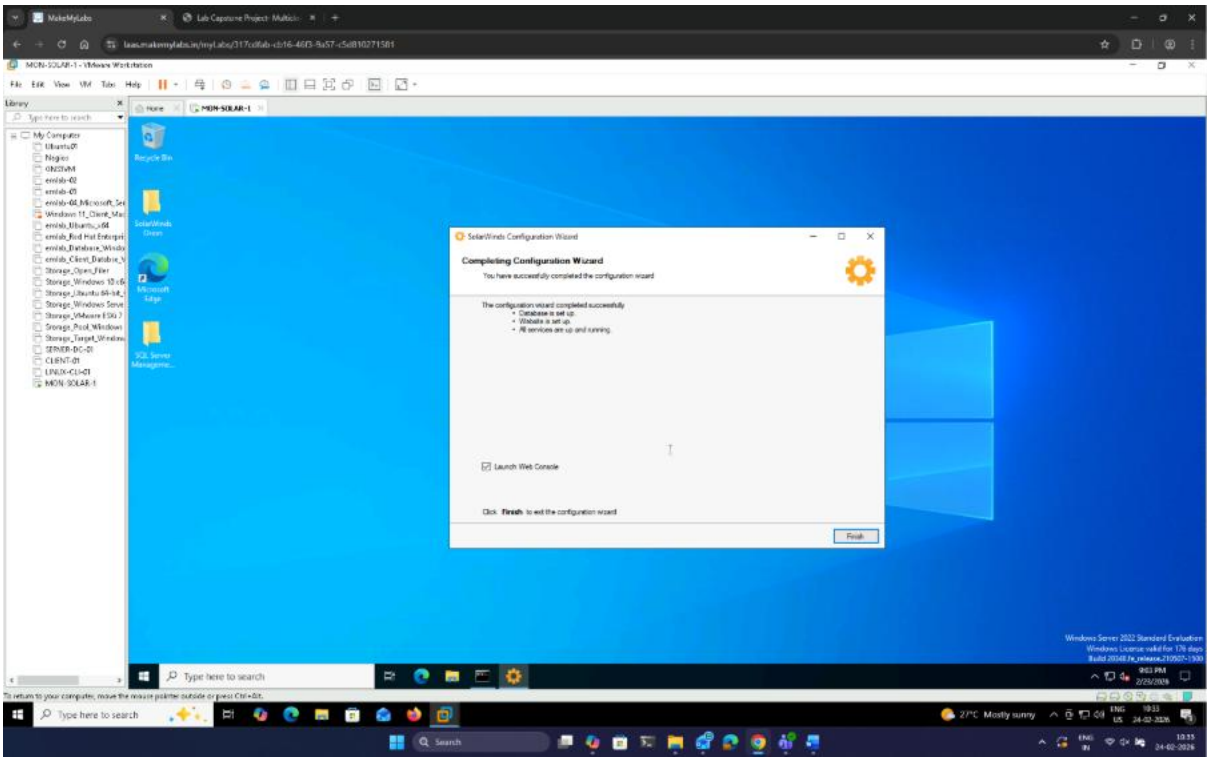




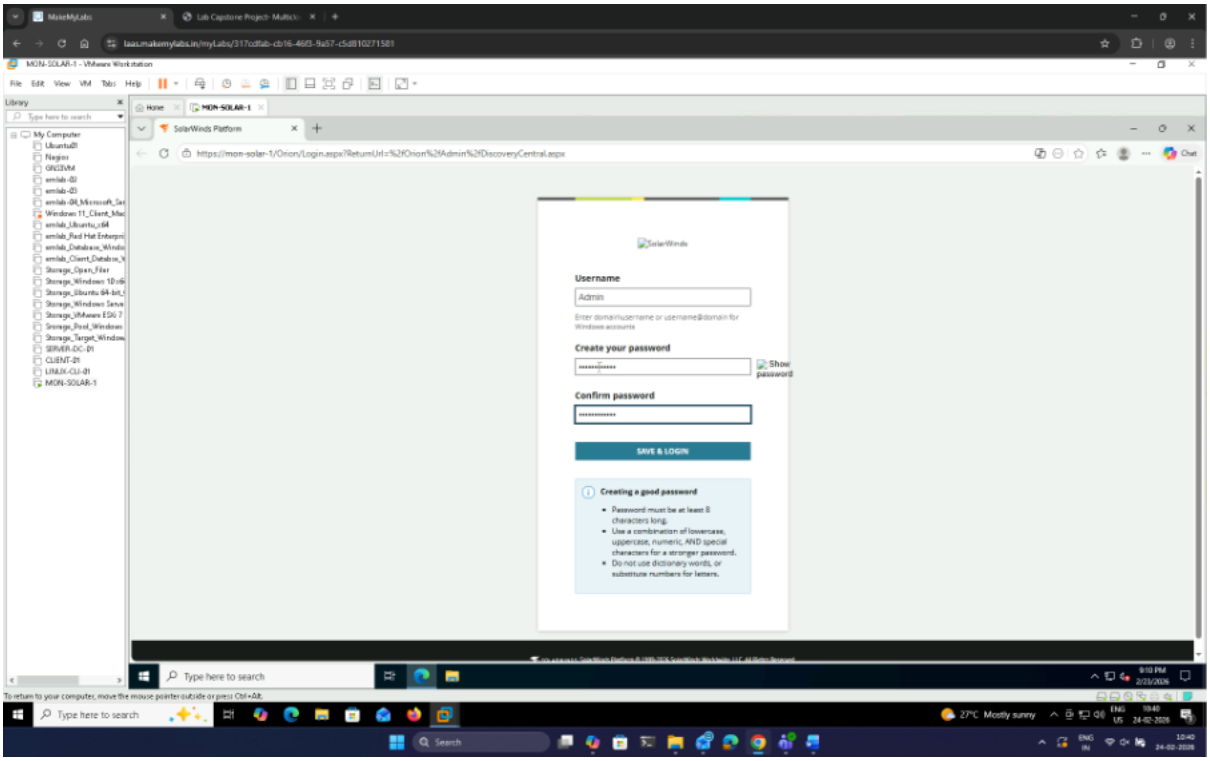
The system recovery partition was located between the primary system partition and the unallocated disk space, preventing volume extension. Using DiskPart, the recovery partition was removed in the lab environment, allowing the C: partition to be extended to utilize the expanded virtual disk capacity.



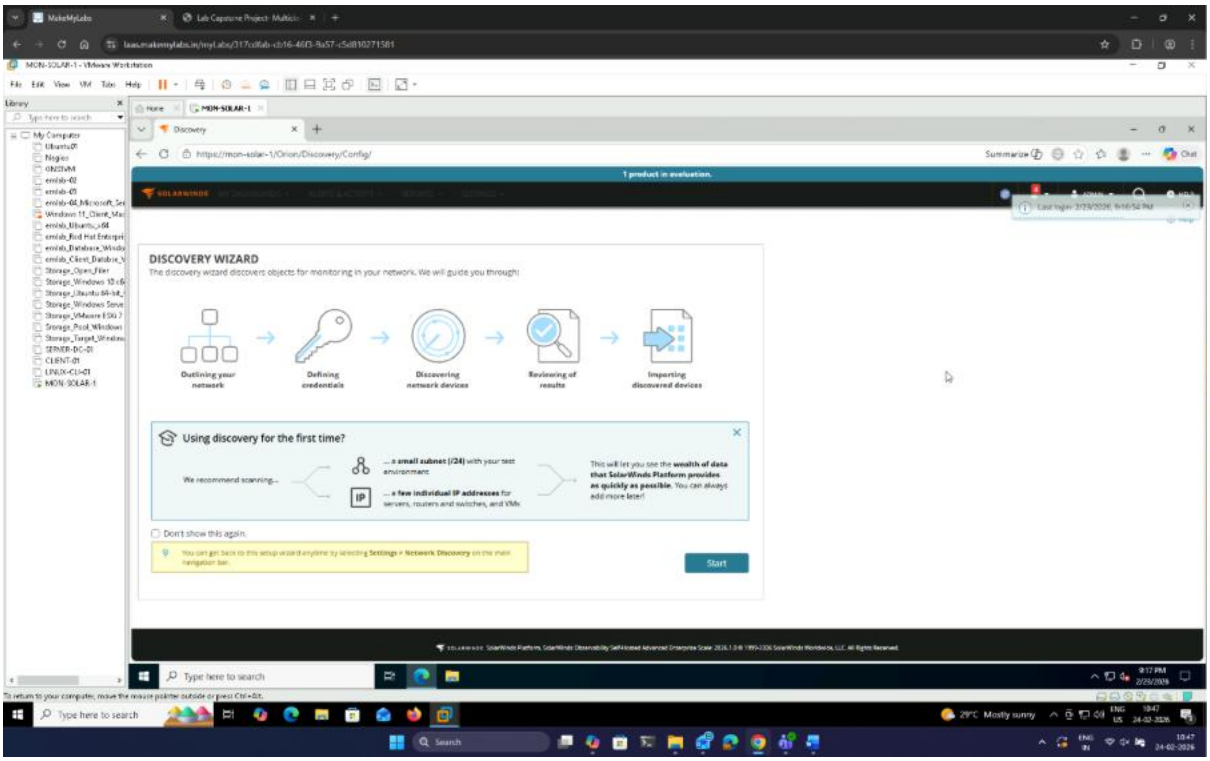
Due to an incomplete initial configuration, the existing SolarWindsOrion database could not be used. The Database was recreated using the SolarWinds Configuration. The database was recreated using the SolarWinds Configuration Wizard to ensure proper schema initialization and platform configuration and the process was repeated.



The SolarWinds Platform was successfully installed and configured on a Windows Server.



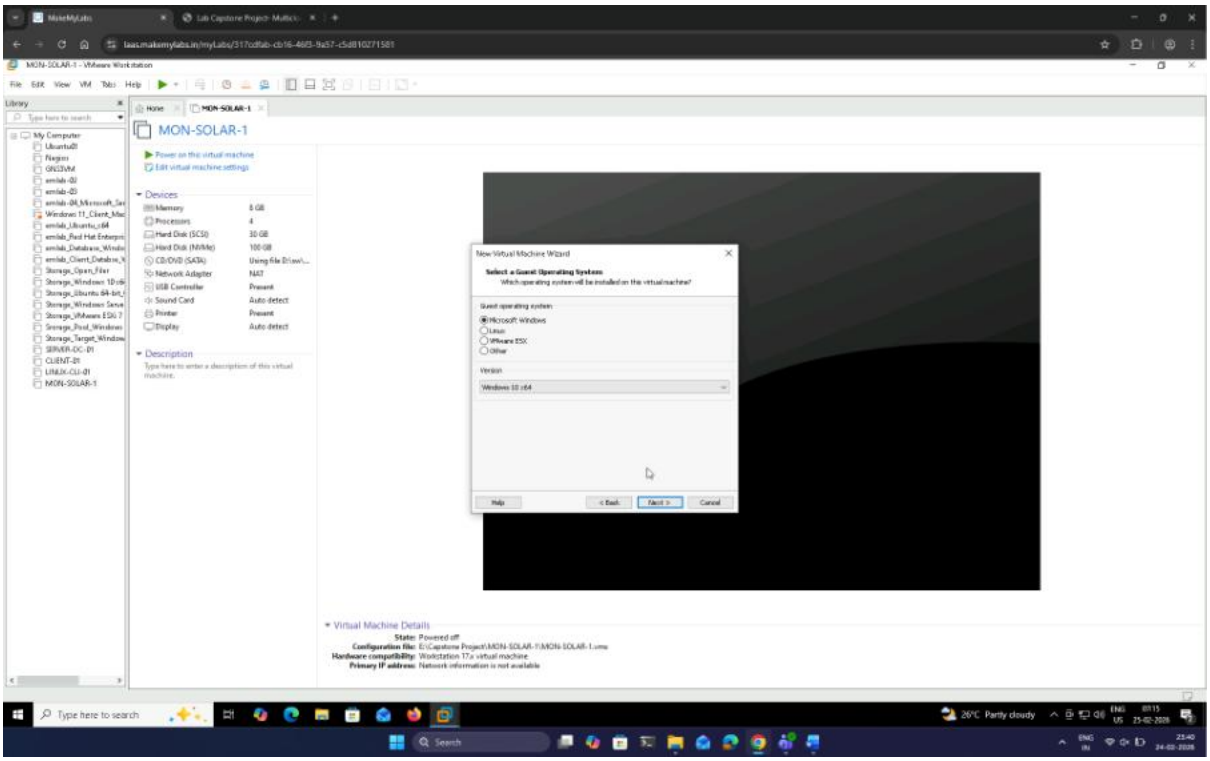
Setup an Admin Password for Authentication.

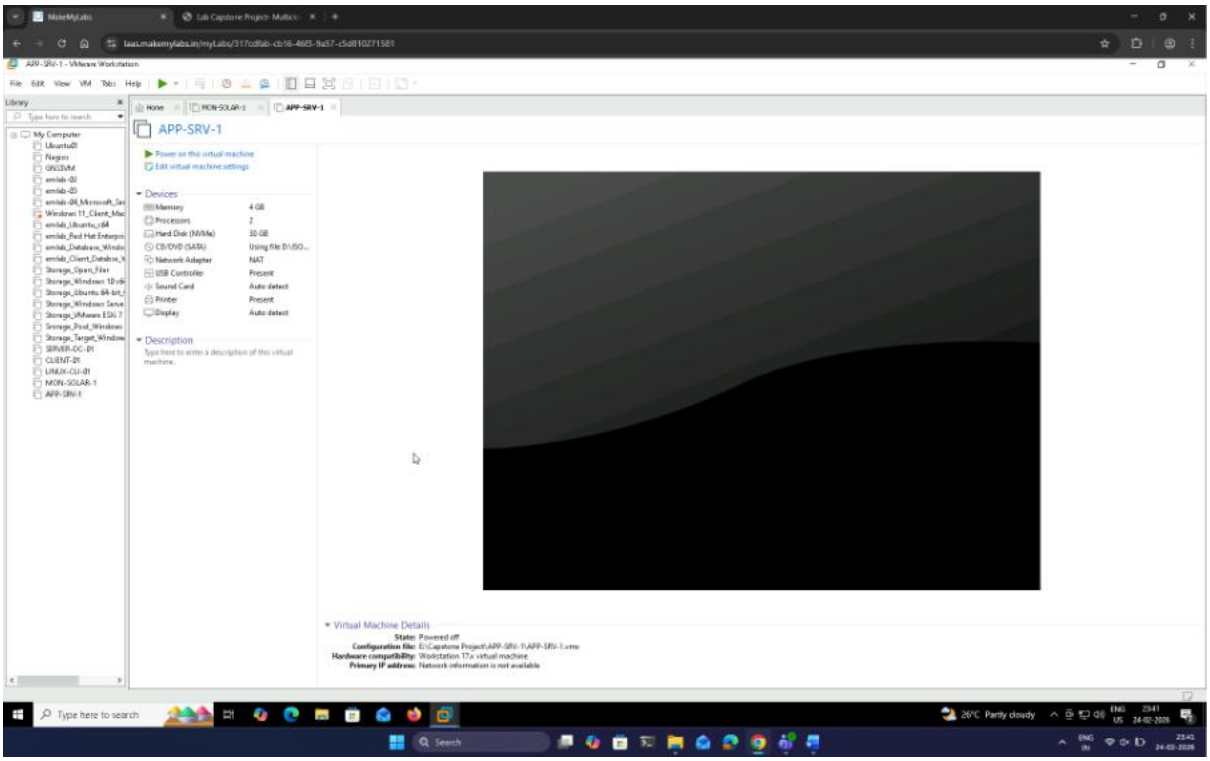


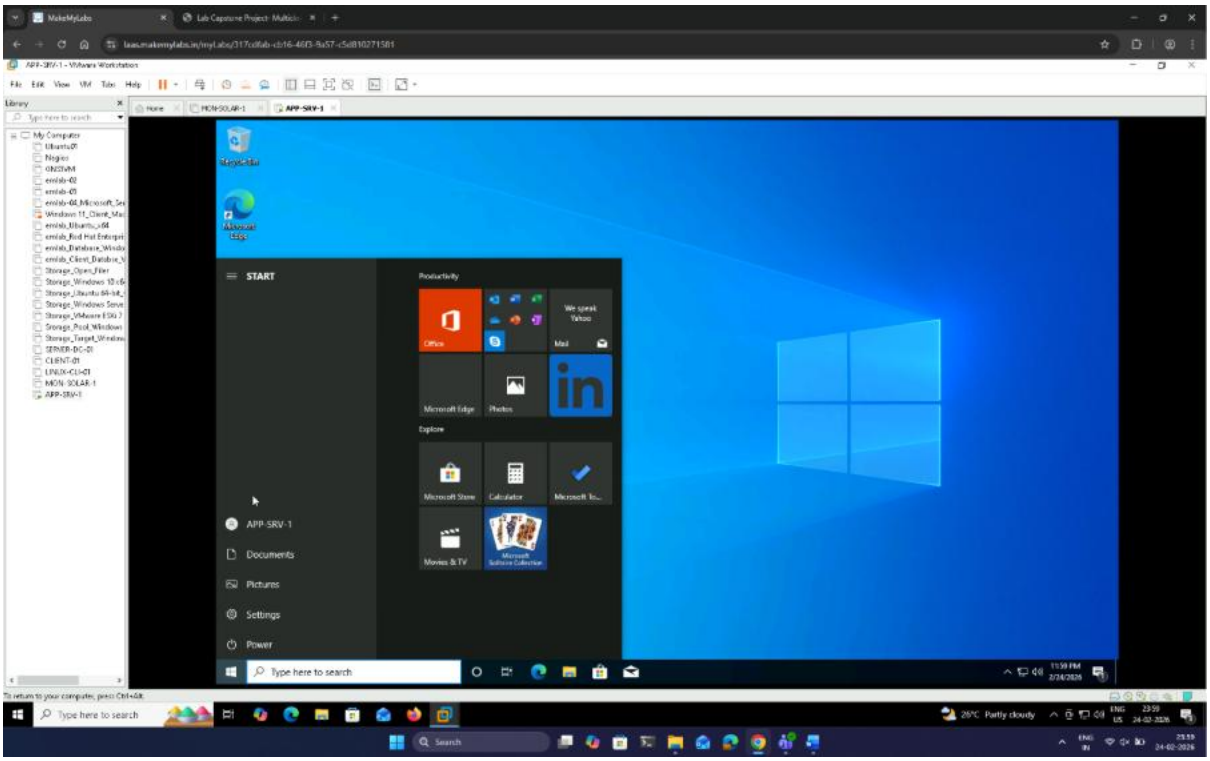
The dashboard is visible as soon as the credentials are verified and the user is logged in.

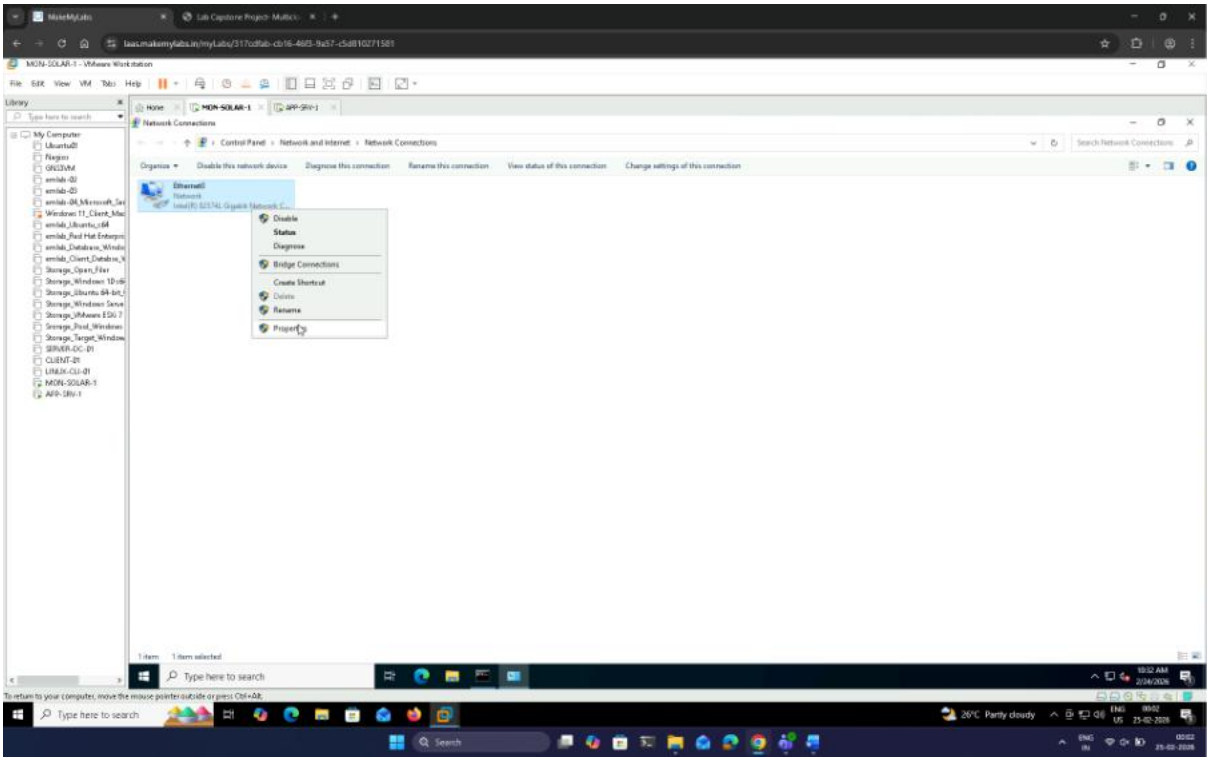
We will now create an VM that we will be adding into the Monitoring network.

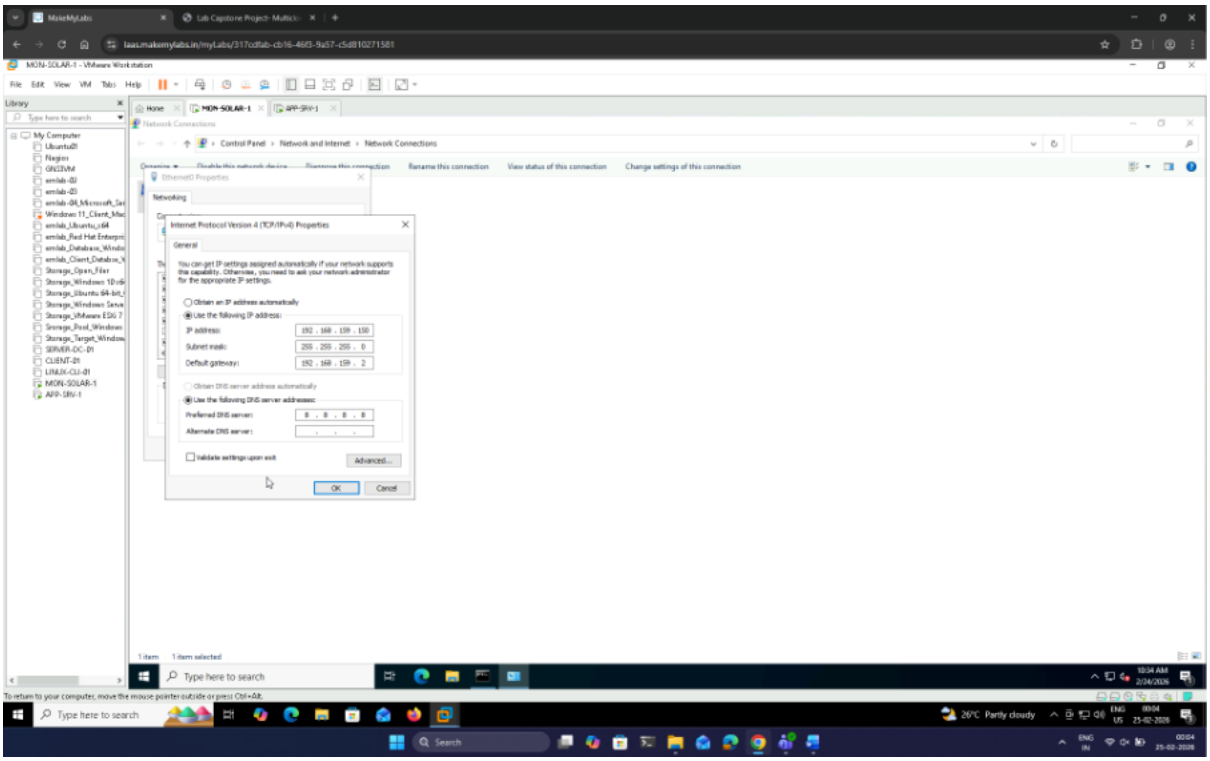
- Name: - APP-SRV-1
- OS: - Windows 10 Client
- RAM: - 4 GB
- CPU: - 2 Cores
- Disk: - 30 GB

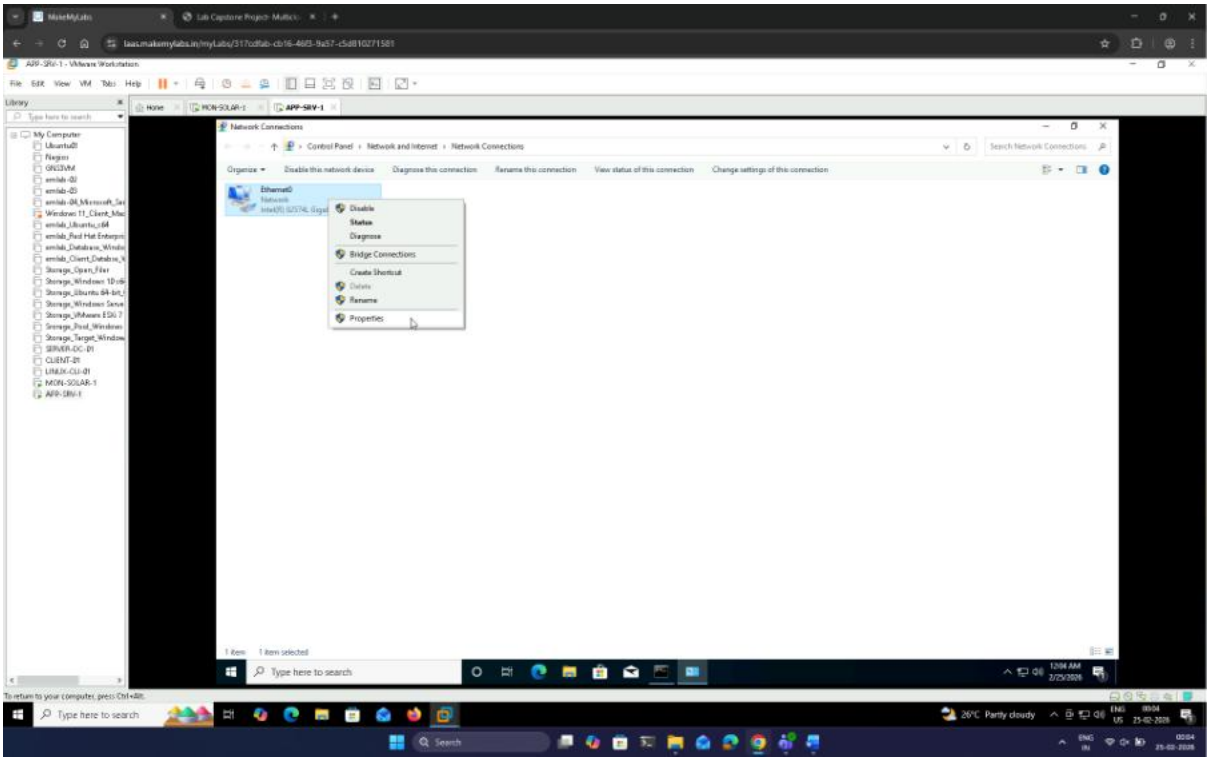

MON-SOLAR-1
New Virtual Machine Wizard
Select a Guest Operating System
Which operating system will be installed on this virtual machine?
Guest operating system
Microsoft Windows
Linux
VMware ESX
Other
Version
Windows 10 x64
Help
< Back
Next >
Cancel
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\MON-SOLAR-1\MON-SOLAR-1.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available

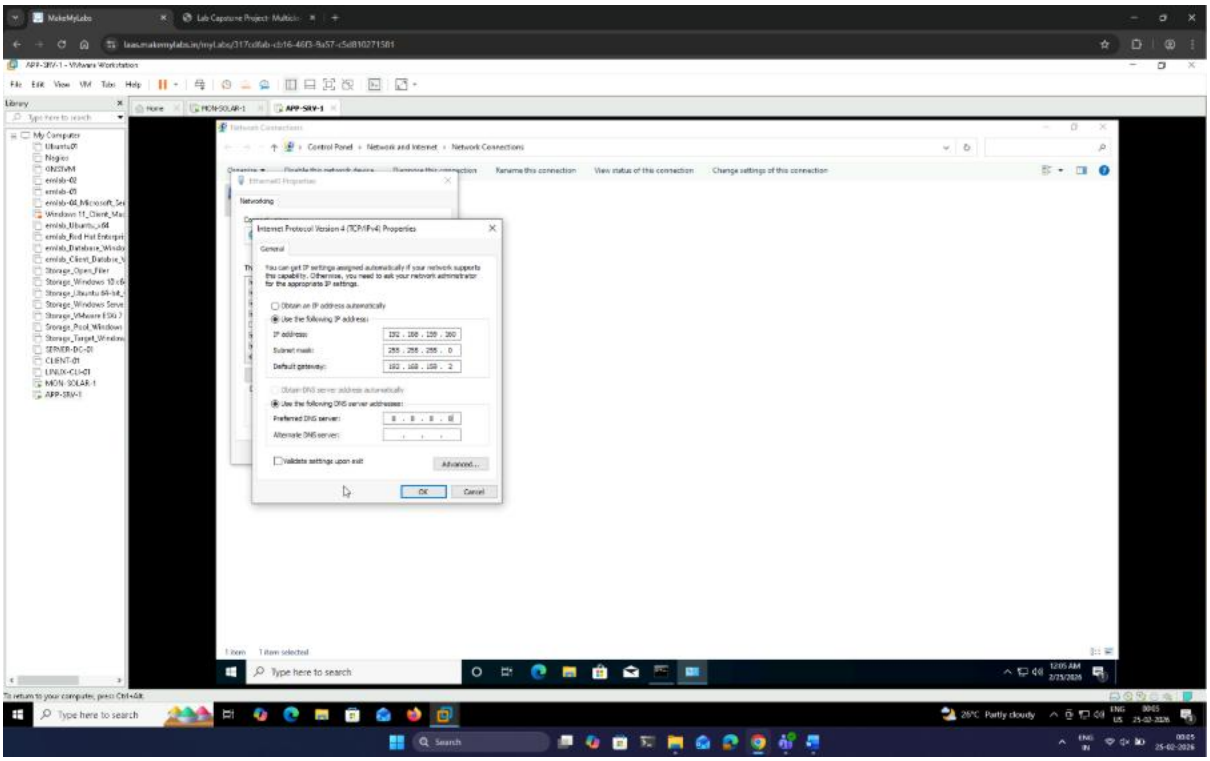

APP-SRV-1
Virtual Machine Details
State: Powered off
Configuration file: E:\Capstone Project\APP-SRV-1\APP-SRV-1.vmx
Hardware compatibility: Workstation 17.x virtual machine
Primary IP address: Network information is not available

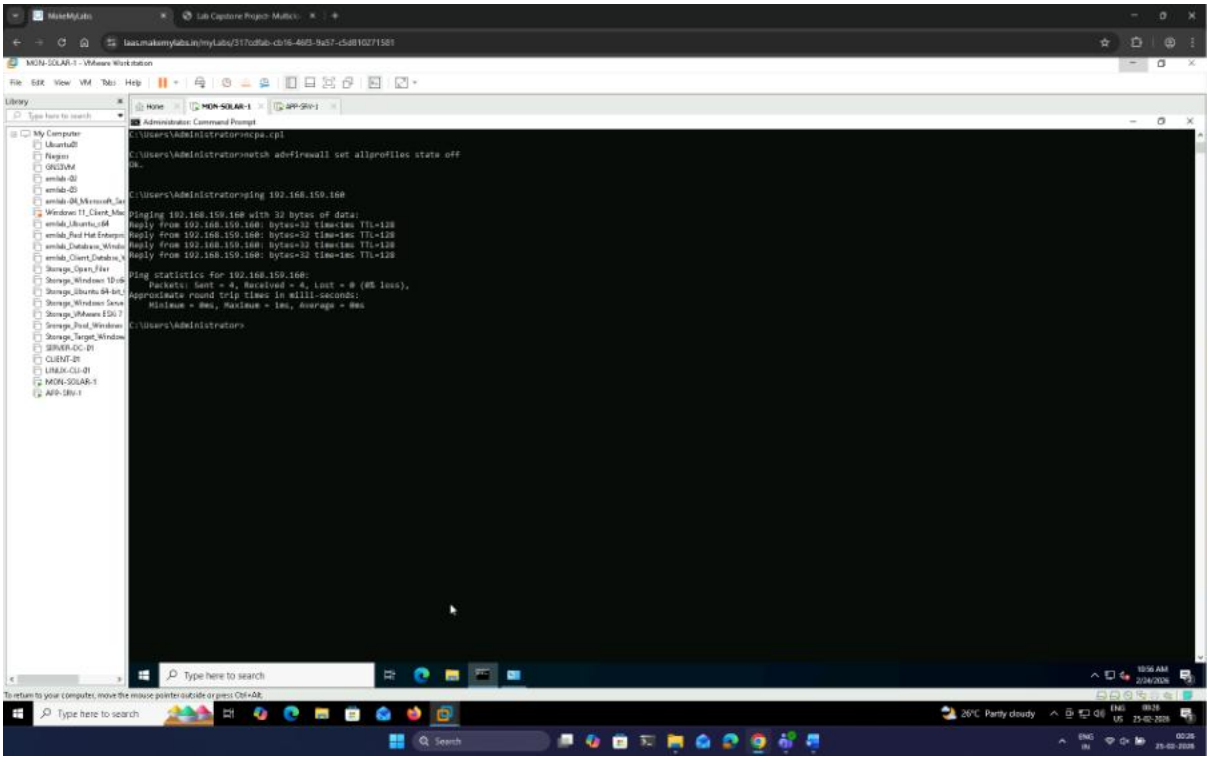

START
APP-SRV-1
Documents
Pictures
Settings
Power
Type here to search

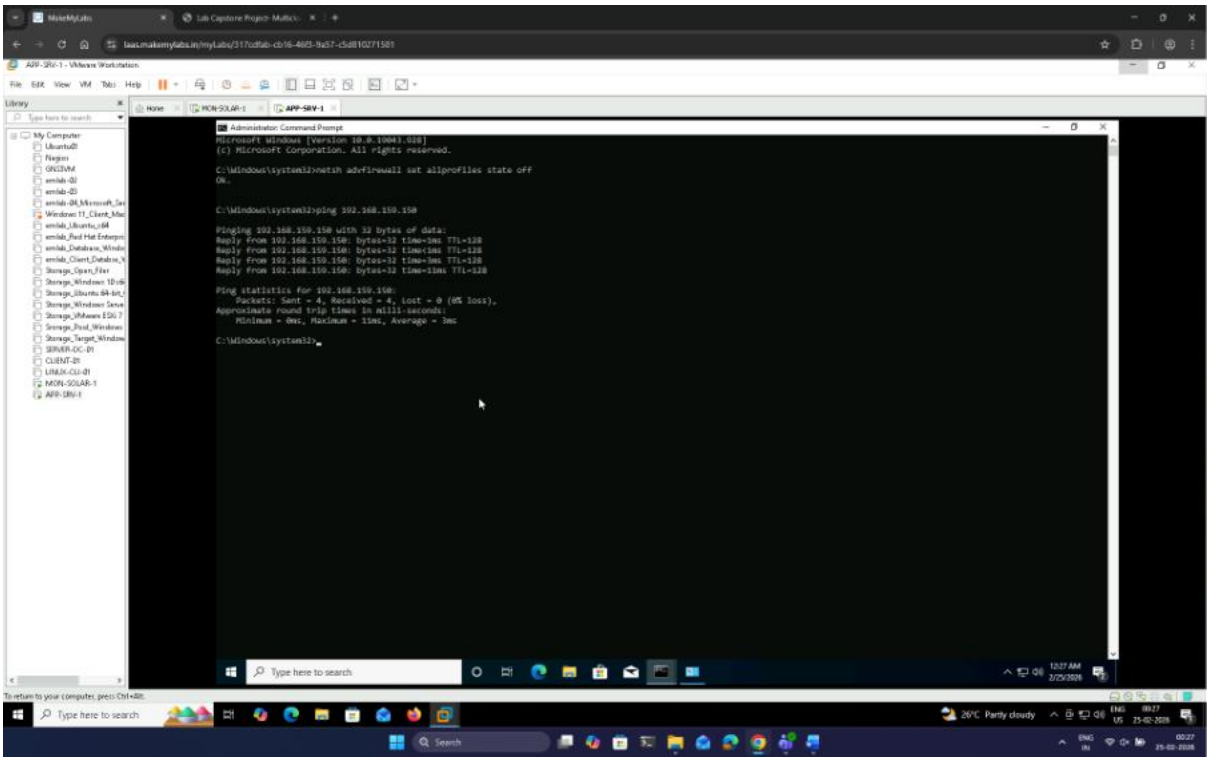

Network Connections
Disable
Status
Diagnose
Bridge Connections
Create Shortcut
Delete
Rename
Properties
Type here to search


Internet Protocol Version 4 (TCP/IPv4) Properties
General
Obtain an IP address automatically
Use the following IP address:
IP address:
Subnet mask:
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
Type here to search

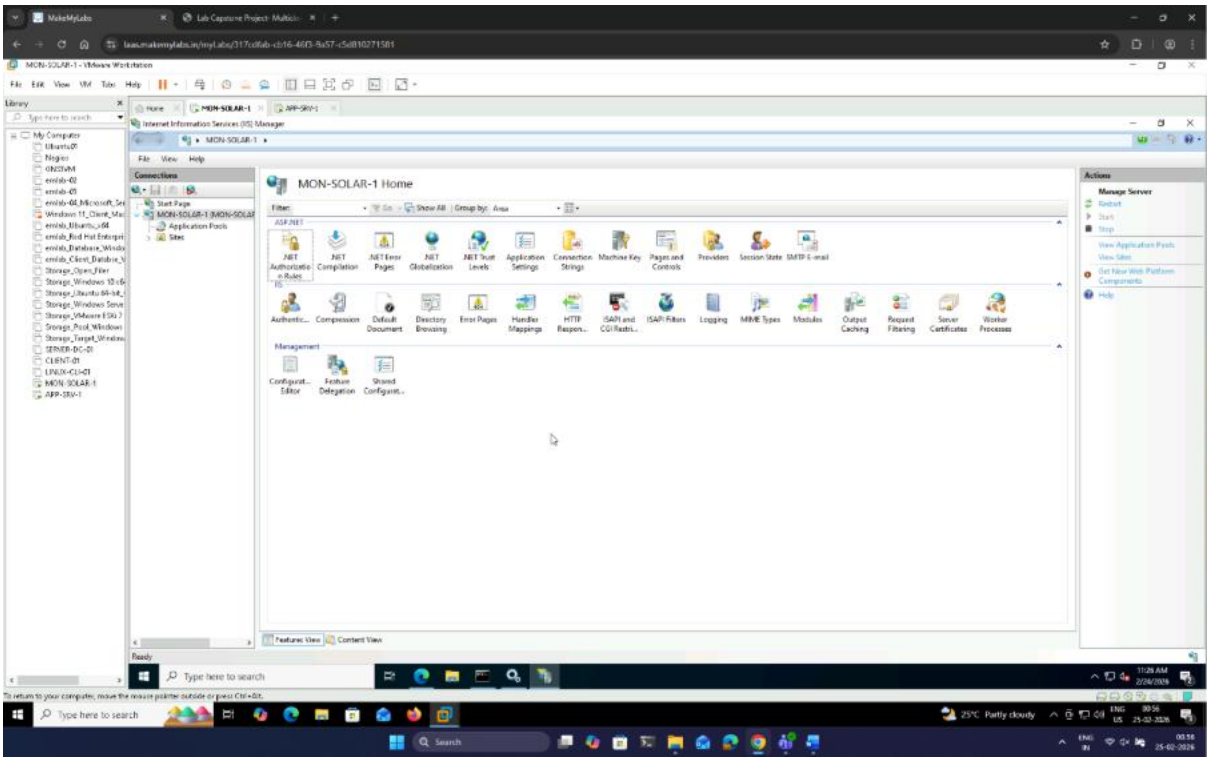

Network Connections
Control Panel > Network and Internet > Network Connections
Organize
Disable this network device
Diagnose this connection
Rename this connection
View status of this connection
Change settings of this connection
Ethernet0
Disable
Status
Diagnose
Bridge Connections
Create Shortcut
Delete
Rename
Properties
Type here to search

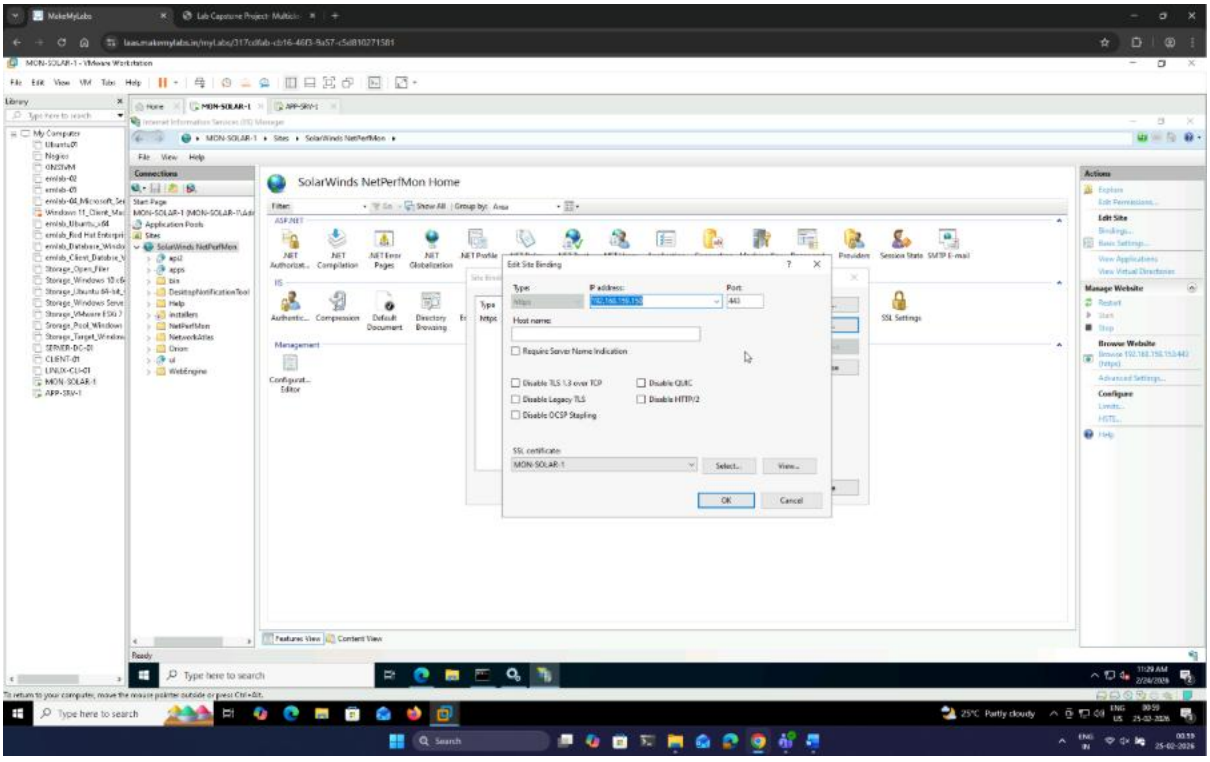

Internet Protocol Version 4 (TCP/IPv4) Properties
Type here to search

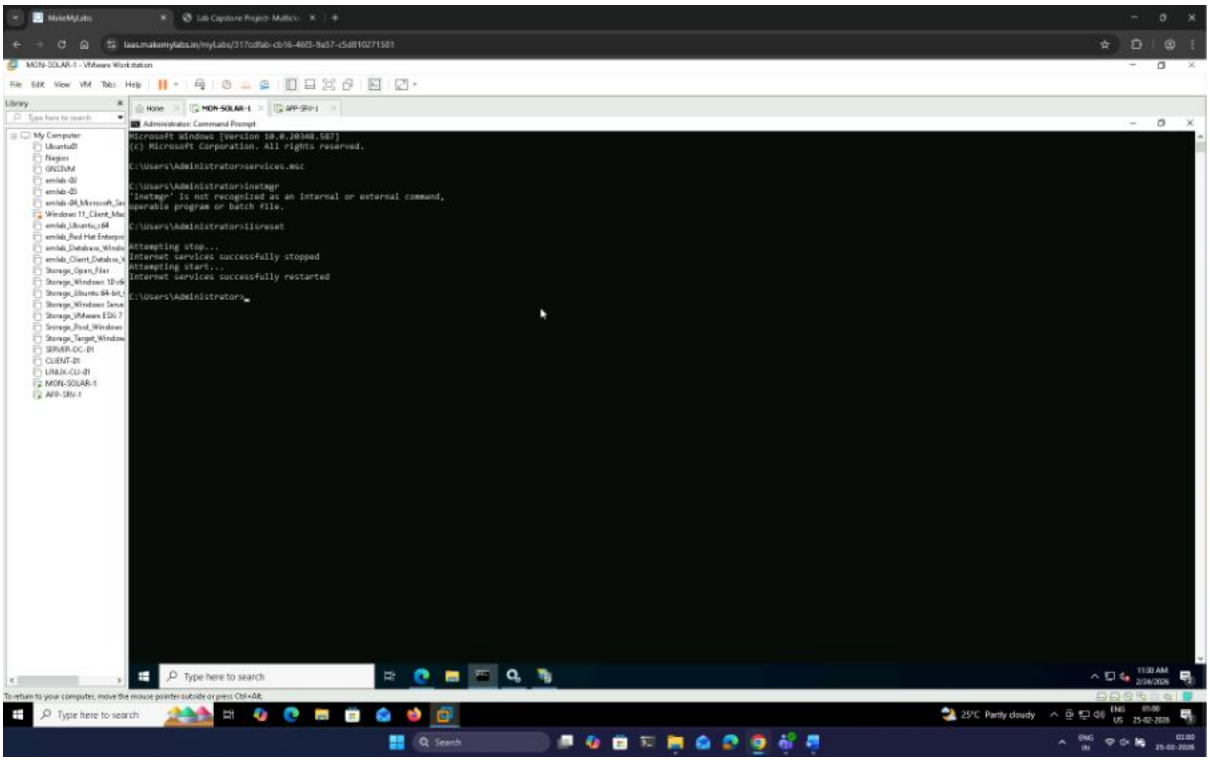

Administrator: Command Prompt
C:\Users\Administrator>ncpa.cpl
C:\Users\Administrator>netsh advfirewall set allprofiles state off
Ok.
C:\Users\Administrator>ping 192.168.159.160
Pinging 192.168.159.160 with 32 bytes of data:
Reply from 192.168.159.160: bytes=32 time<1ms TTL=128
Reply from 192.168.159.160: bytes=32 time<1ms TTL=128
Reply from 192.168.159.160: bytes=32 time<1ms TTL=128
Reply from 192.168.159.160: bytes=32 time<1ms TTL=128
Ping statistics for 192.168.159.160:
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 0ms, Maximum = 1ms, Average = 0ms
C:\Users\Administrator>
Type here to search

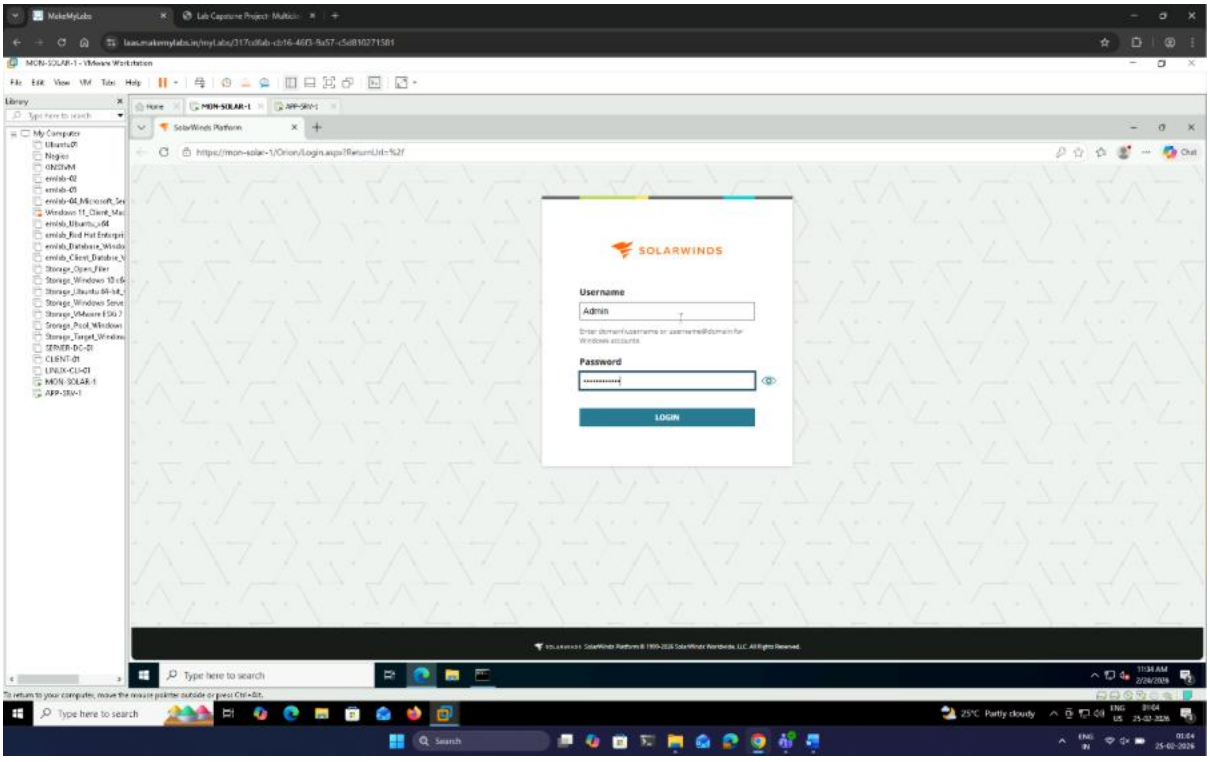


After assigning static IP addresses, initial connectivity testing failed due to Windows Firewall restrictions blocking ICMP traffic. Firewall rules were temporarily adjusted to allow ping communication between monitored systems, enabling proper monitoring communication.


MON-SOLAR-1 Home

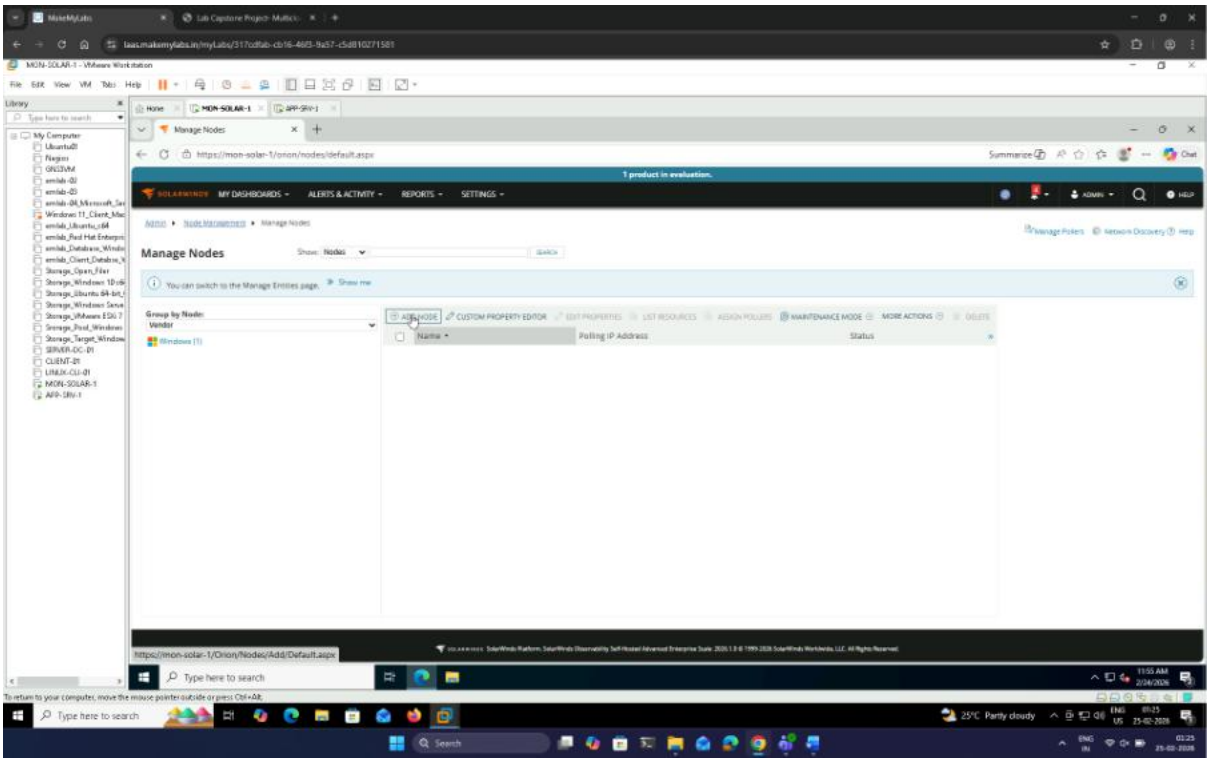

SolarWinds NetPerfMon Home
Edit Site Binding

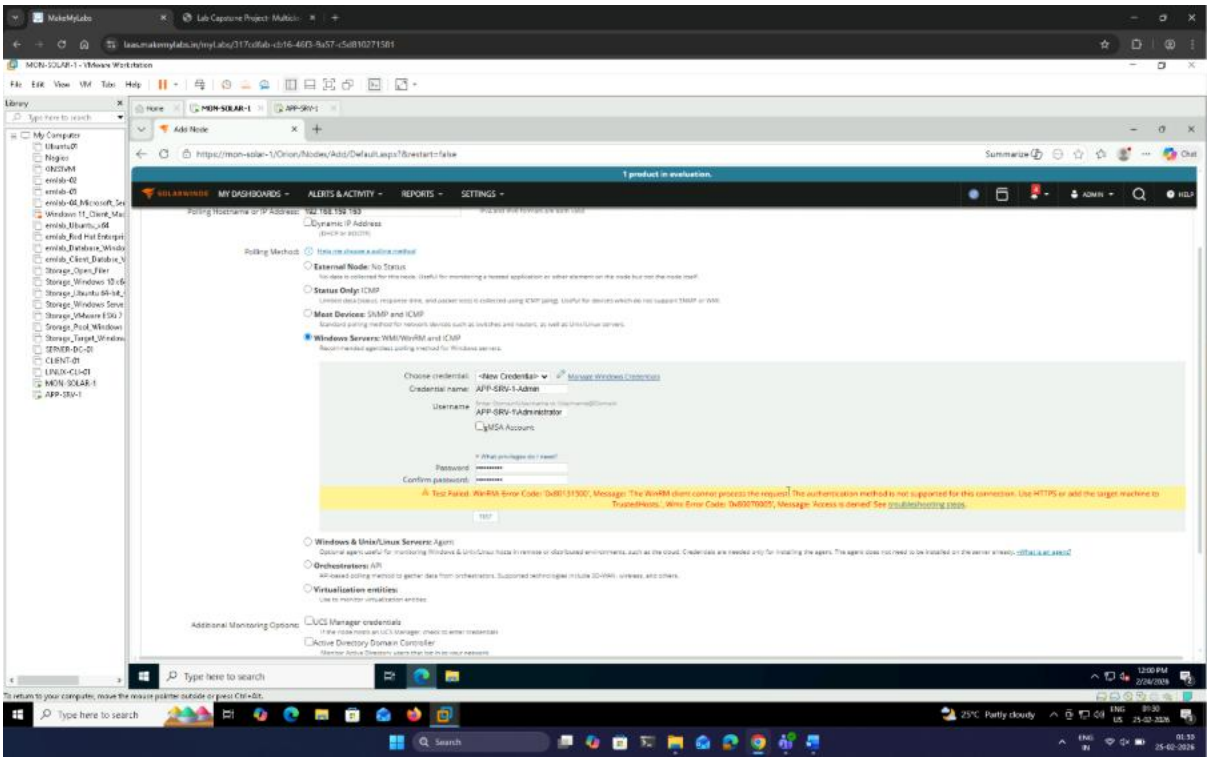




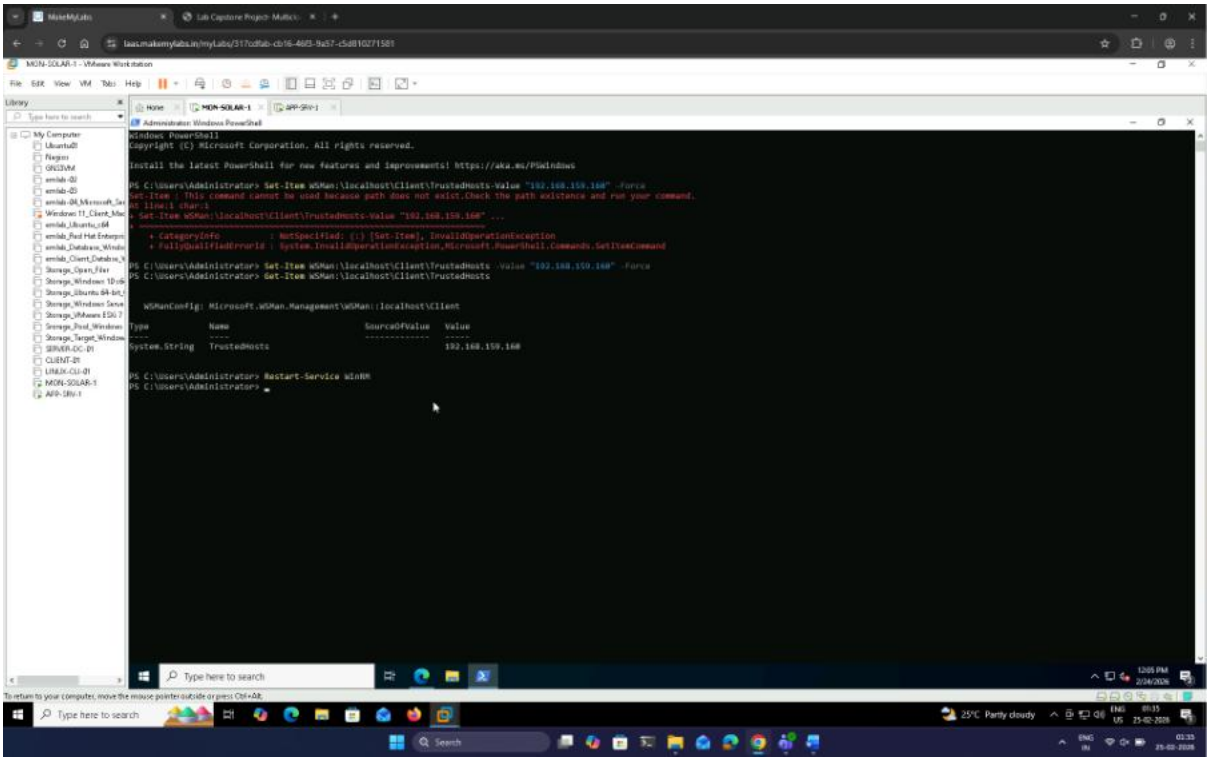
After changing the monitoring server's IP address to a static configuration, the SolarWinds web console became inaccessible and displayed as "ERR_CONNECTION_RESET" error. Upon investigation using IIS Manager, it was found that the HTTPS binding for the

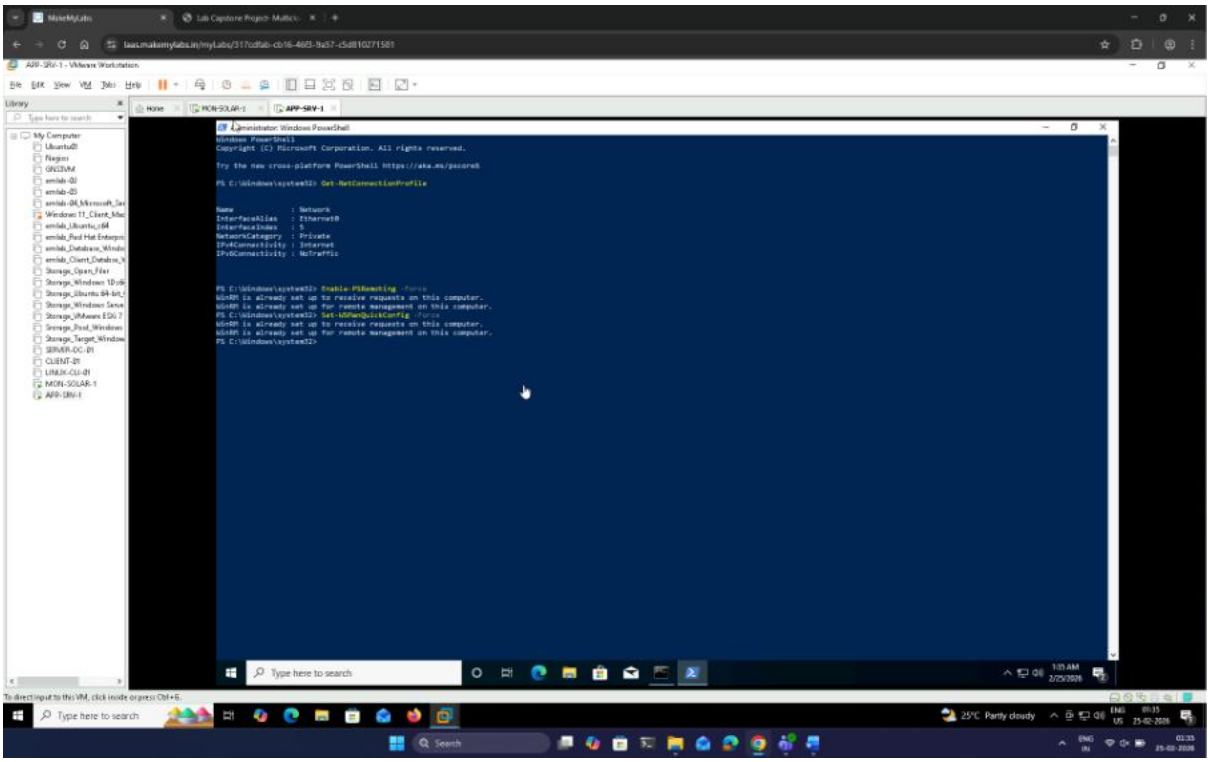
SolarWinds NetPerfMon site was still configured with the previous IP address. Because IIS was listening on an outdated IP address, the web console could not be reached. The binding was modified to the new IP address instead of the old address, and IIS was restarted using the iisreset command. After this correction, the SolarWinds web console became accessible via both hostname and IP address.

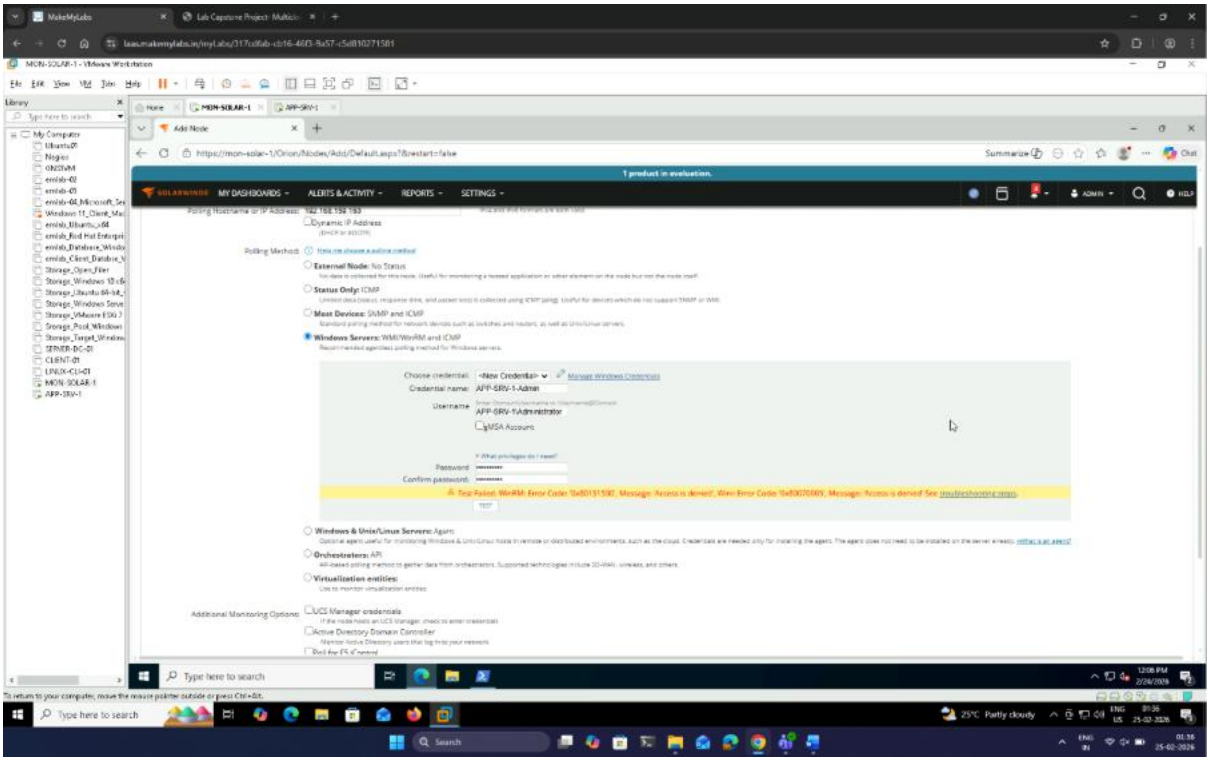




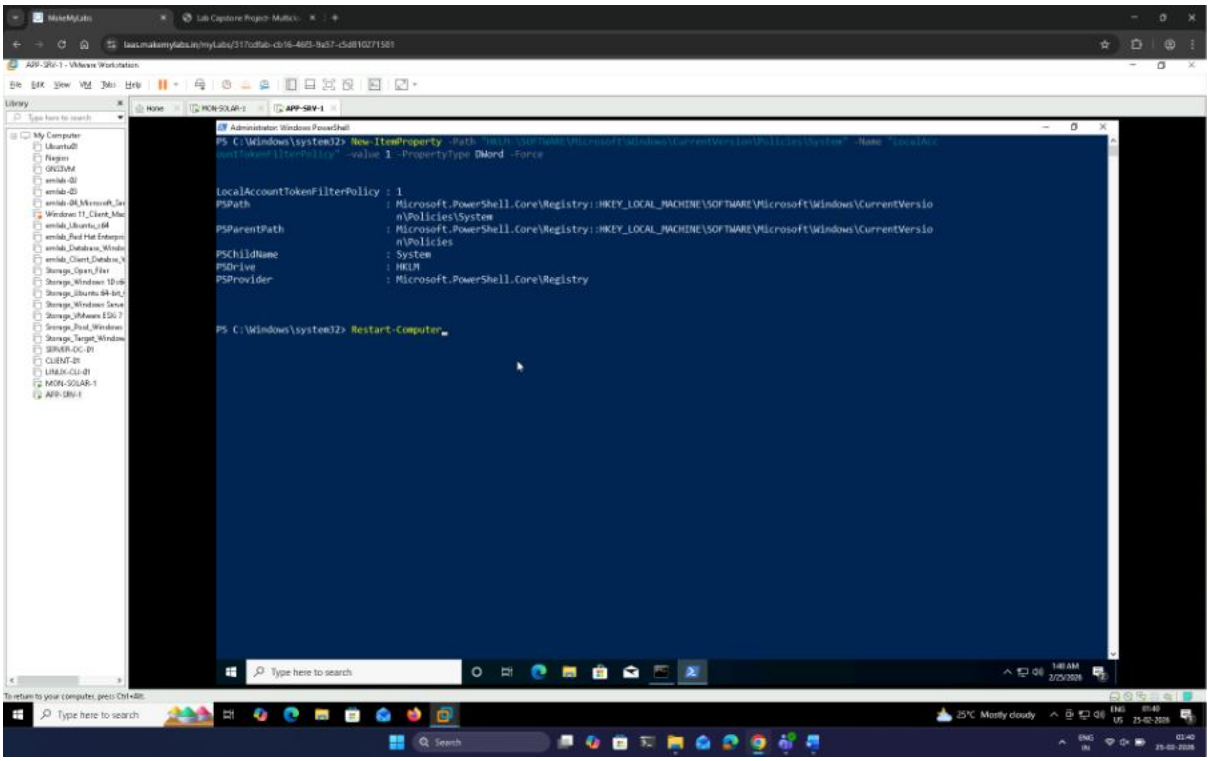
This is a common issue which happens when both the machines are not in a Workgroup or are not domain joined and WinRM blocks authentication by default between workgroup machines.

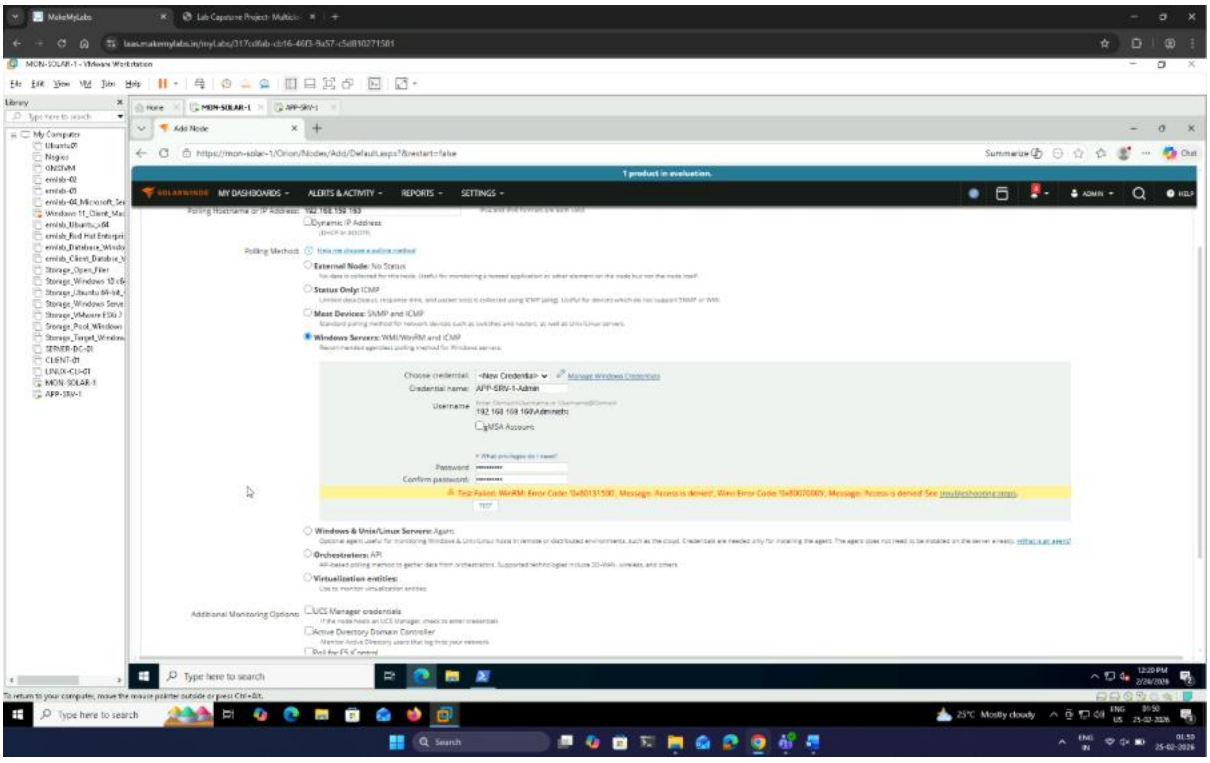


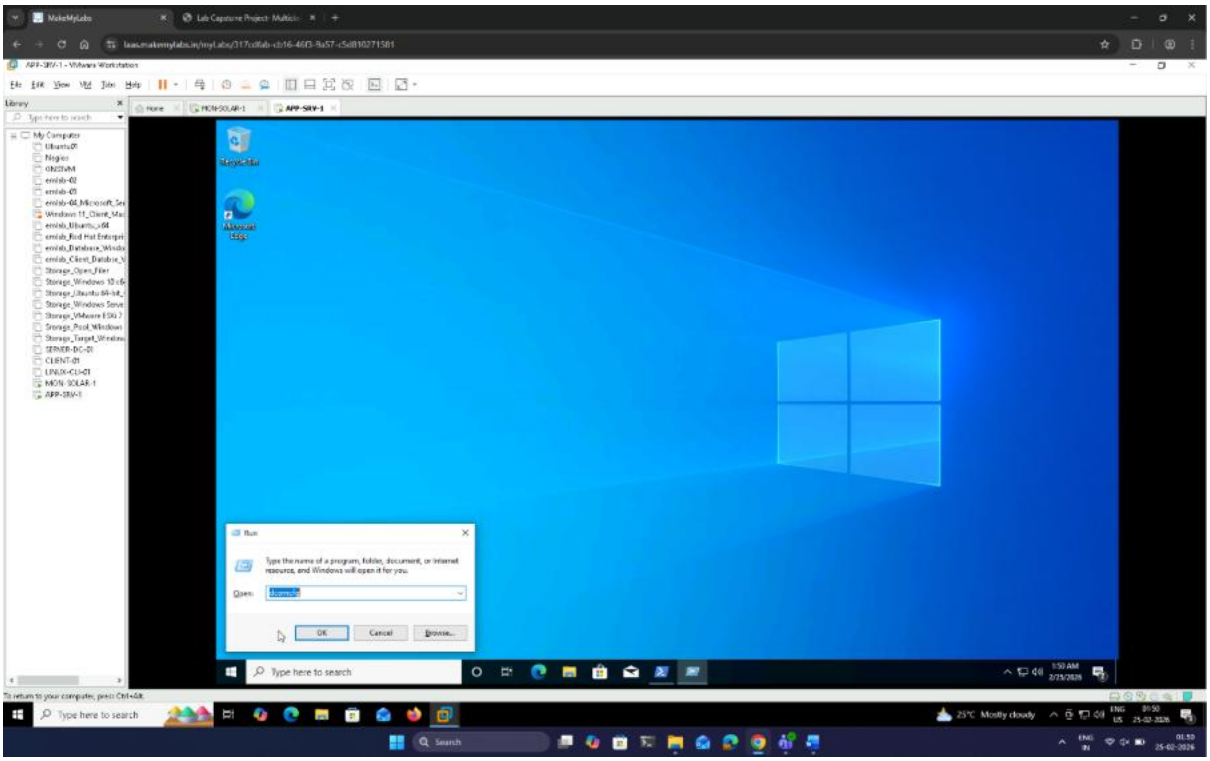




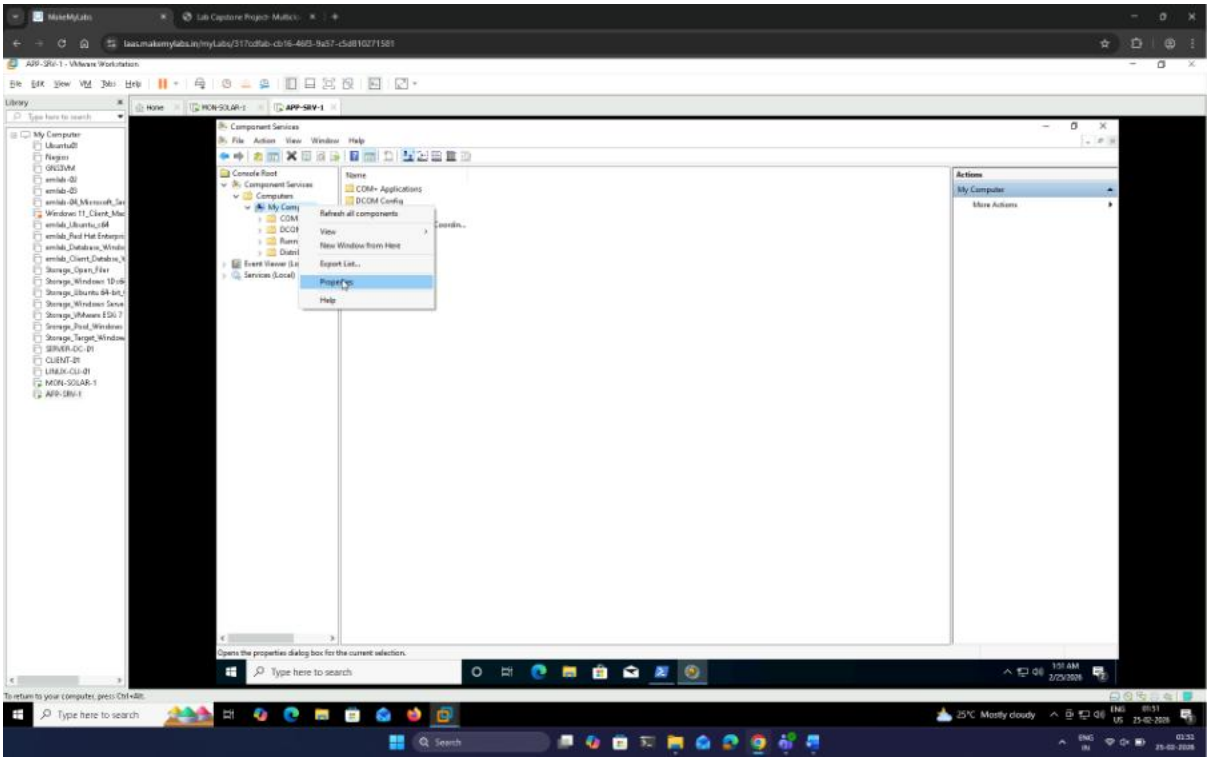
This error indicates that Authentication is reaching the machine by Windows is blocking the remote admin access.

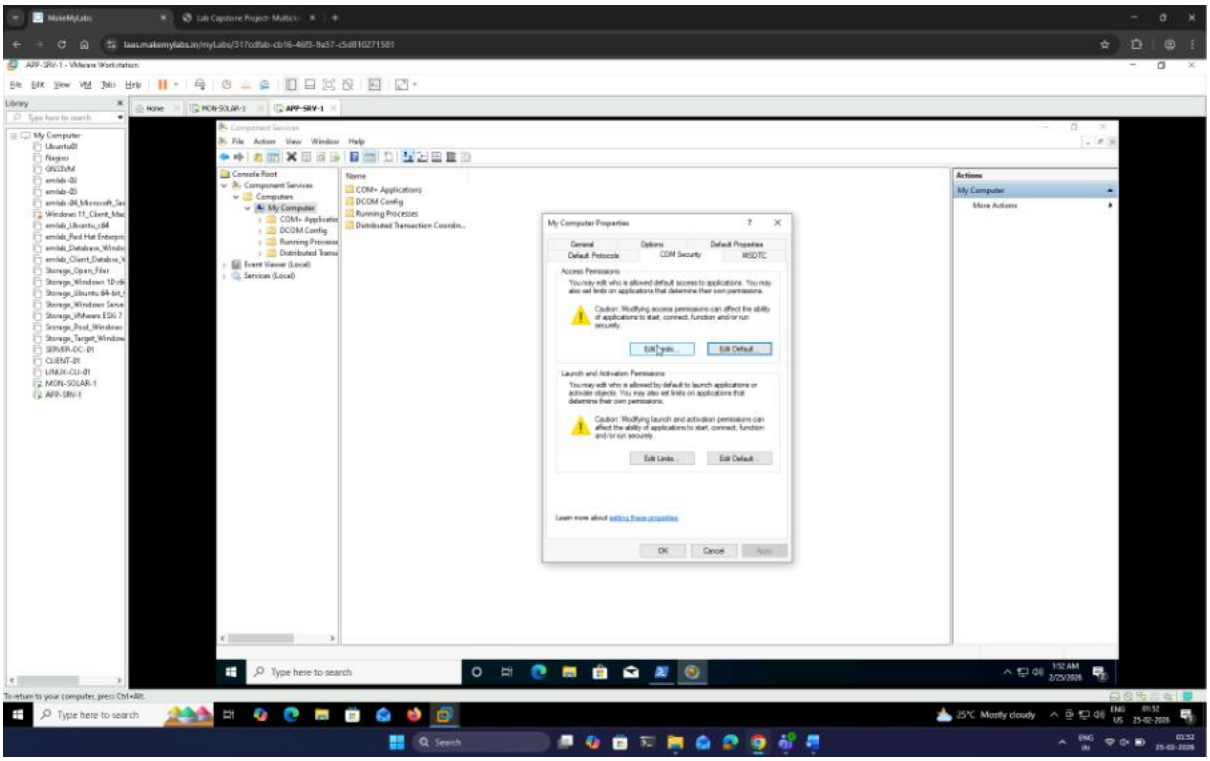

Administrator: Windows PowerShell
PS C:\Windows\system32> New-ItemProperty -Path "HKLM:\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System" -Name "LocalAccountTokenFilterPolicy" -value 1 -PropertyType DWord -Force
LocalAccountTokenFilterPolicy : 1
PSPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies\System
PSParentPath : Microsoft.PowerShell.Core\Registry::HKEY_LOCAL_MACHINE\SOFTWARE\Microsoft\Windows\CurrentVersion\Policies
PSChildName : System
PSDrive : HKLM
PSProvider : Microsoft.PowerShell.Core\Registry
PS C:\Windows\system32> Restart-Computer

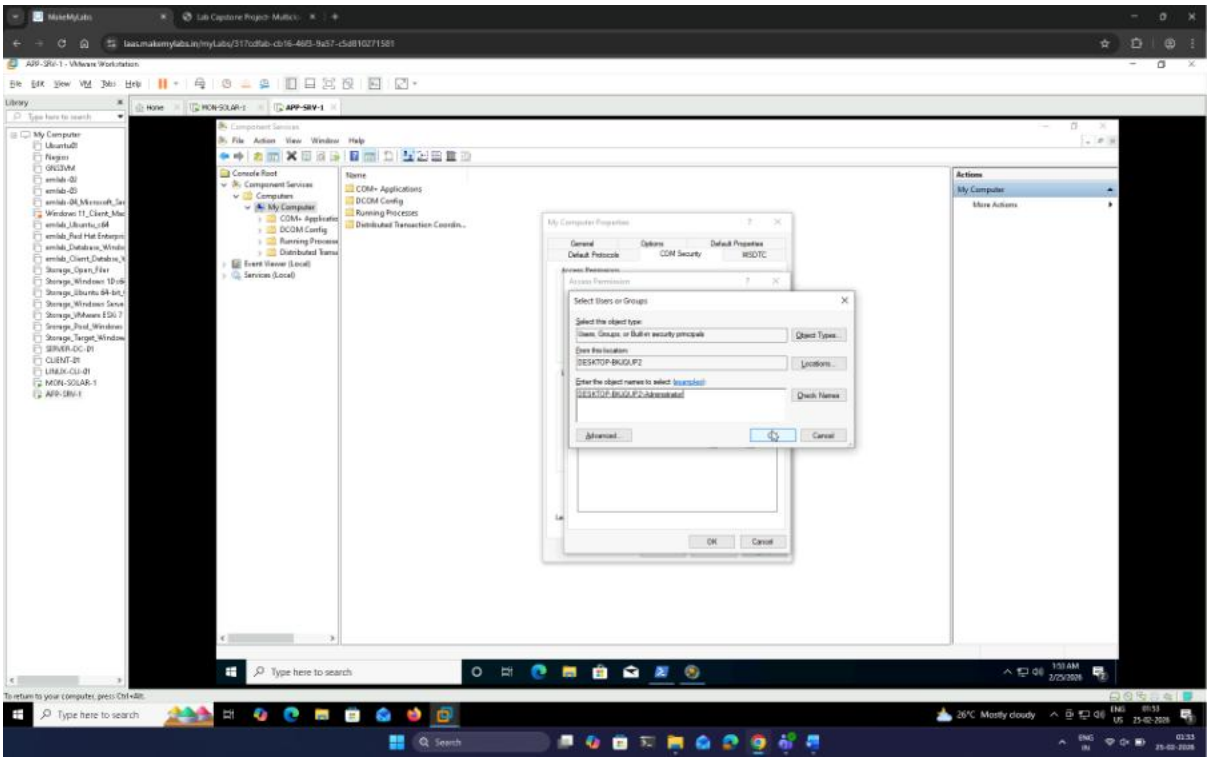

1 product in evaluation.
MY DASHBOARDS
ALERTS & ACTIVITY
REPORTS
SETTINGS
Windows Servers: WMI/WinRM and ICMP
Credential name: APP-SRV-1-Admin
Windows & Unix/Linux Servers: Agent
Orchestrators: API
Virtualization entities:

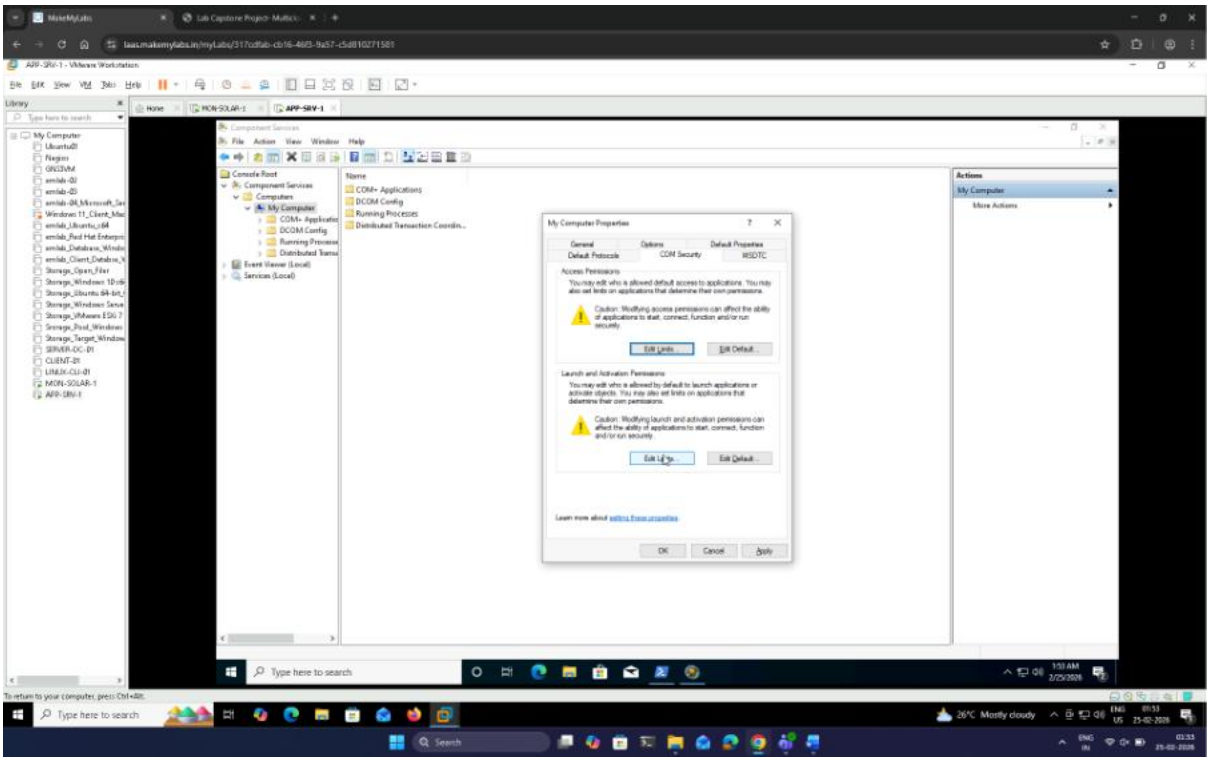


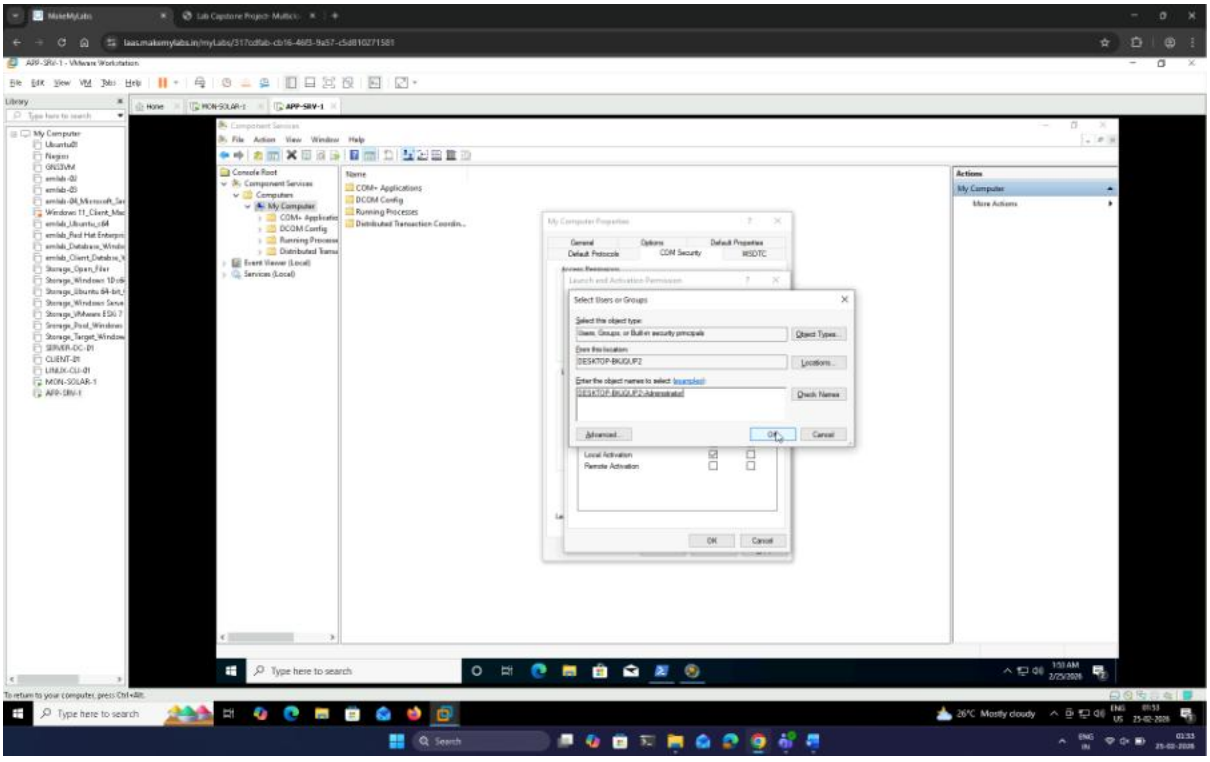
The command is dcomcnfg.

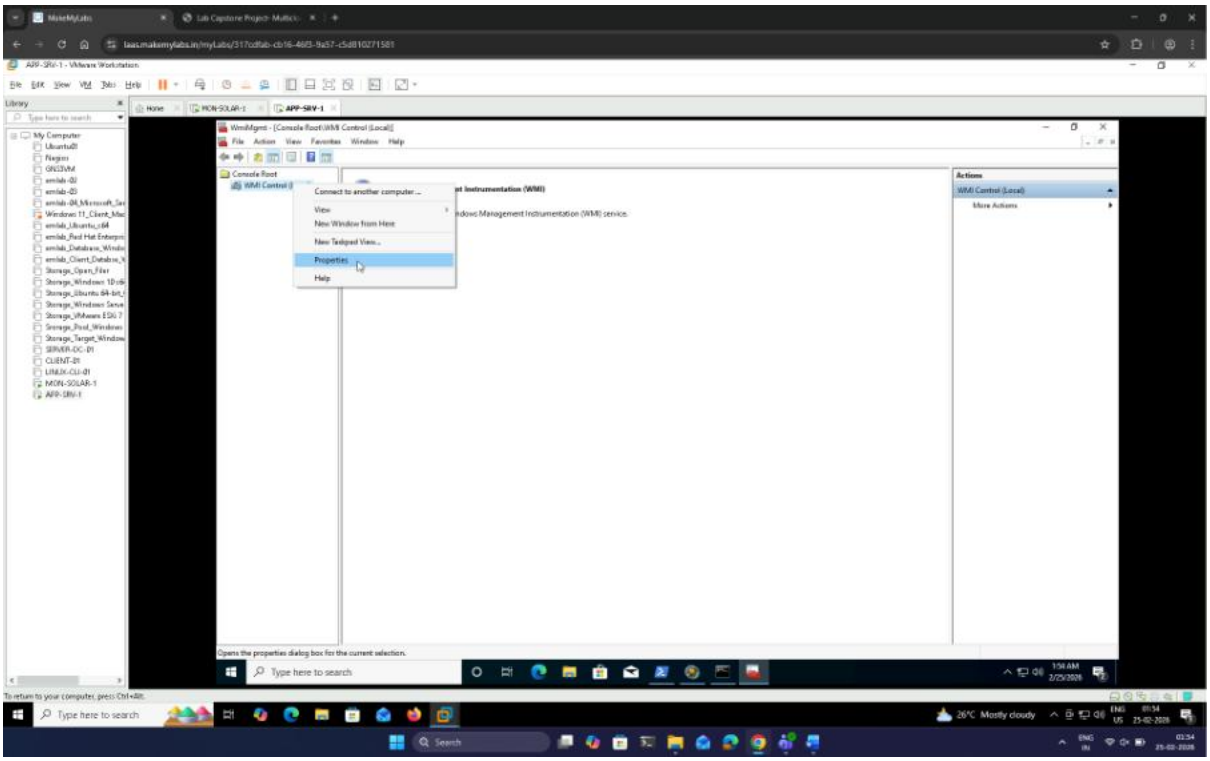


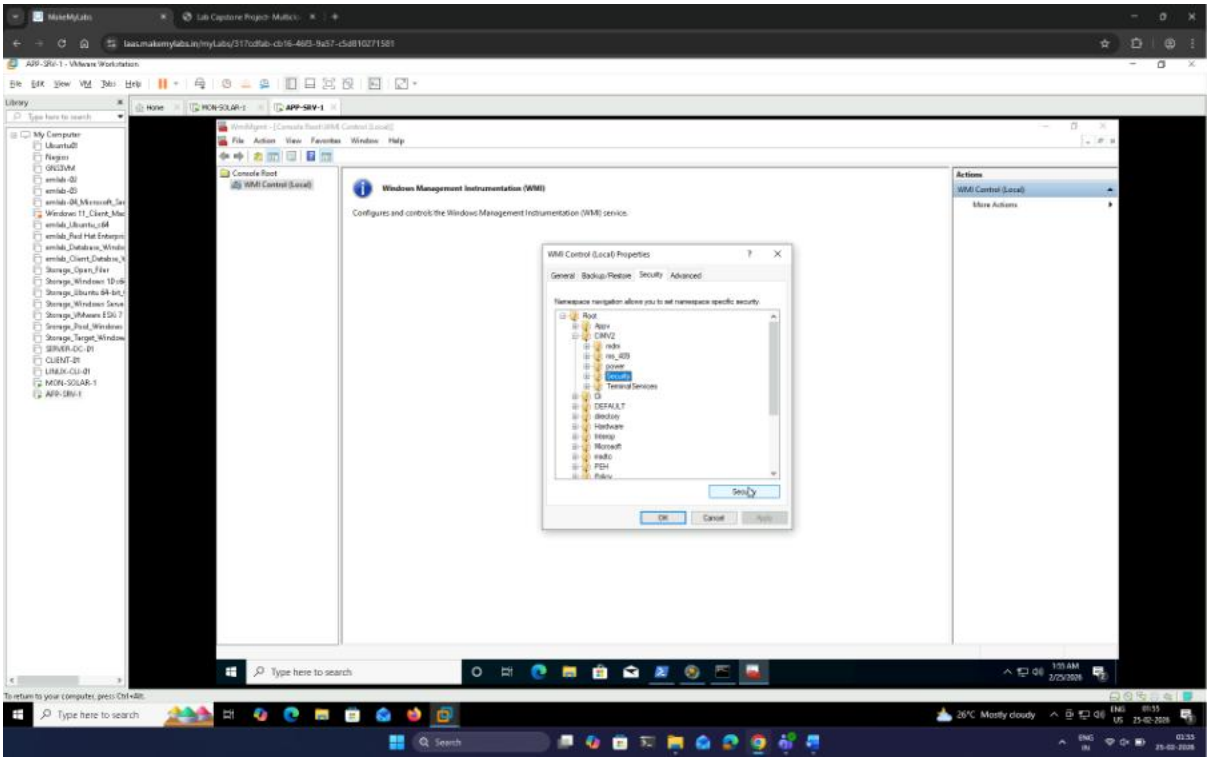


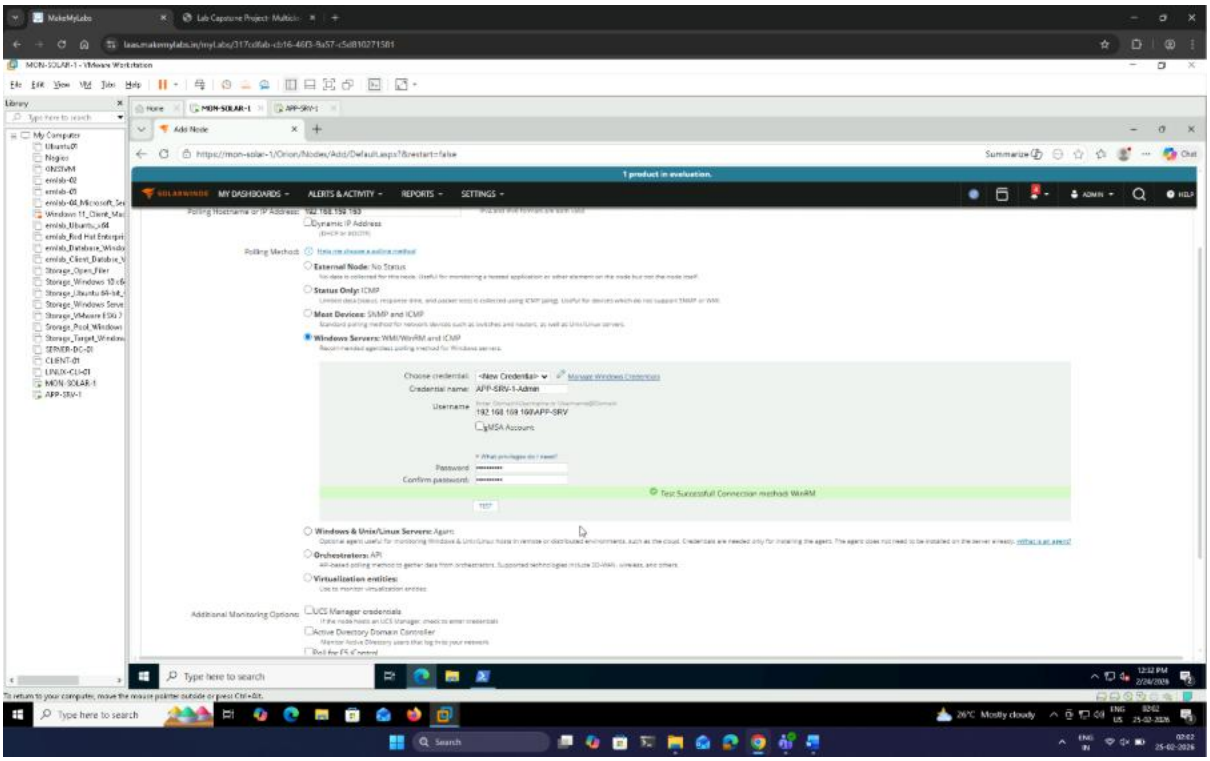







Connect to another computer...
View
New Window from Here
New Taskpad View...
Properties
Help
Opens the properties dialog box for the current selection.


Windows Management Instrumentation (WMI)
Configures and controls the Windows Management Instrumentation (WMI) service.
WMI Control (Local) Properties
General
Backup/Restore
Security
Advanced
OK
Cancel



During the configuration of SolarWinds to monitor the Windows 10 virtual machine (APP-SRV-1), the credential test initially failed with the error:

WinRM Error Code: 0x80131500

WMI Error Code: 0x80070005 – Access is Denied

Although basic network connectivity between MON-SOLAR-1 and APP-SRV-1 was verified using ICMP (ping), SolarWinds was unable to authenticate and collect WMI Data from the remote system.

To resolve this issue, a systematic troubleshooting approach was followed:

First, WinRM was configured and verified on both machines. The command ***Enable-PSRemoting -Force*** was executed on the target machine to enable remote management. The WinRM service status was verified, and connectivity was tested from the monitoring server using the ***Test-WsMan*** command, which confirmed that WinRM communication was functioning correctly.

Since the environment was configured in a workgroup (non-domain) setup, TrustedHosts was configured on the SolarWinds server using:

Set-Item WSMan:\localhost\Client\TrsutedHosts -Value "*"

The WInRM service was restarted to apply the changes. This ensured that NTLM authentication could function between workgroup systems.

Next, User Account Control (UAC) remote restrictions were addressed by verifying and configuring the registry key:

LocalAccountTokenFilterPolicy = 1

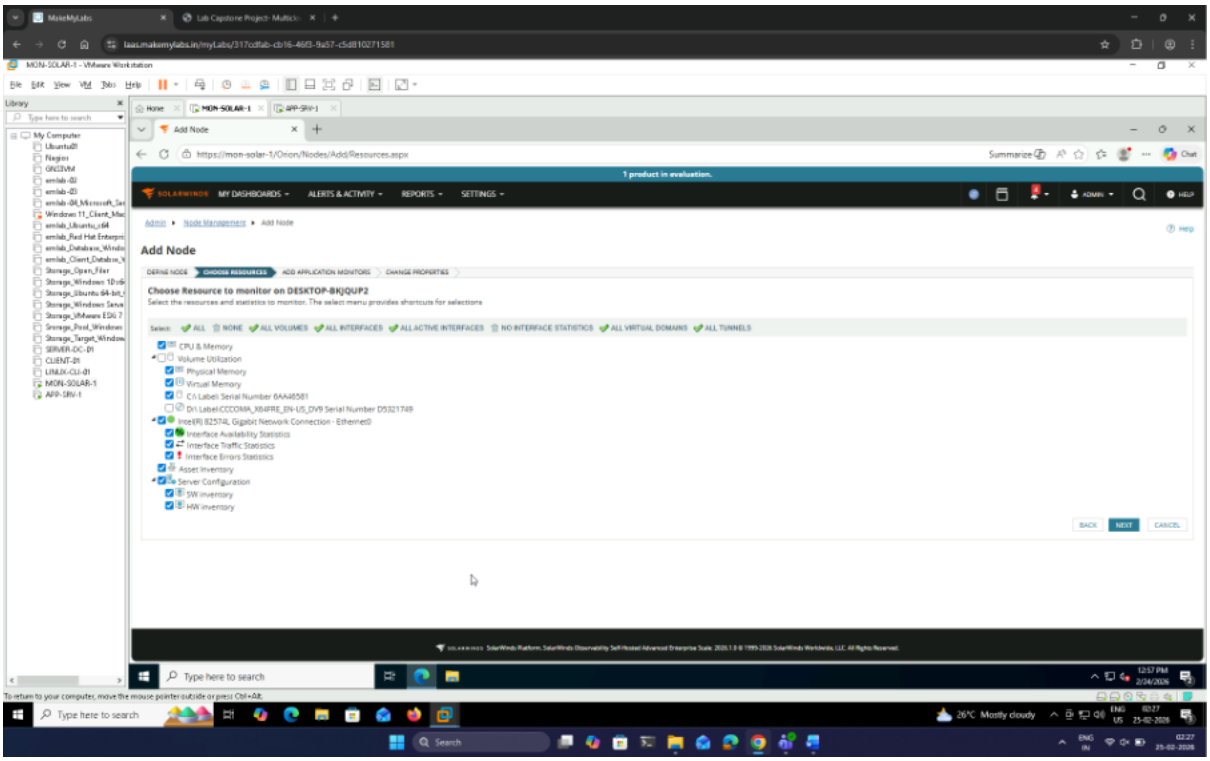
This modification allowed local administrative accounts to perform remote administrative tasks without being blocked by UAC filtering. Despite these configurations, access was still denied. Further analysis identified that Distribured COM (DCOM) and VMI namespace security permissions were restricting remote access.

The following additional steps were performed on APP-SRV-1: -

- Enabled Distributed COM via ***dcomcmfg.***
- Configured COM security settings to allow Remote Access, Remote Launch and Remote Activation permissions for the Administrative account.
- Modified VMI Control (***wmimgmt.msc***) namespace security under ***Root\CIMV2*** to grant Remote Enable and Execute Methods permissions.
- Verified and enabled necessary firewall rules for Windows Management Instrumentation (WMI).

After applying these configurations and restarting the system, SolarWinds was successfully able to authenticate and establish WMI-Based monitoring using Windows credentials.

This troubleshooting process demonstrated the importance of properly configuring WinRM, WMI, DCOM, firewall rules and local security policies while implementing agentless monitoring in a workgroup-based environment.


Add Node
Add Node
Choose Resource to monitor on DESKTOP-BKJQUP2
CPU & Memory
Volume Utilization
Physical Memory
Virtual Memory
Interface Availability Statistics
Interface Traffic Statistics
Interface Errors Statistics
Asset Inventory
Server Configuration
SW inventory
HW inventory
BACK
NEXT
CANCEL